



Towards Quantitative Visual Analytics with Structured Brushing and Linked Statistics

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Mario Đuras², Eduard Gröller^{3,4}, and Helwig Hauser⁴

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²AVL Zagreb, Croatia

³TU Wien, Austria

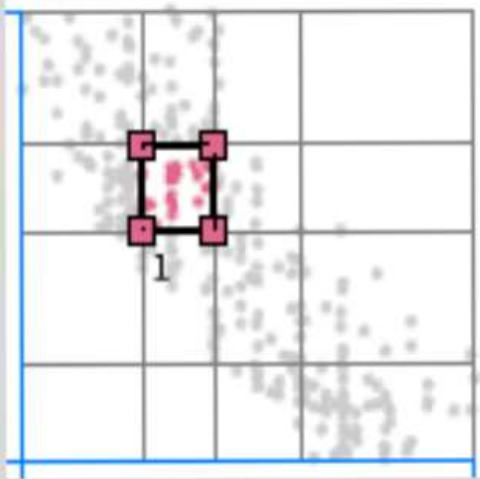
⁴University of Bergen, Norway



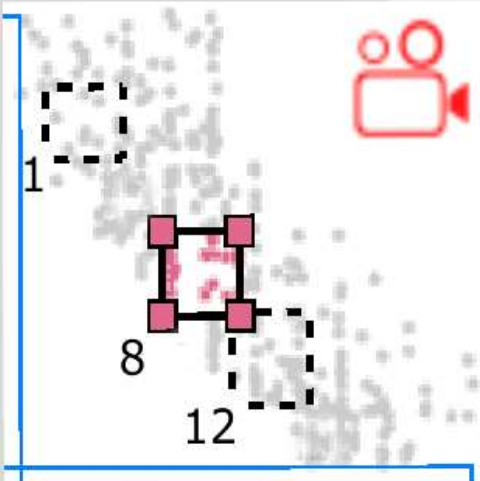
Extensions for Reproducible and Quantitative Visual Analytics

Structured Brushing

Constrained

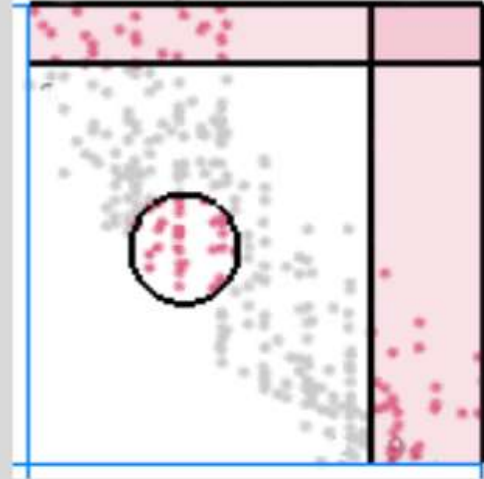


Automatic

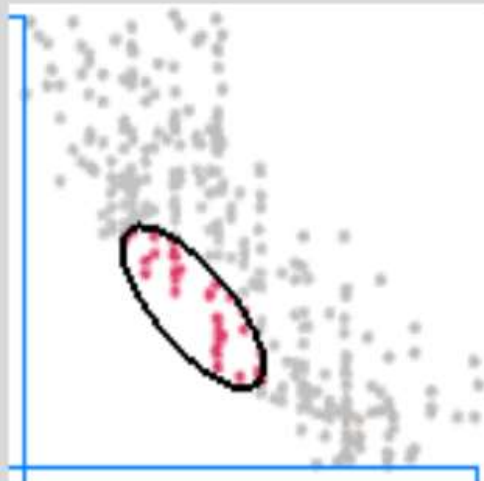


Percentile Brushes

Circle/Rectangle

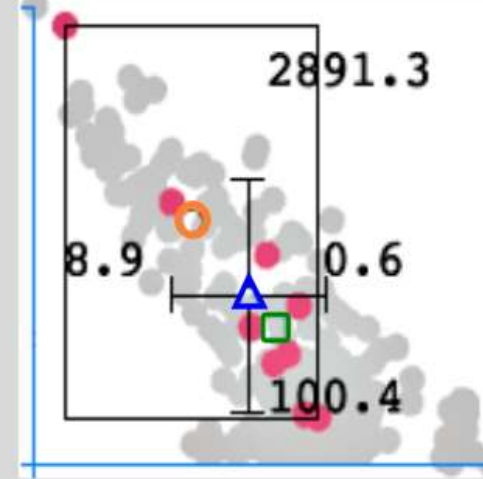


Mahalanobis



Linked Statistics

Overlays

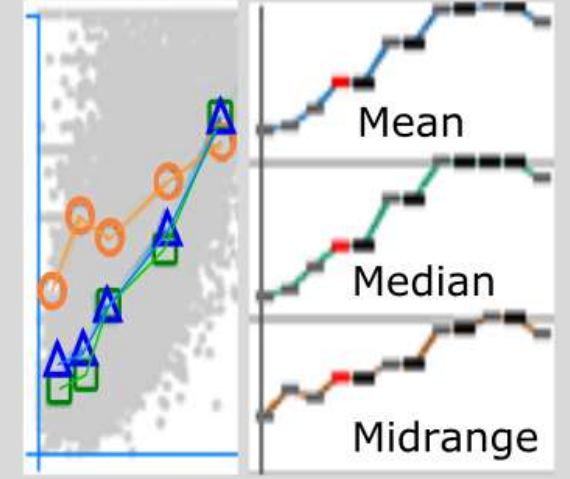


Numerical Tables

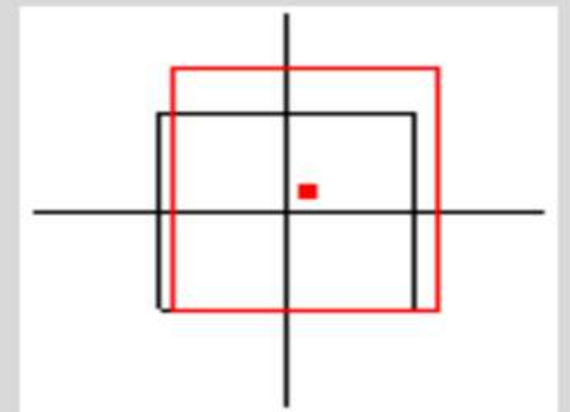
	X	Y
△ Mean	55.8	2933.6
□ Median	57.9	2355.0
○ Midrange	51.0	4329.0
Spread	6.4	2175.1

Change Visualization

Traces/Graphs



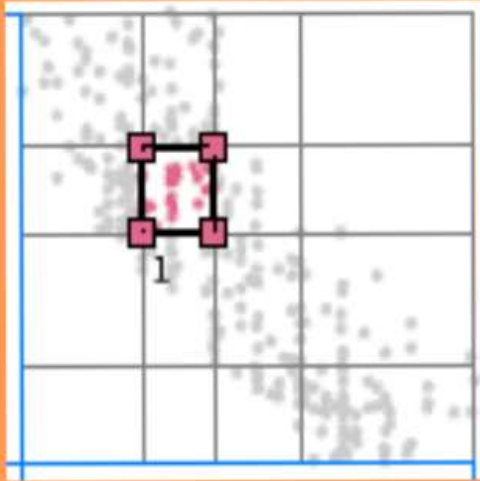
Difference-Plot



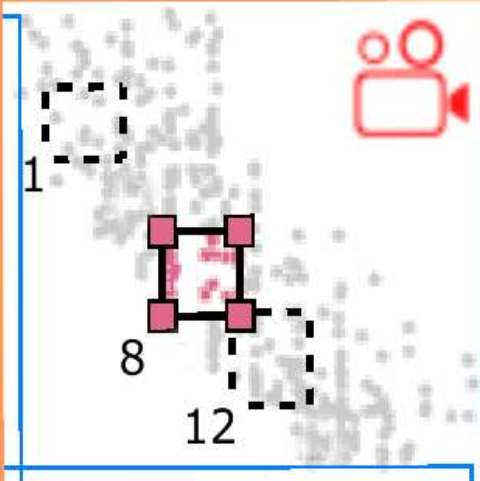
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Structured Brushing

Constrained

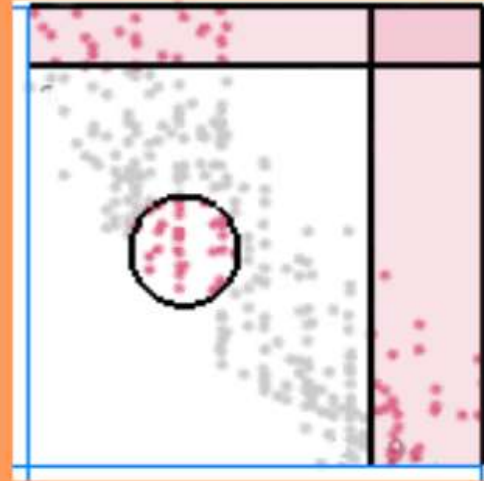


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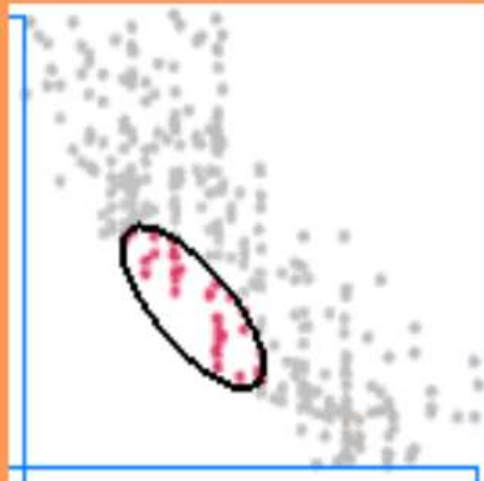


Percentile Brushes

Circle/Rectangle

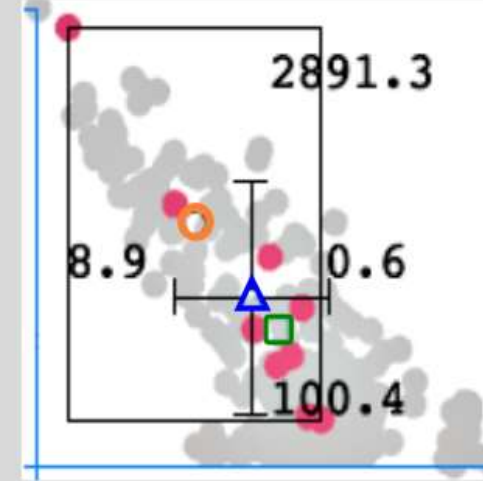


Mahalanobis



Linked Statistics

Overlays

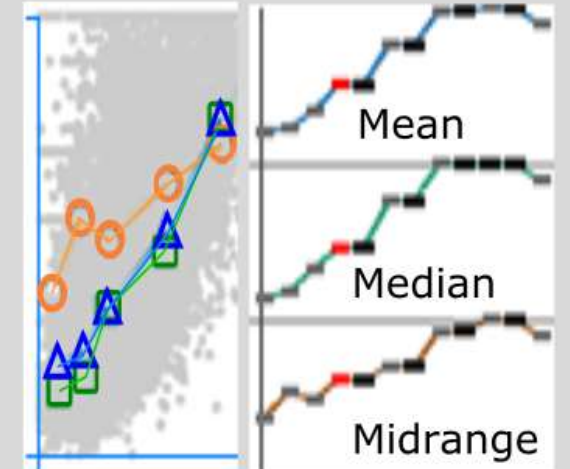


Numerical Tables

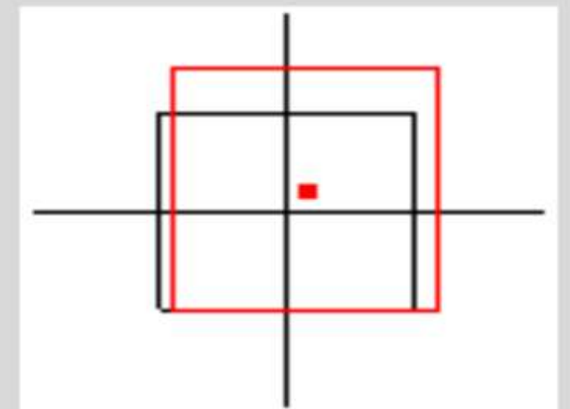
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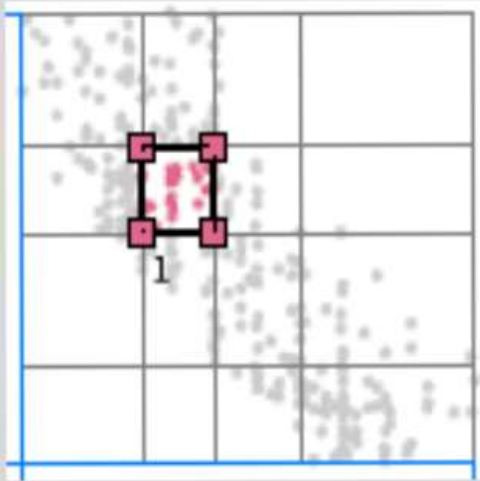
Difference-Plot



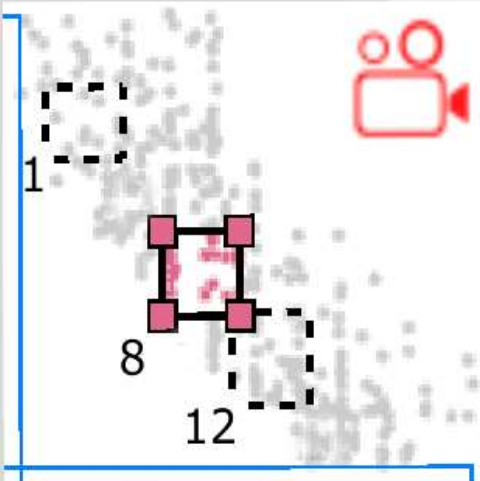
Extensions for Reproducible and Quantitative Visual Analytics

Structured Brushing

Constrained

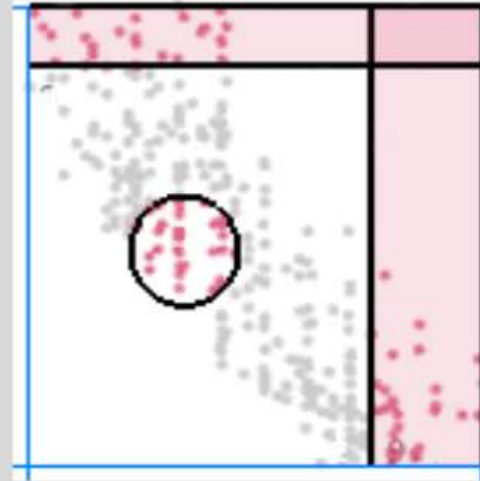


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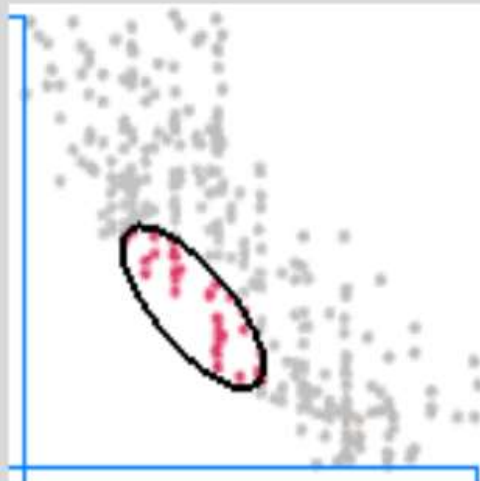


Percentile Brushes

Circle/Rectangle

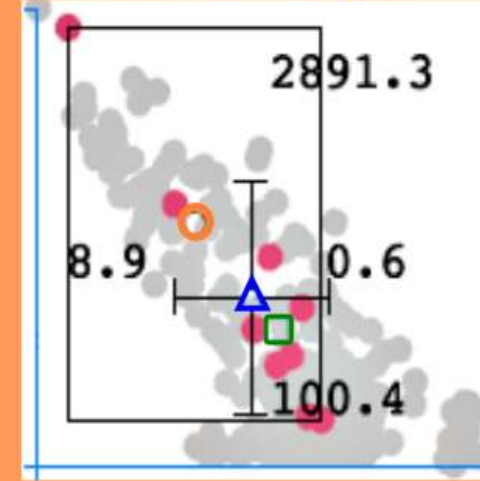


Mahalanobis



Linked Statistics

Overlays

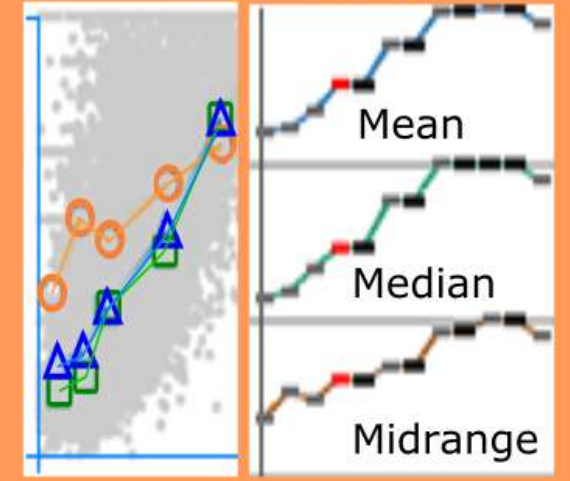


Numerical Tables

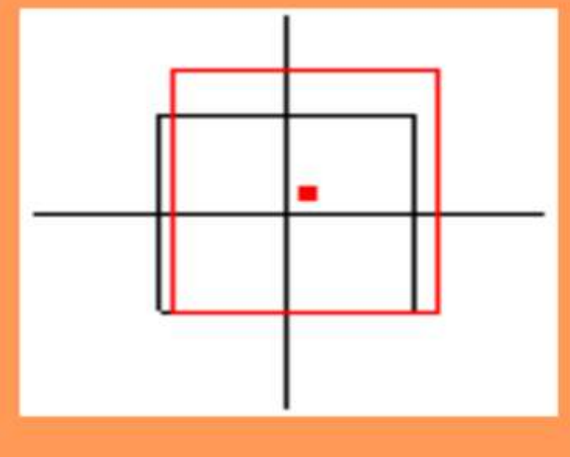
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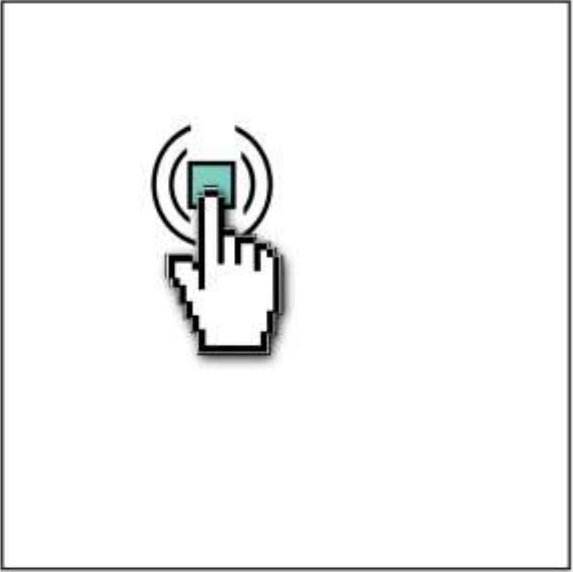
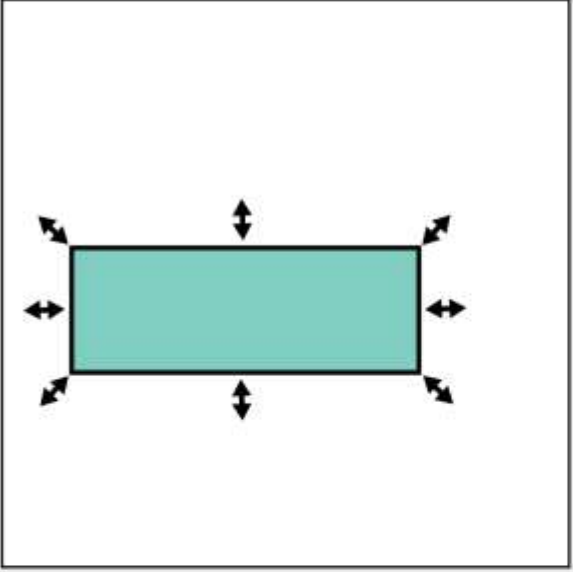
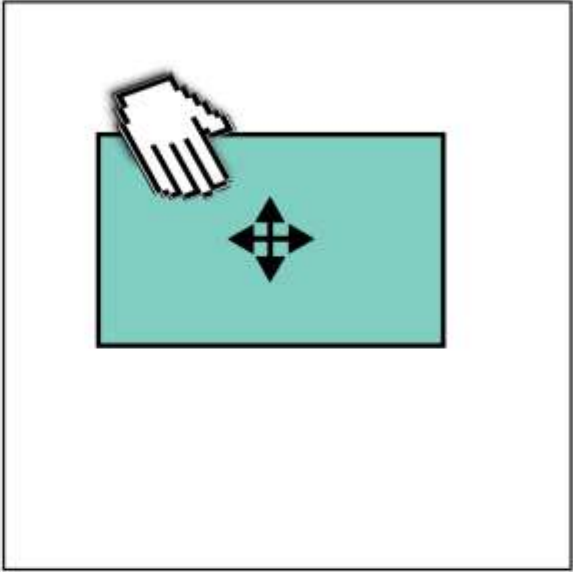
Change Visualization

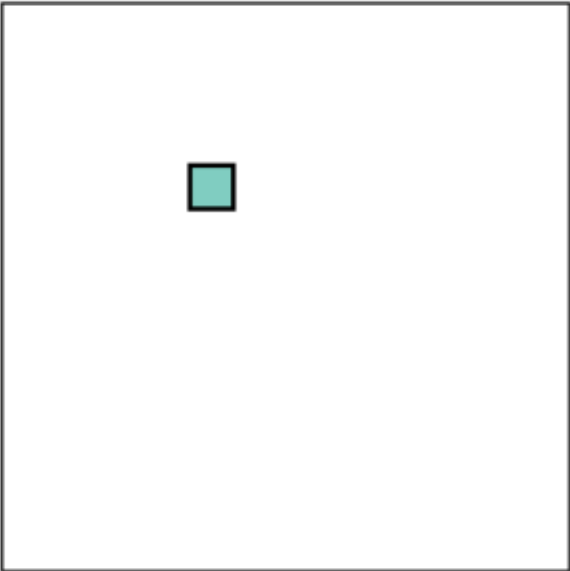
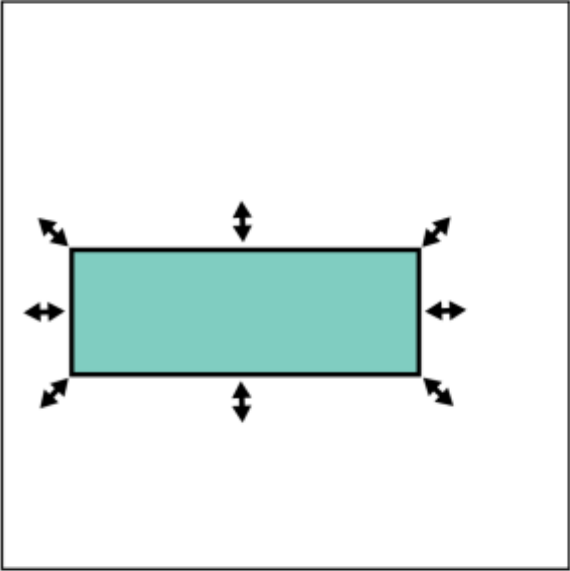
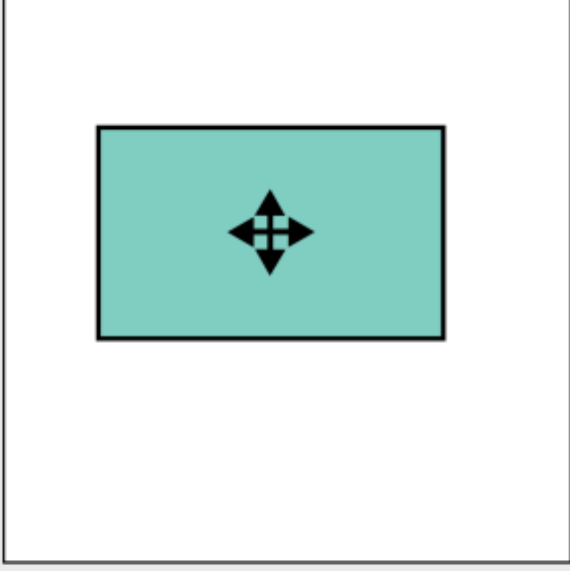
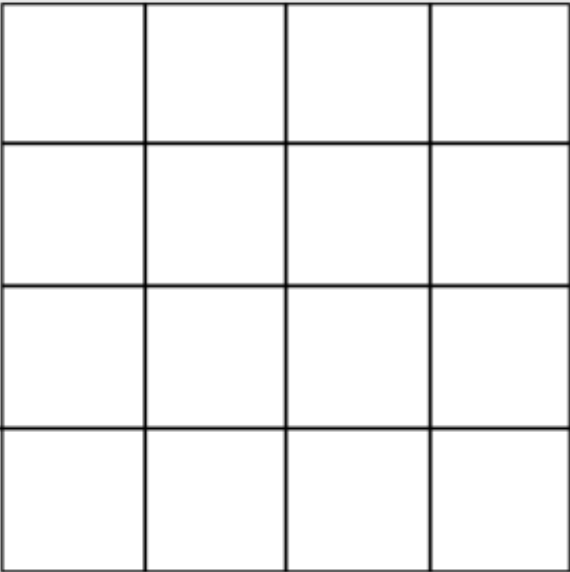
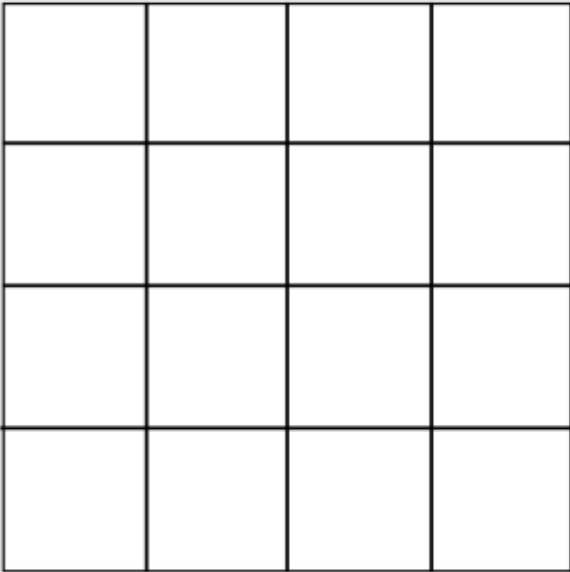
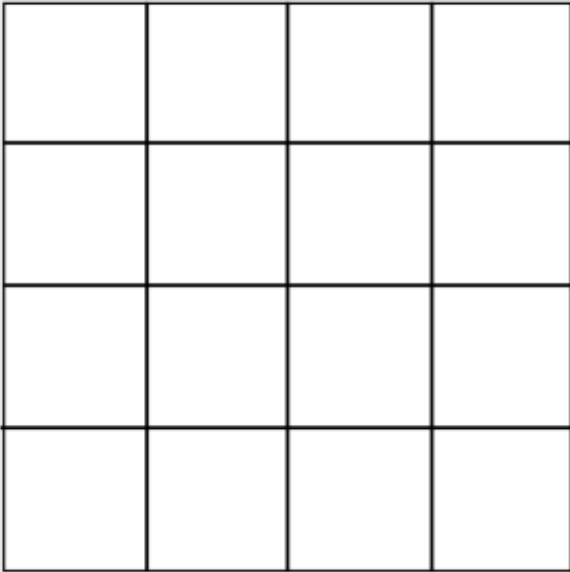
Traces/Graphs

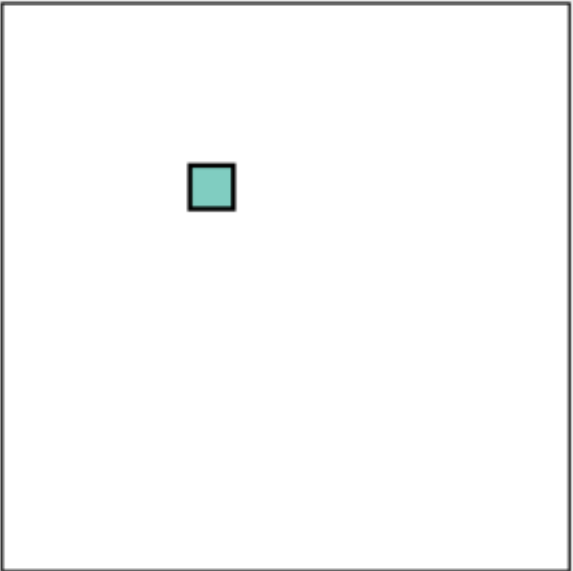
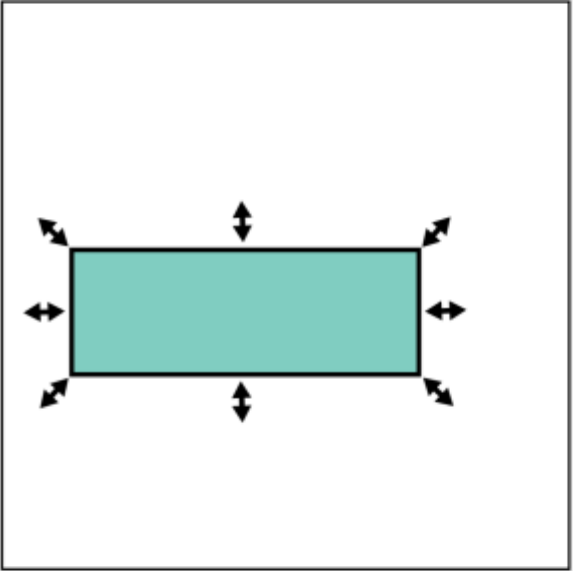
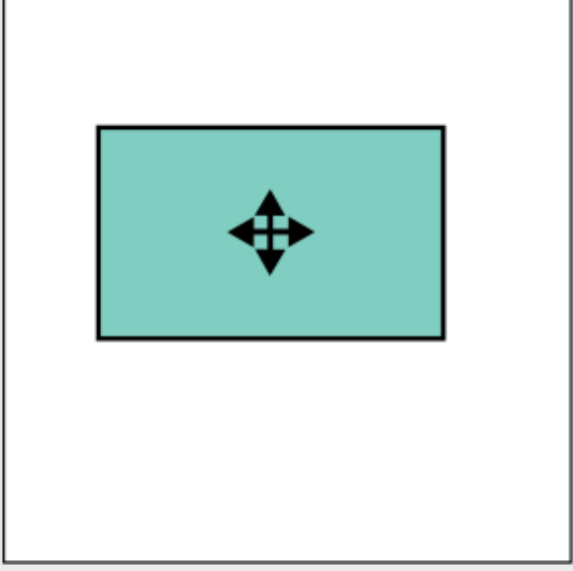
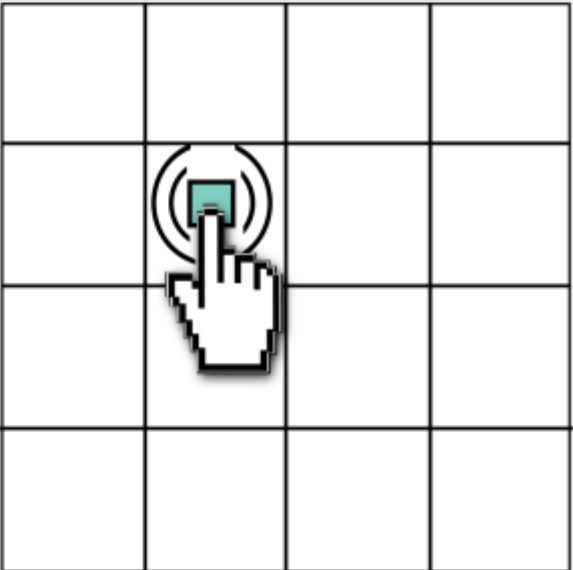
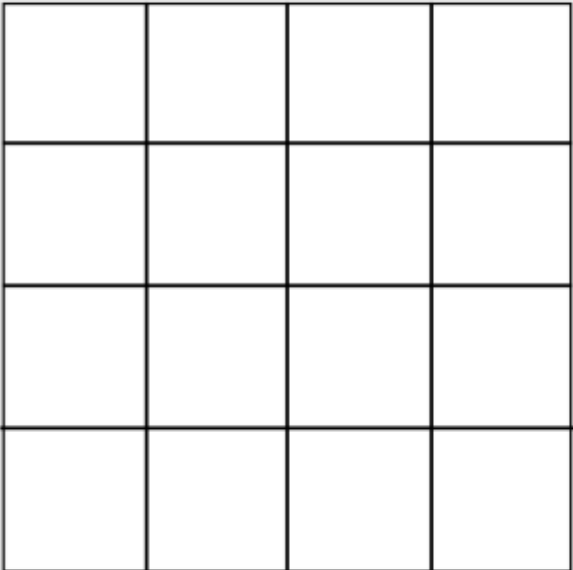
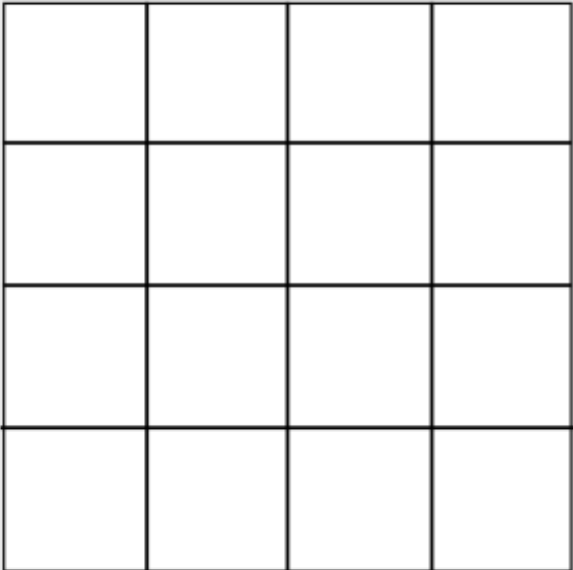


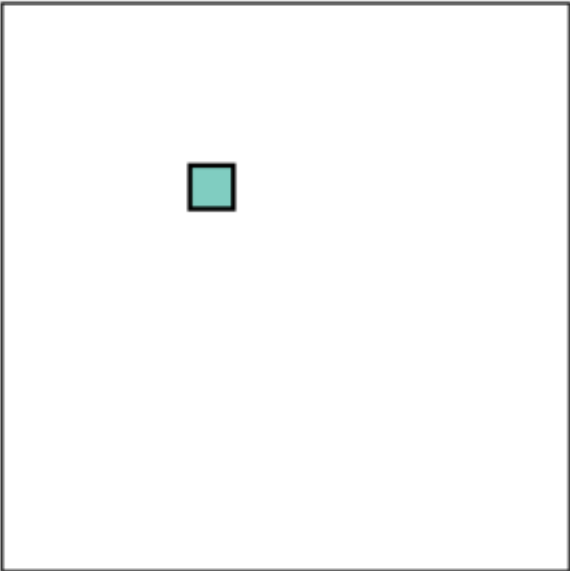
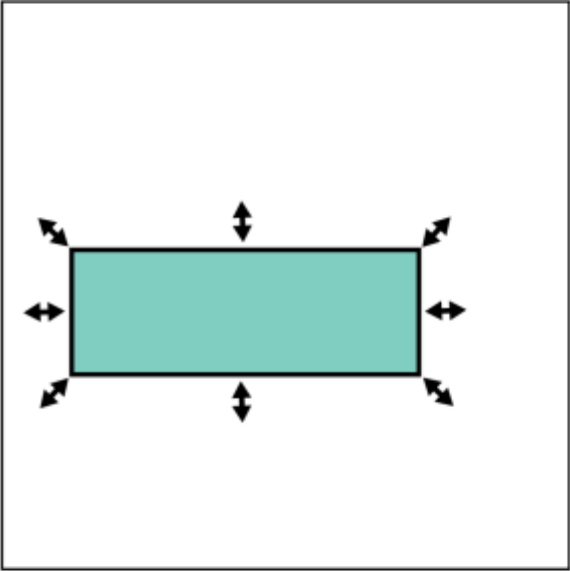
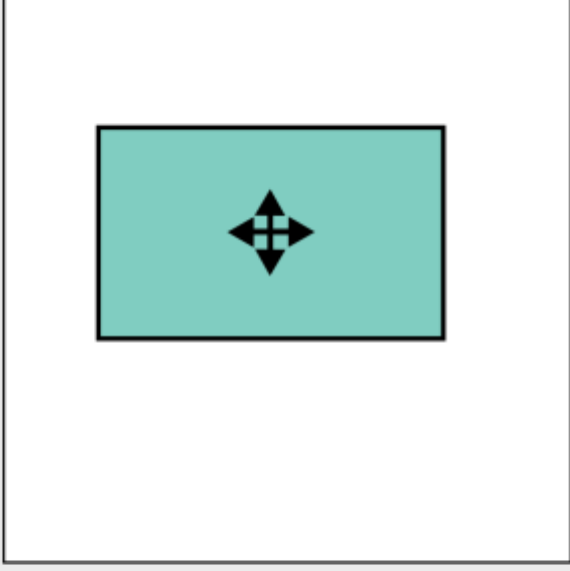
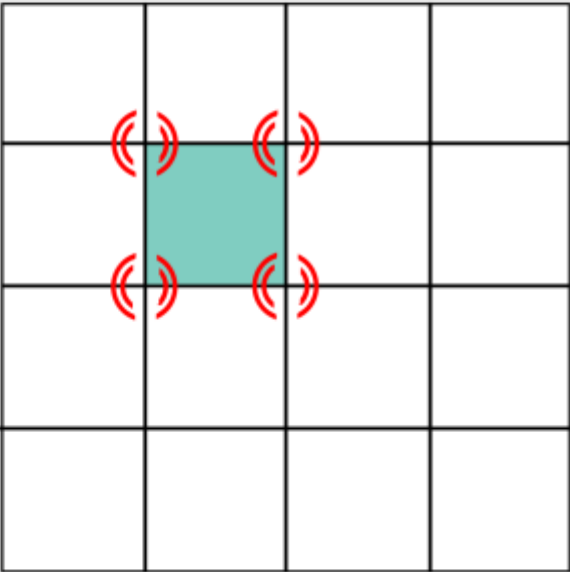
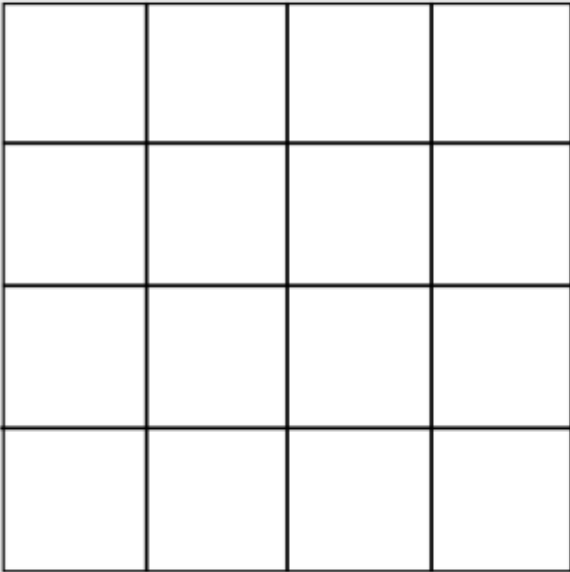
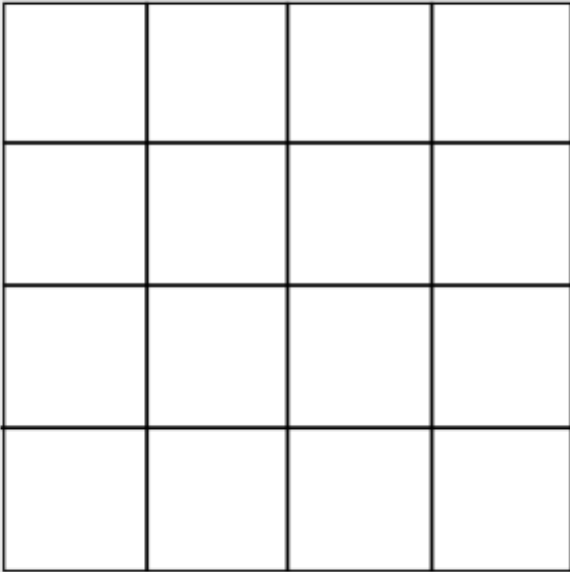
Difference-Plot

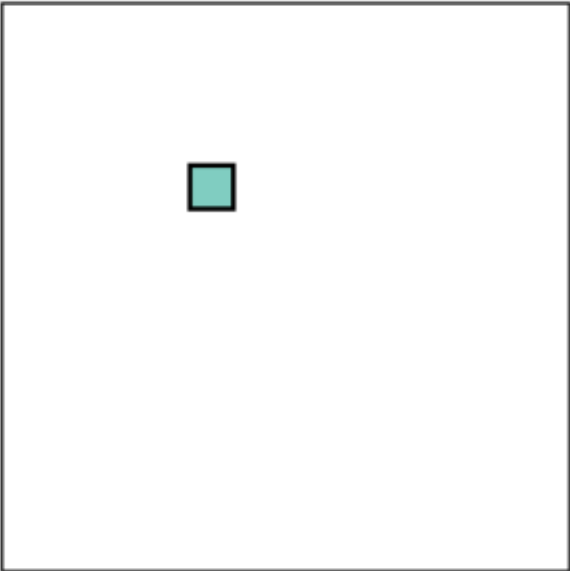
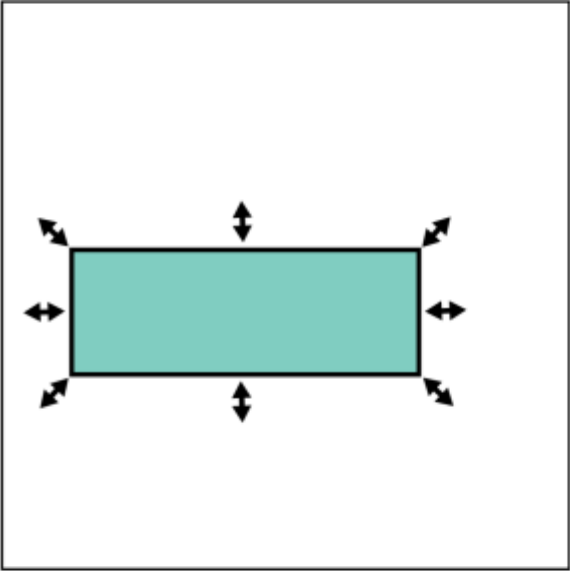
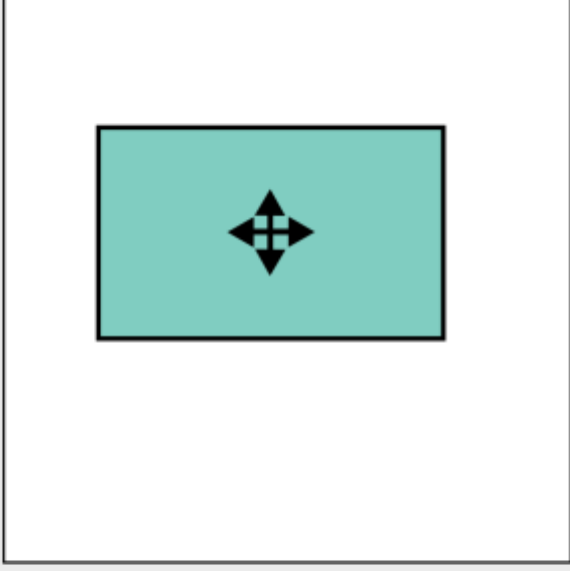
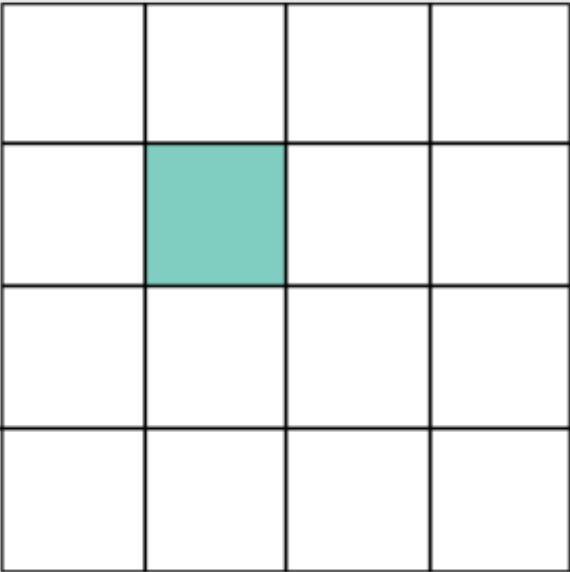
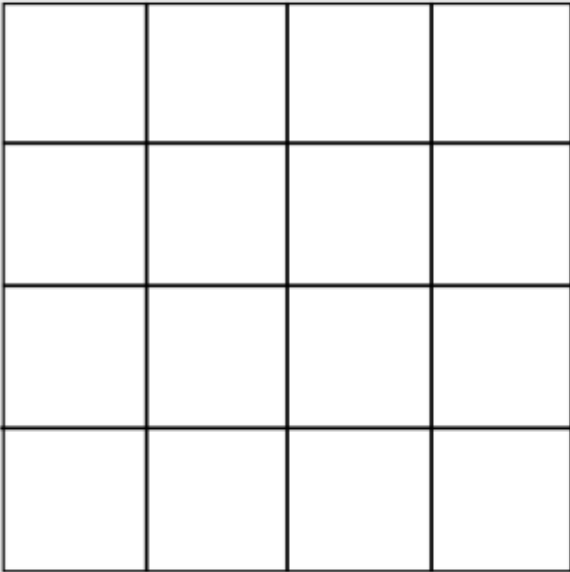
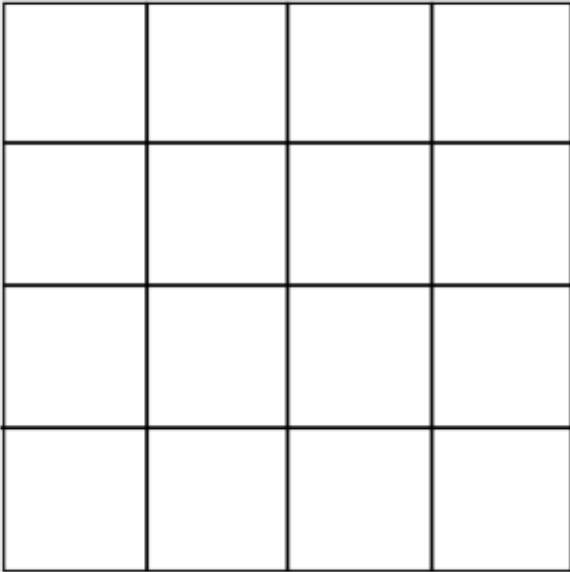


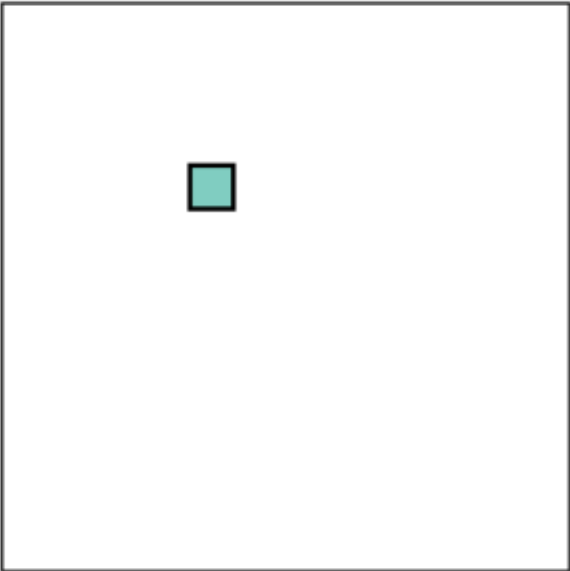
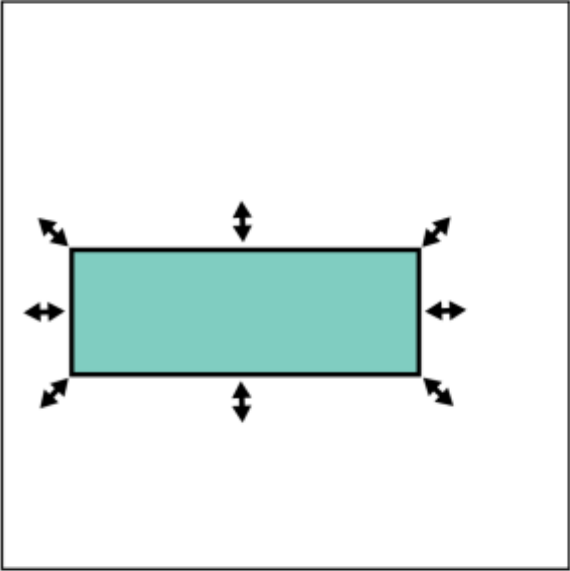
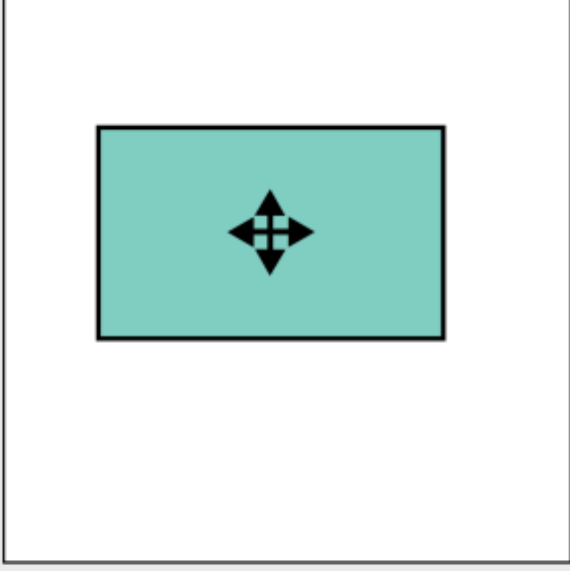
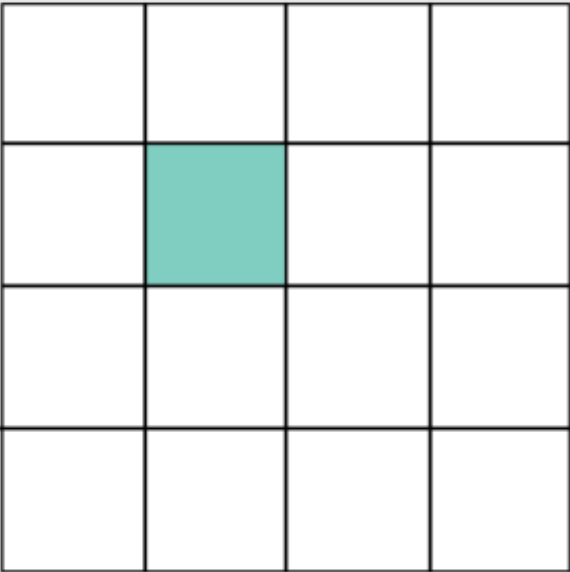
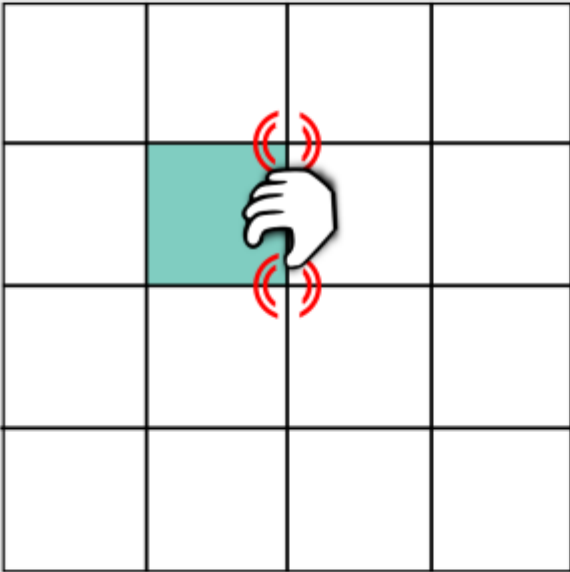
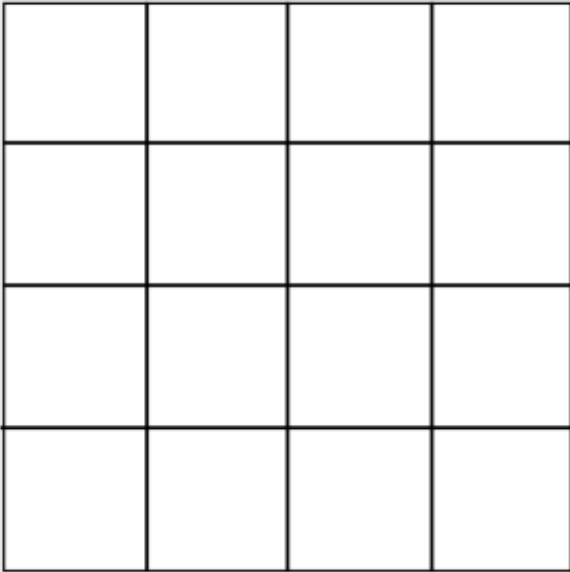
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Constrained			
Automatic			

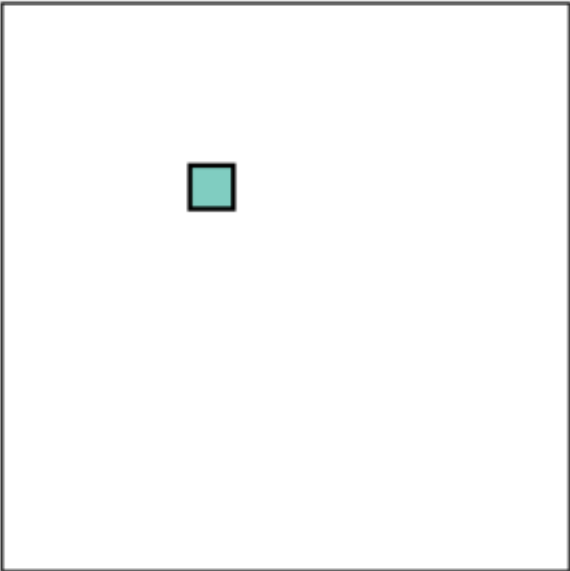
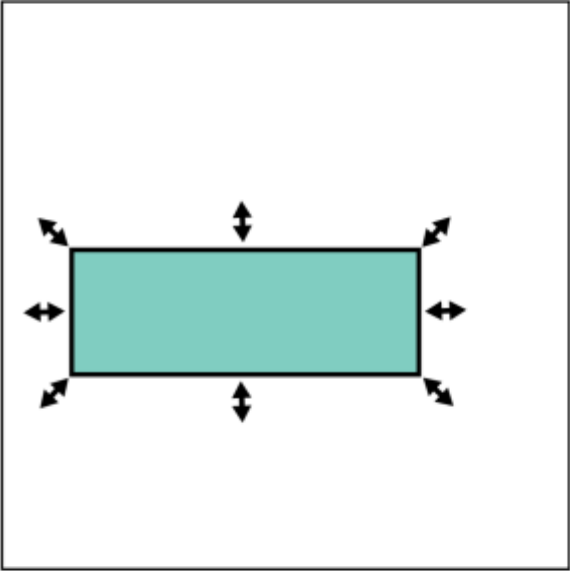
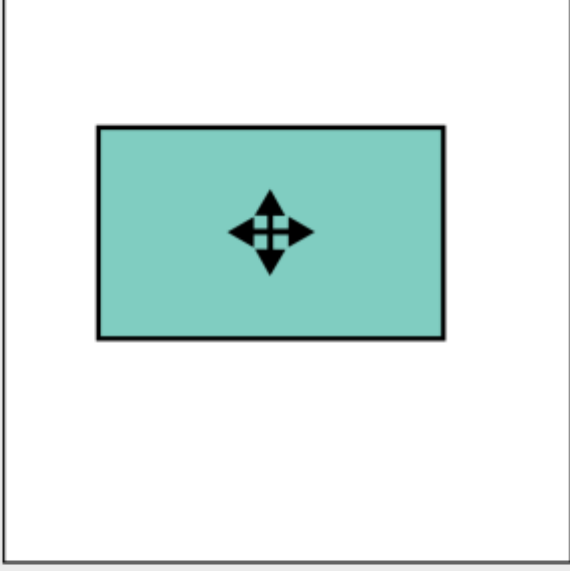
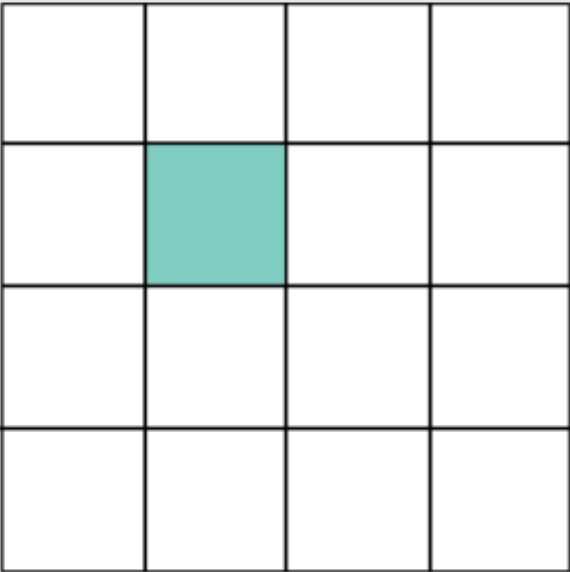
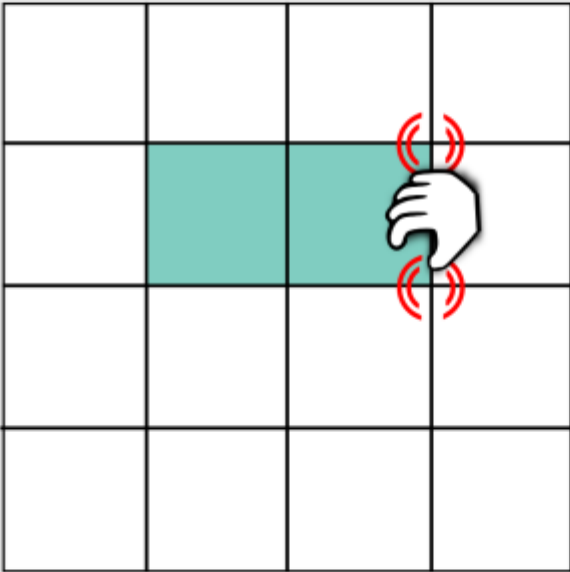
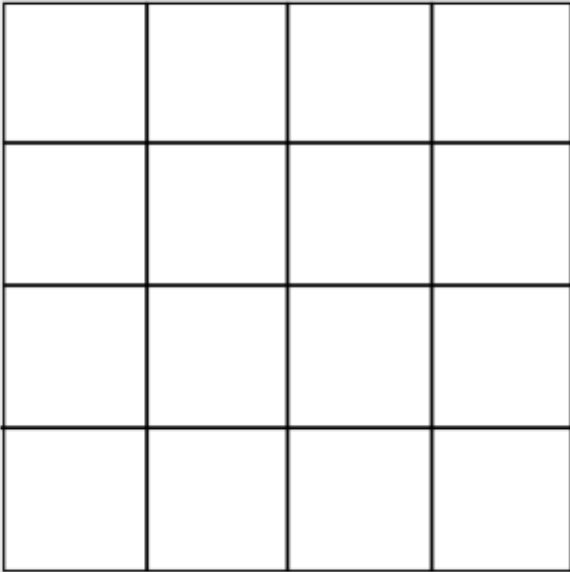
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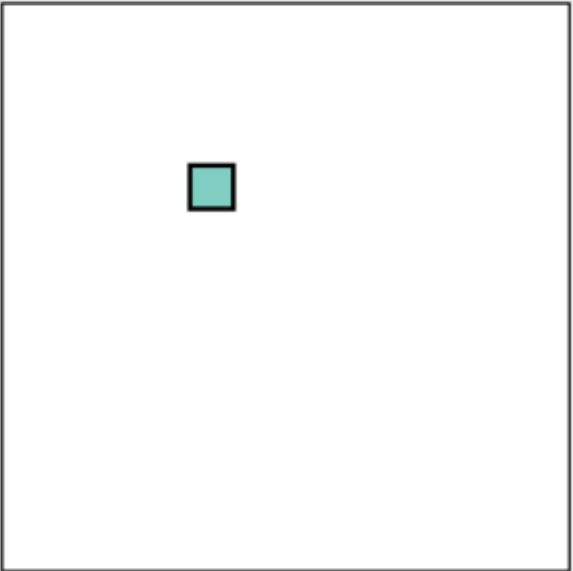
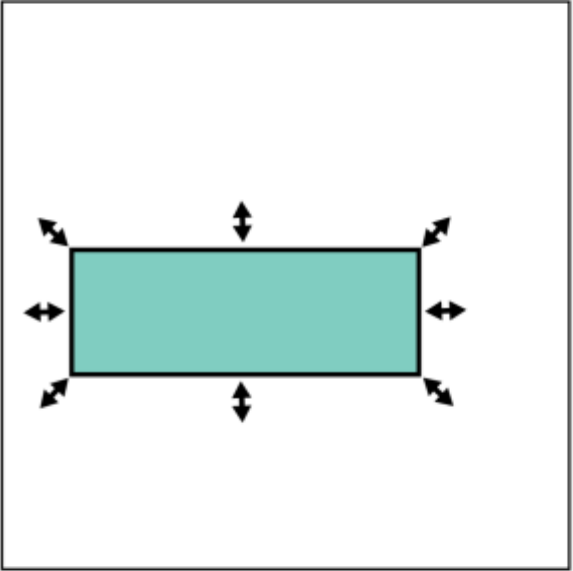
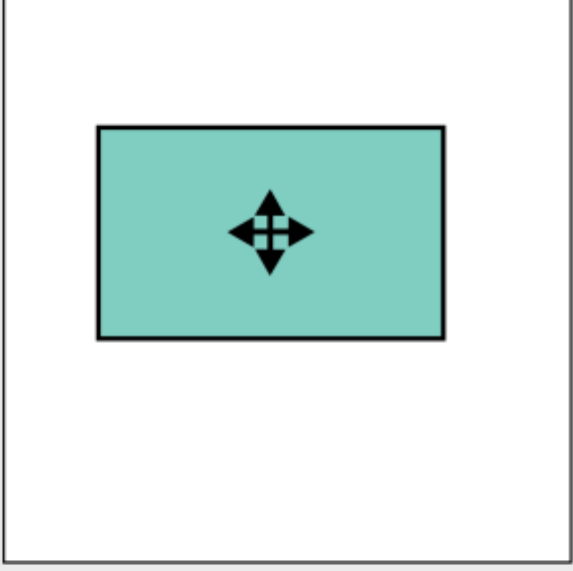
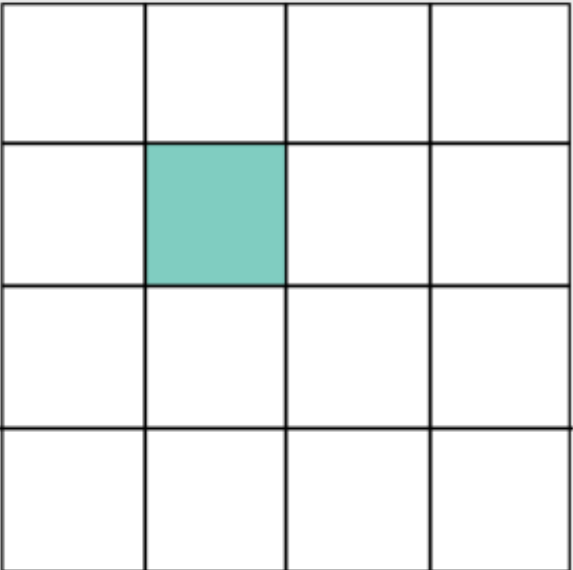
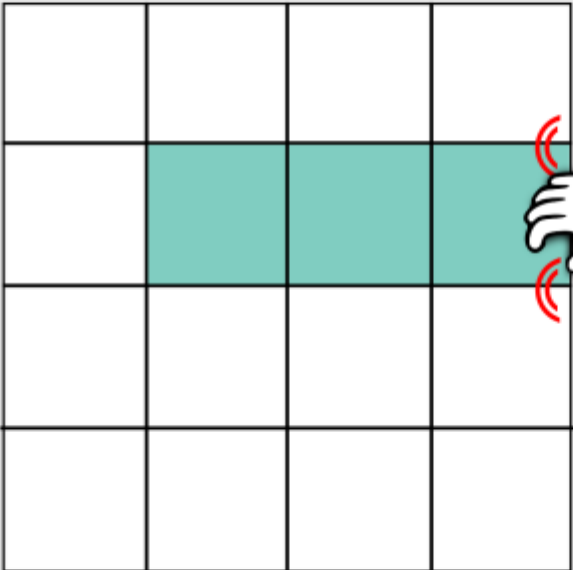
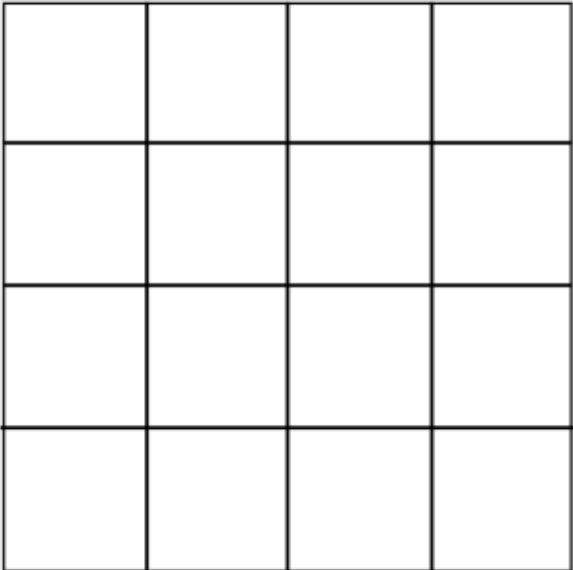
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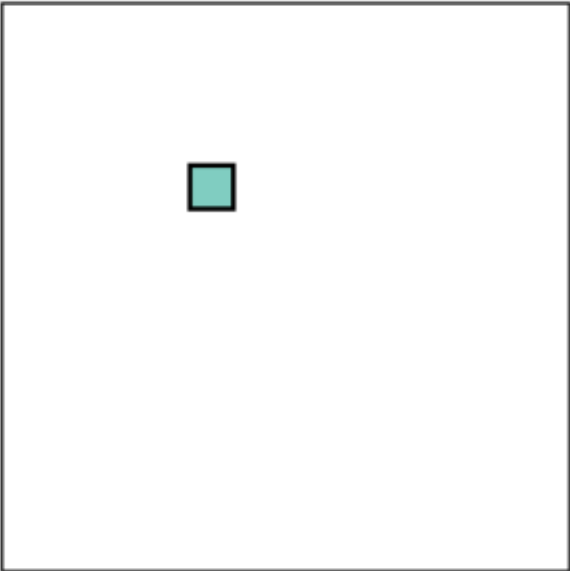
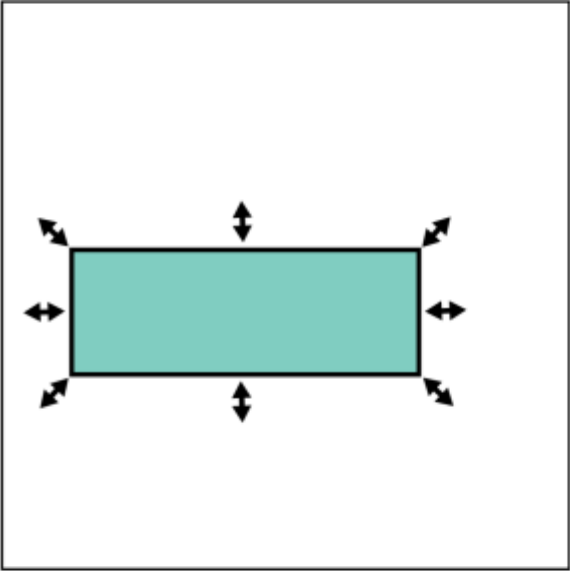
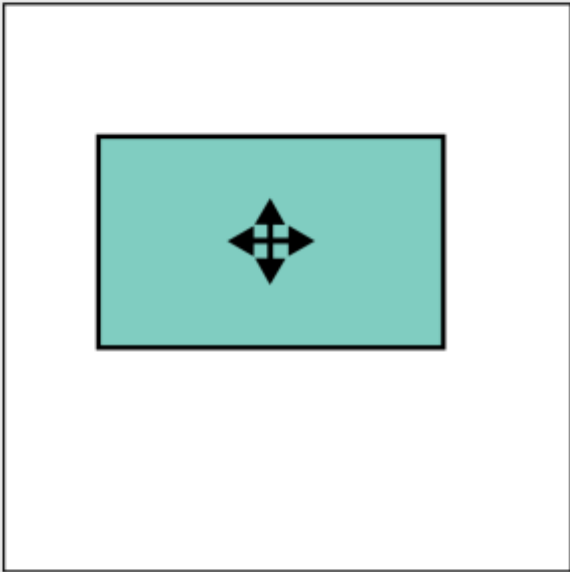
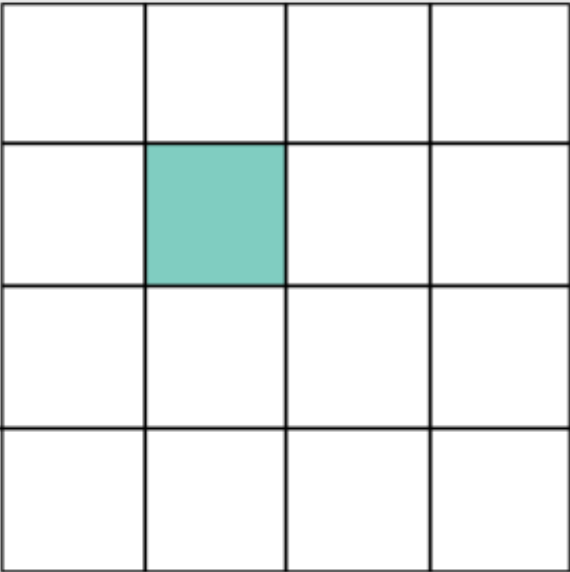
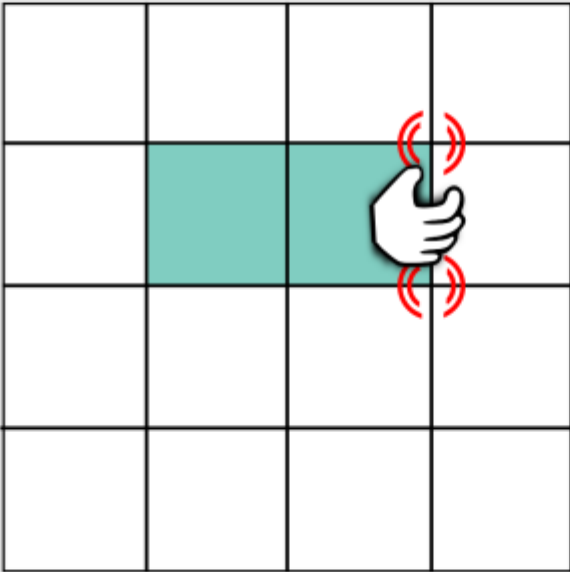
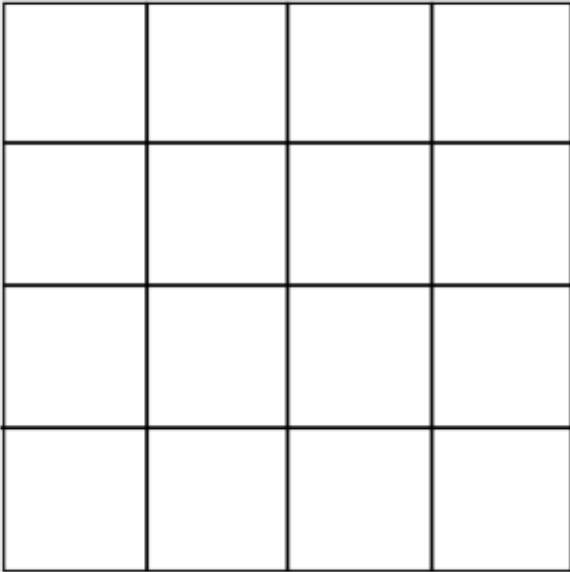
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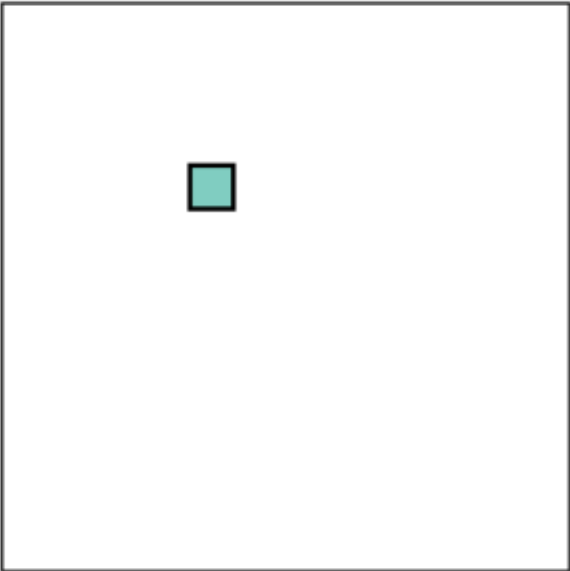
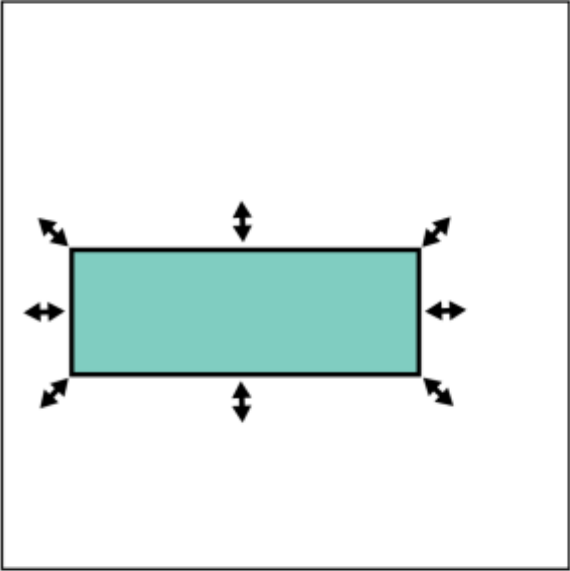
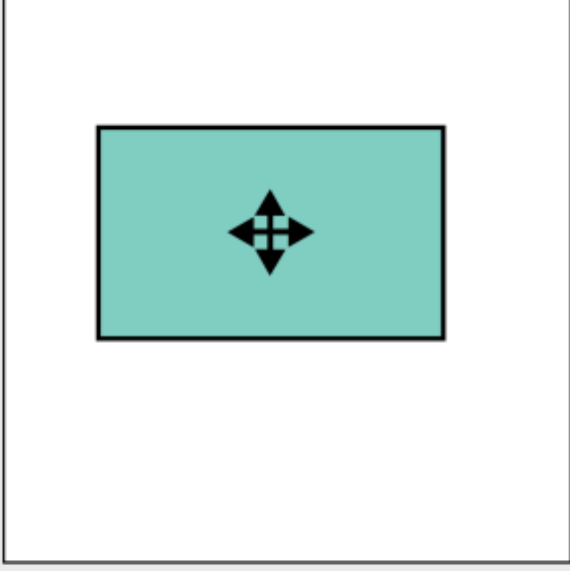
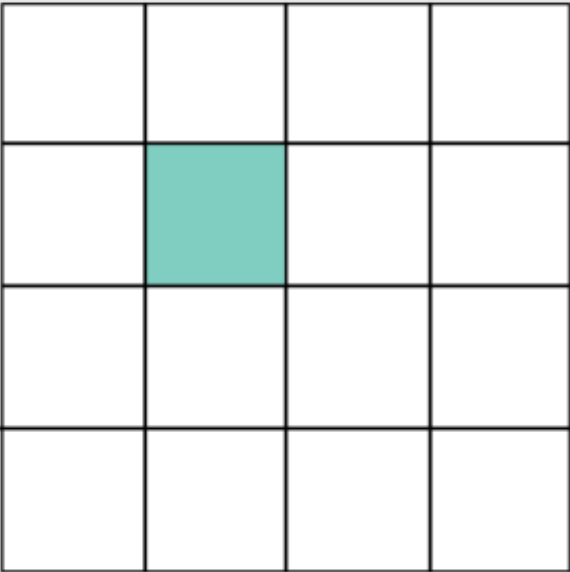
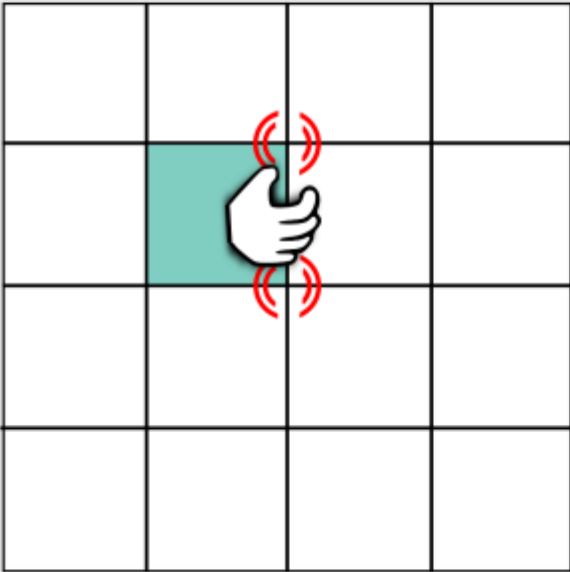
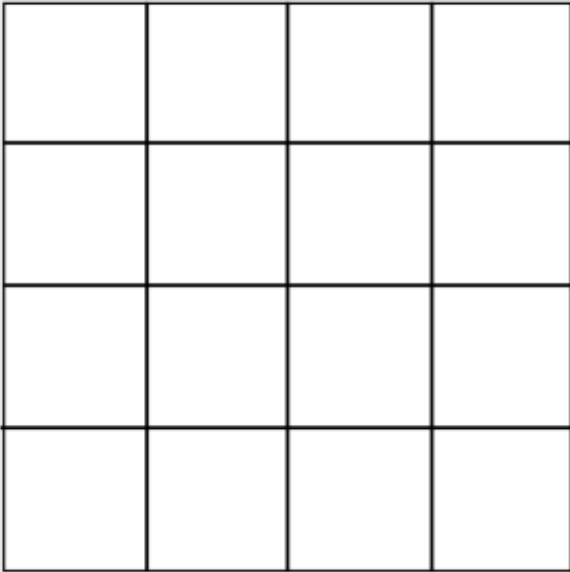
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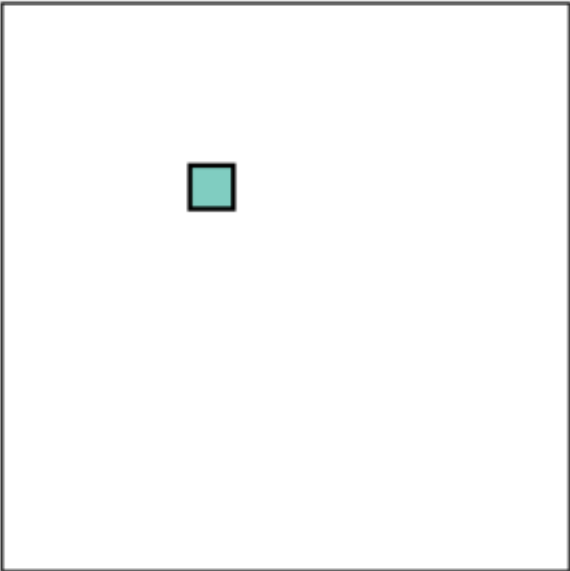
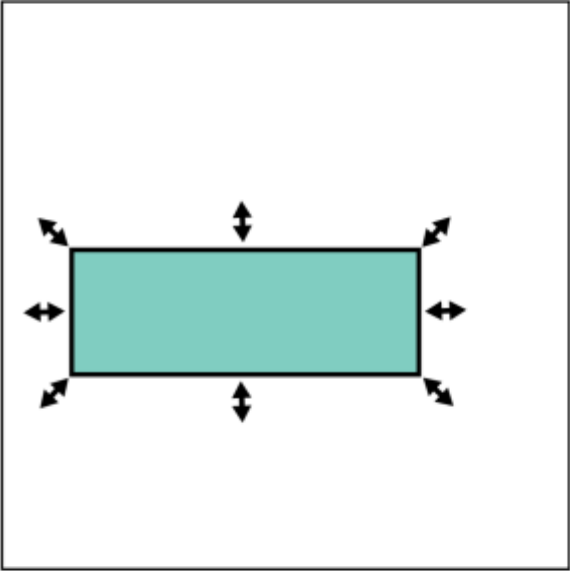
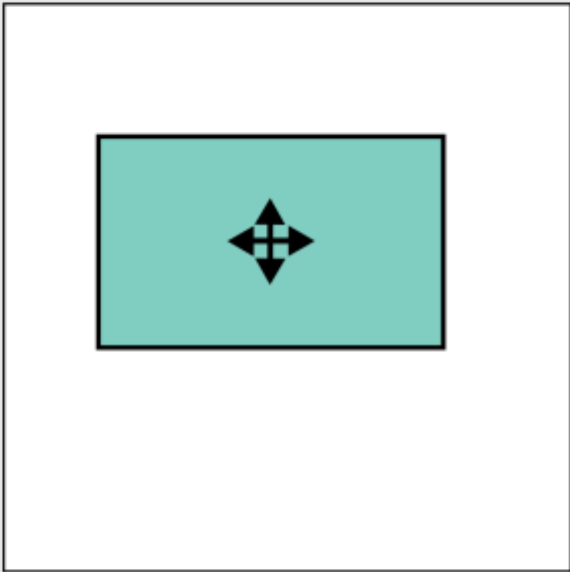
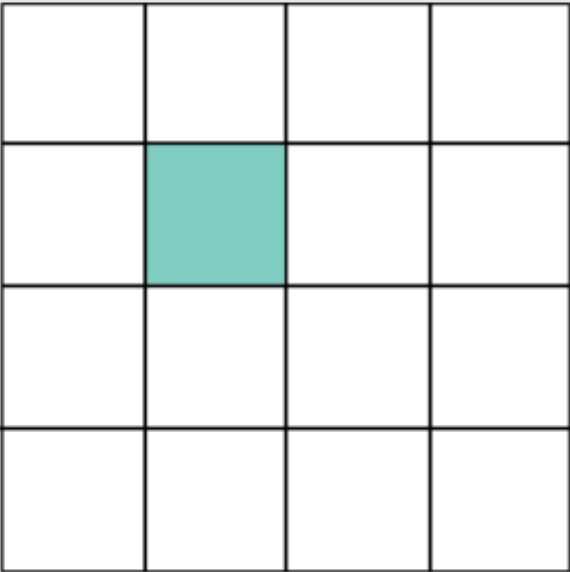
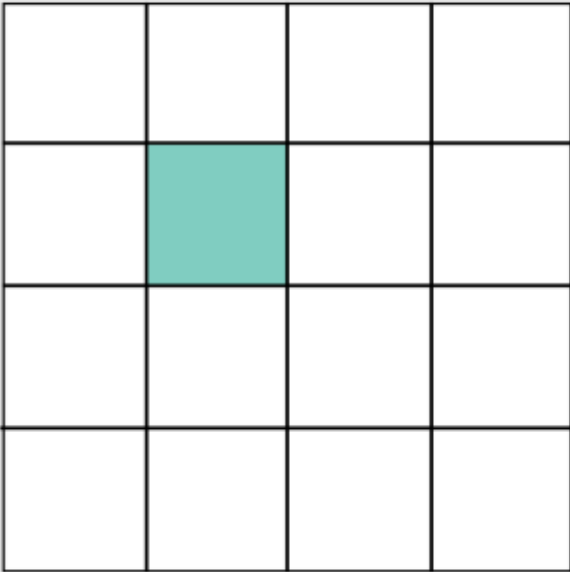
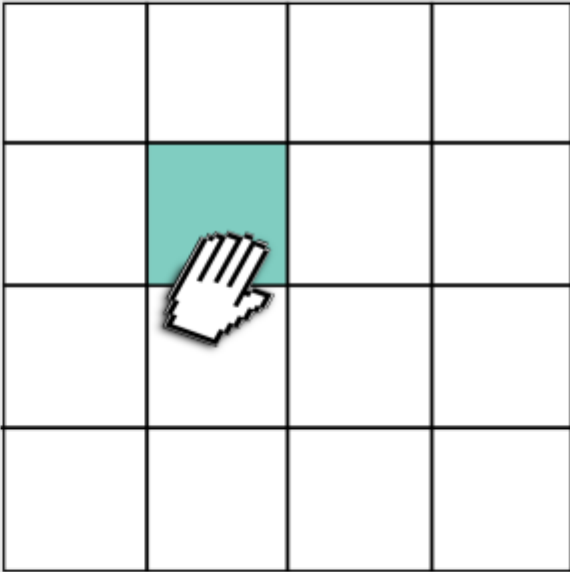
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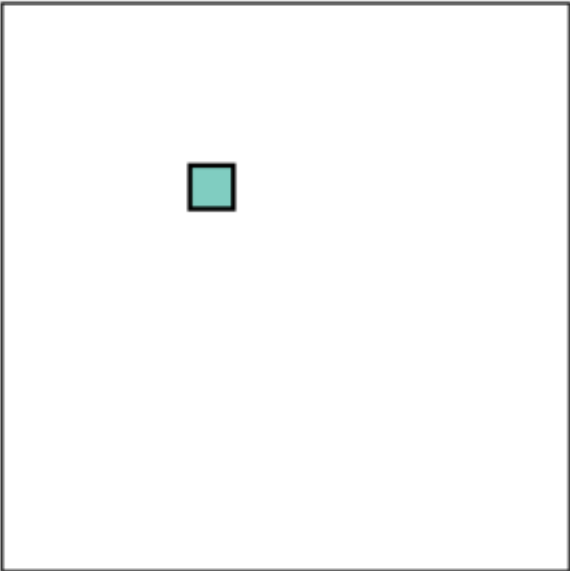
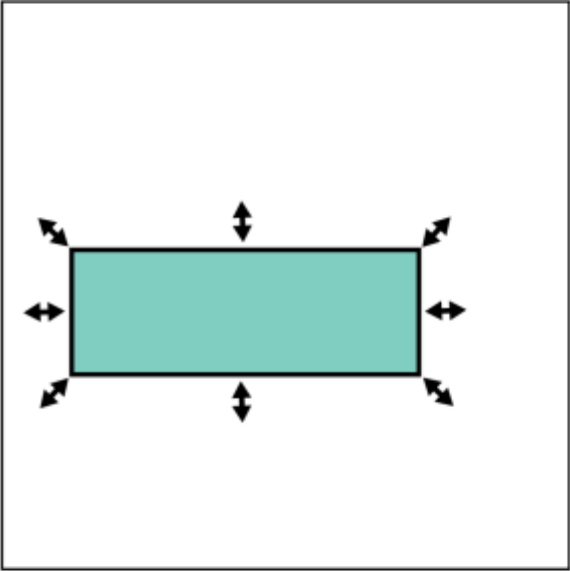
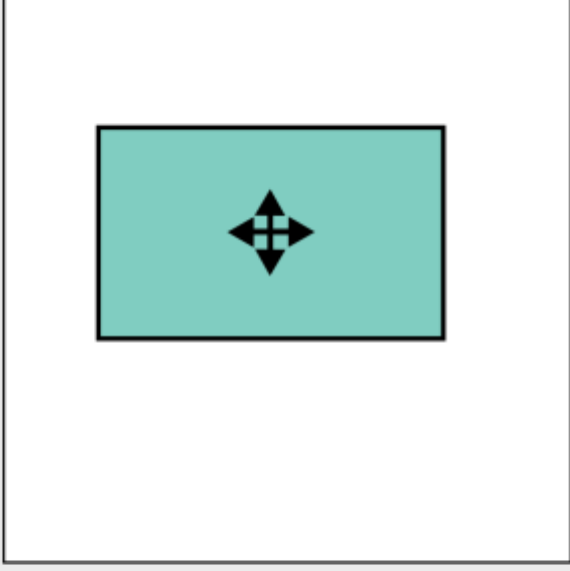
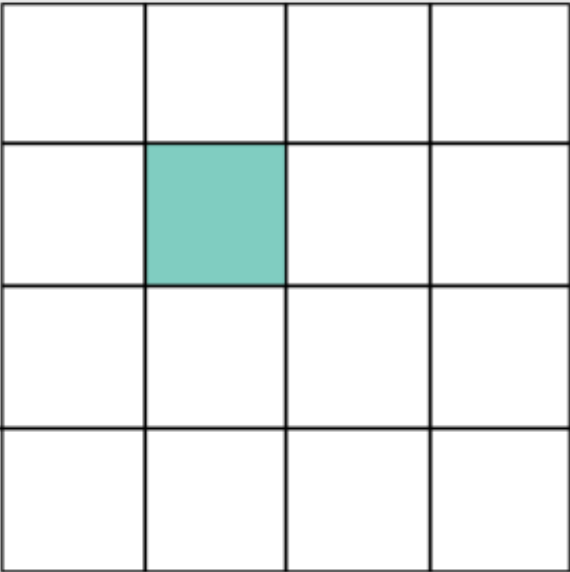
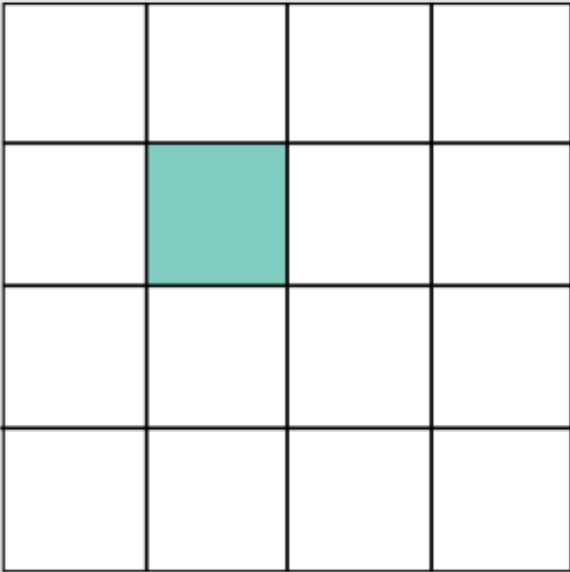
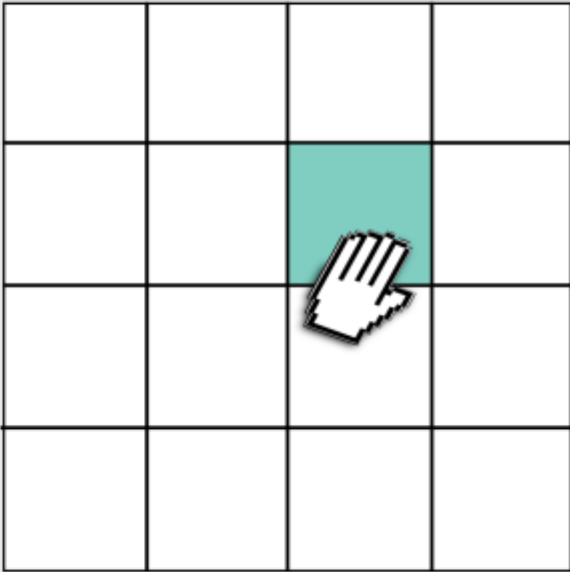
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Constrained			
Automatic			

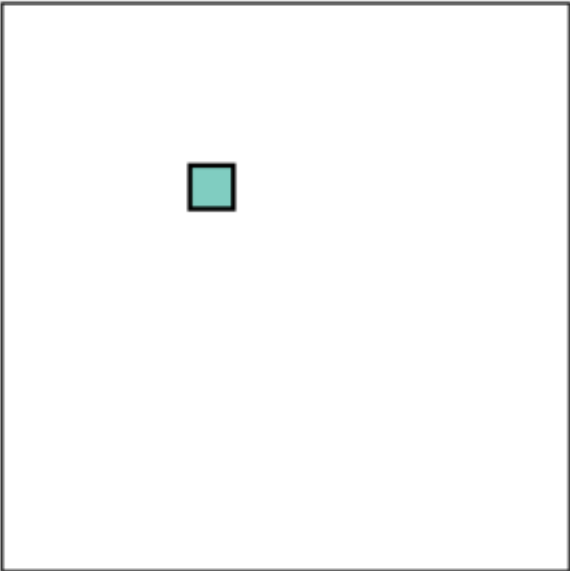
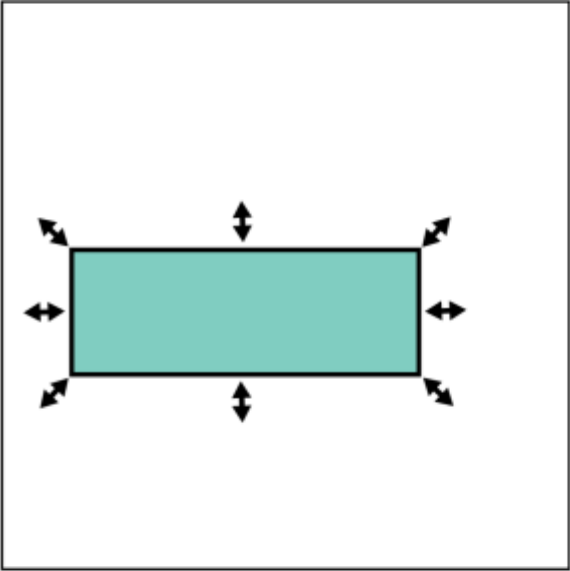
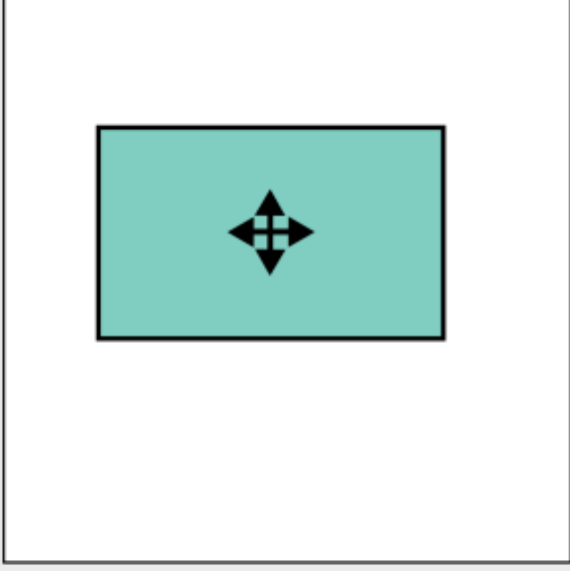
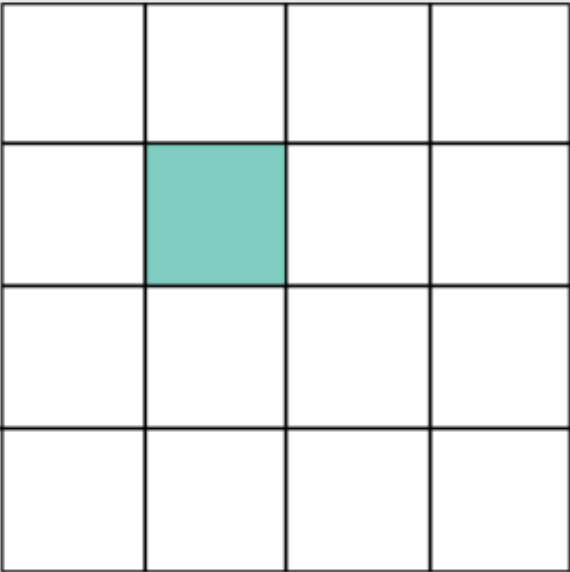
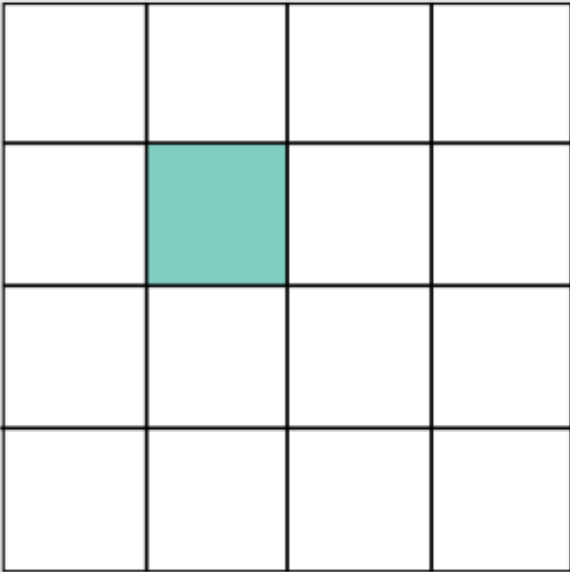
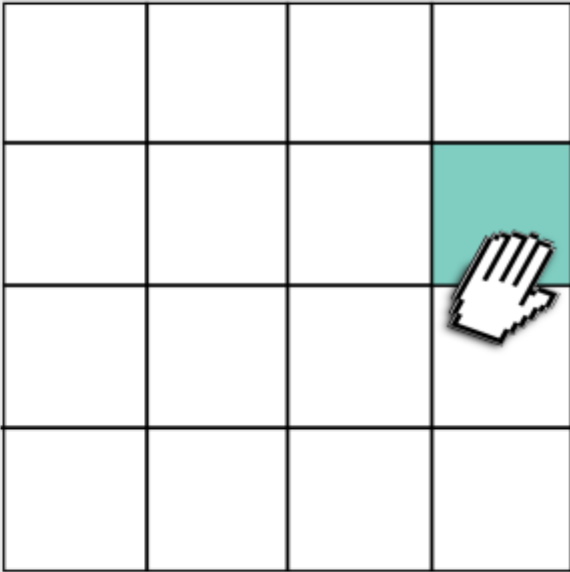
	Brush Anchoring	Brush Extent	Brush Movement
Unconstrained			
Constrained			
Automatic			

	Brush Anchoring	Brush Extent	Brush Movement
Unconstrained			
Constrained			
Automatic			

	Brush Anchoring	Brush Extent	Brush Movement
Unconstrained			
Constrained			
Automatic			

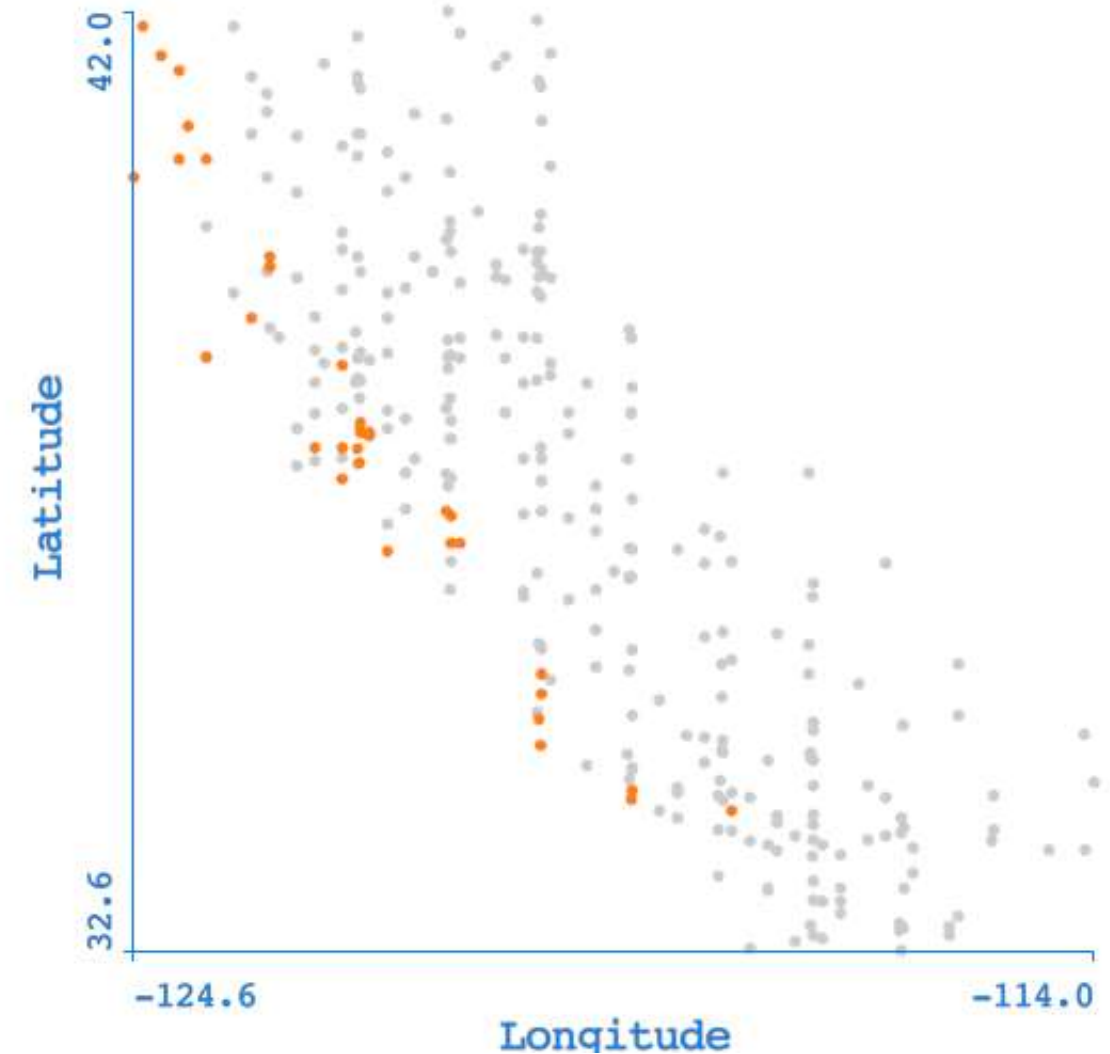
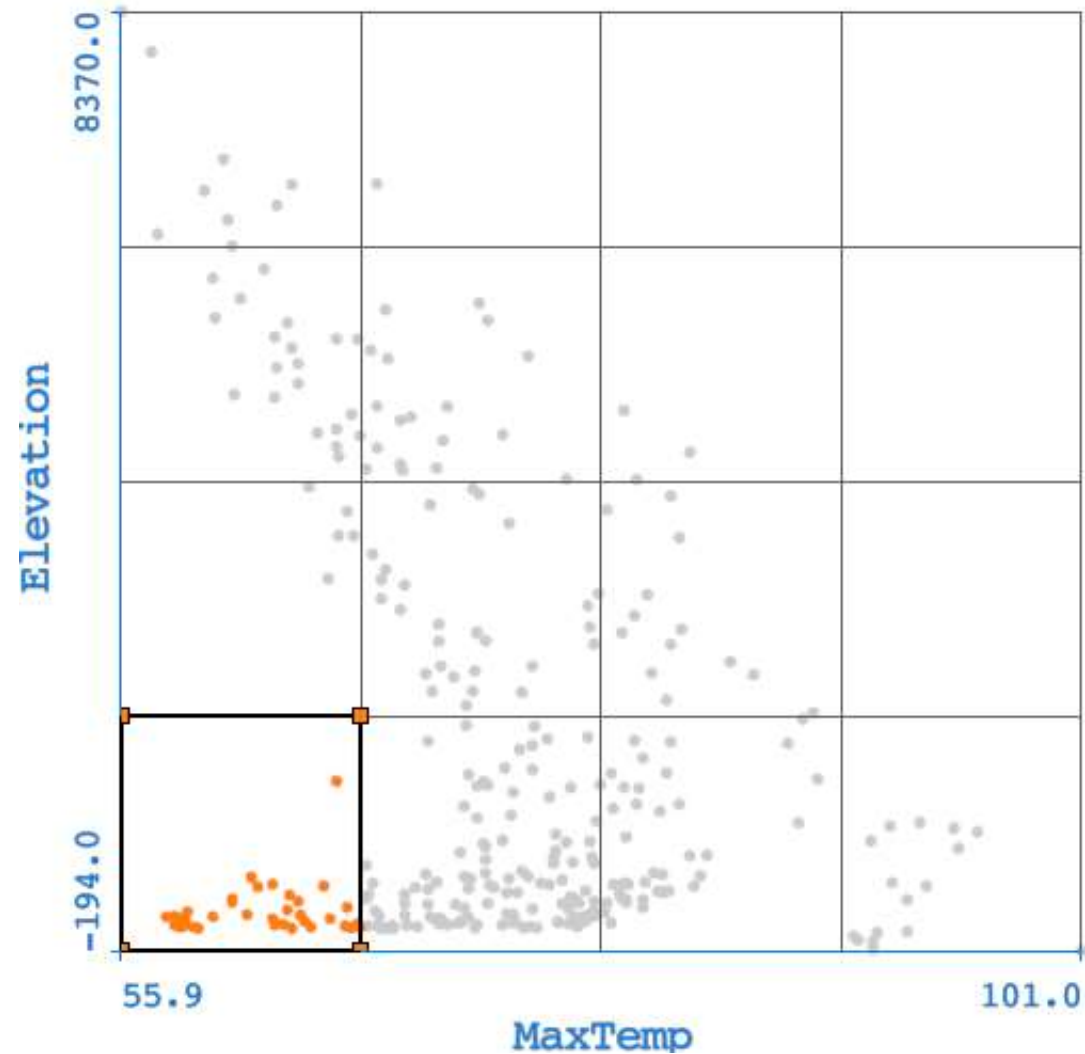
	Brush Anchoring	Brush Extent	Brush Movement
Unconstrained			
Constrained			
Automatic			

	Brush Anchoring	Brush Extent	Brush Movement
Unconstrained			
Constrained			
Automatic			

	Brush Anchoring	Brush Extent	Brush Movement
Unconstrained			
Constrained			
Automatic			

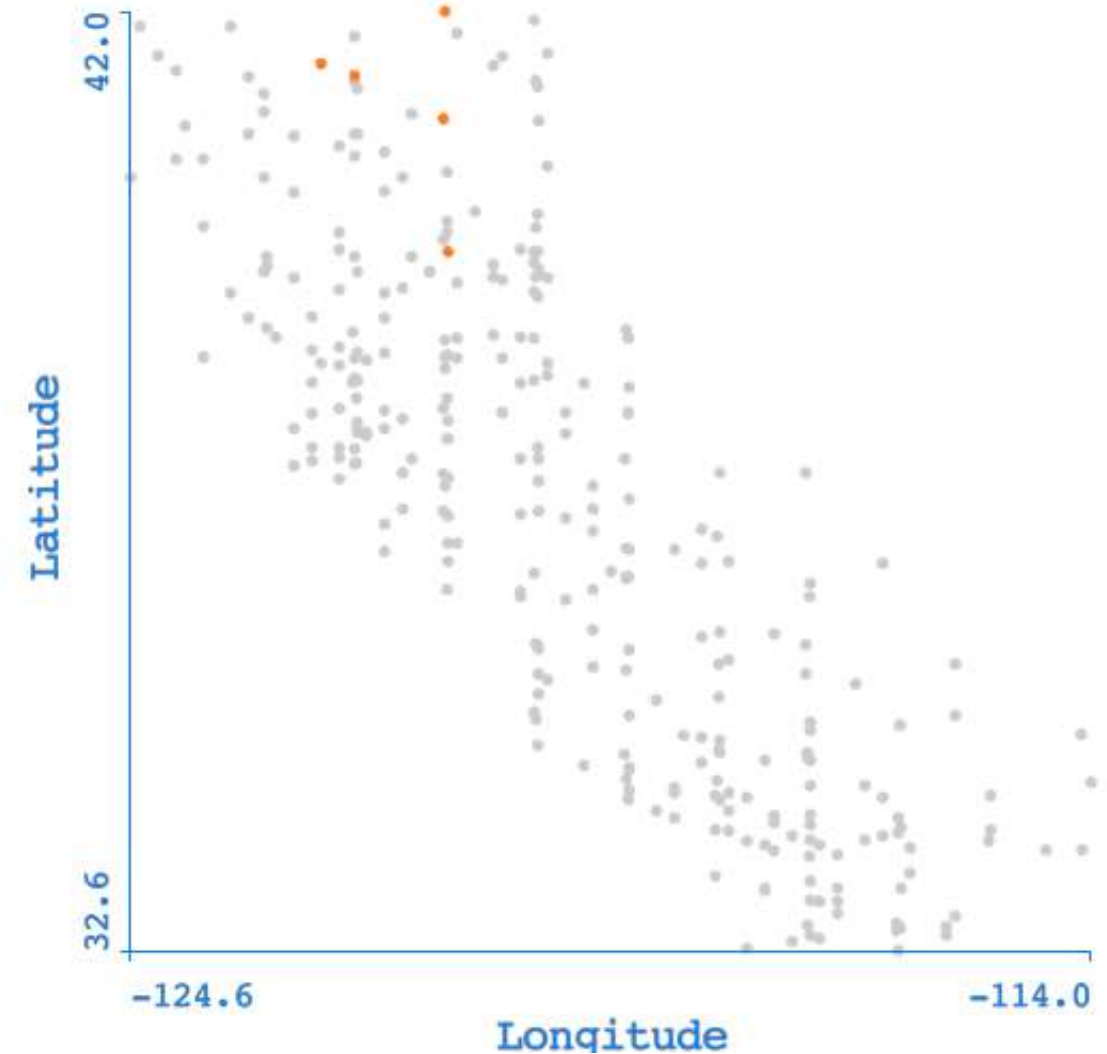
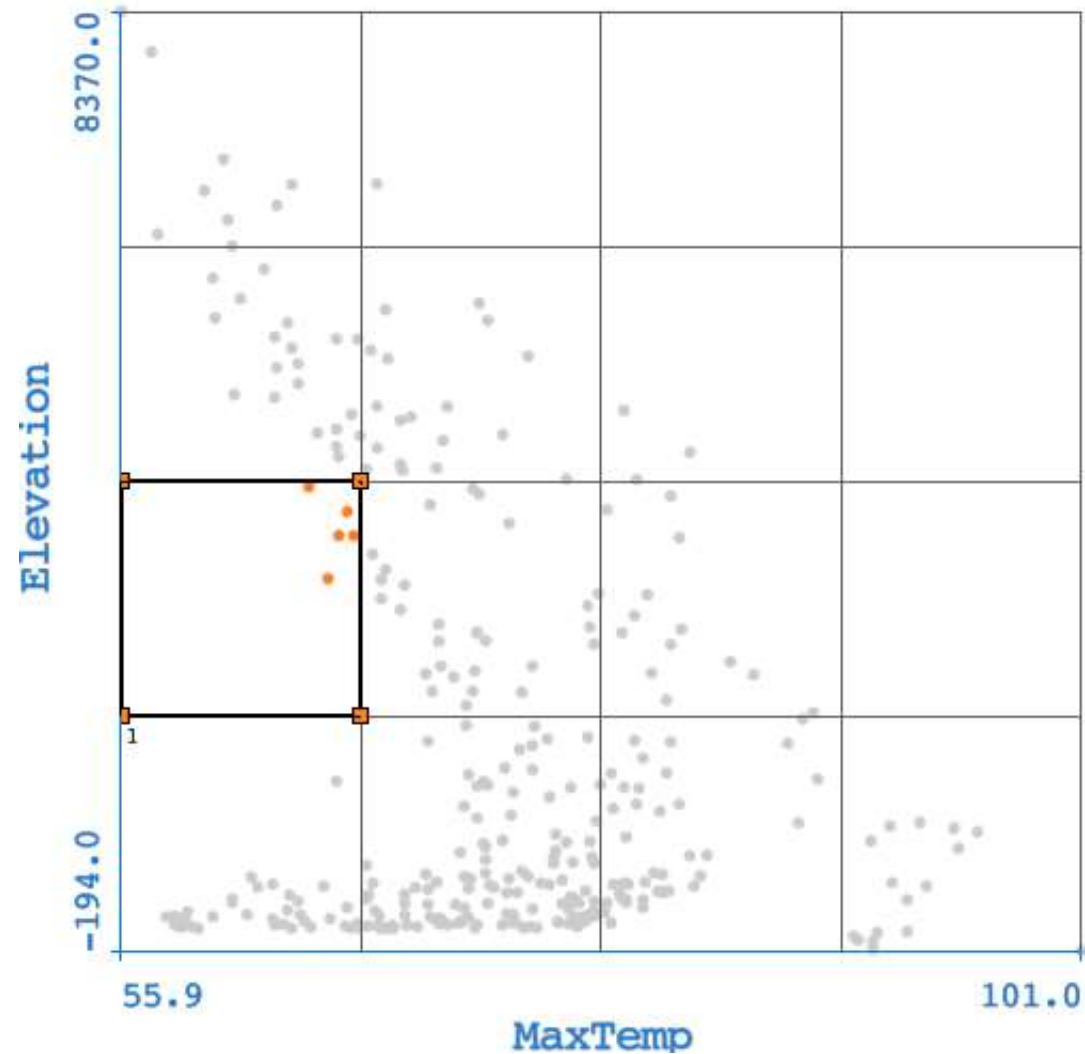
Snap-to-grid

- Constrained brush movement



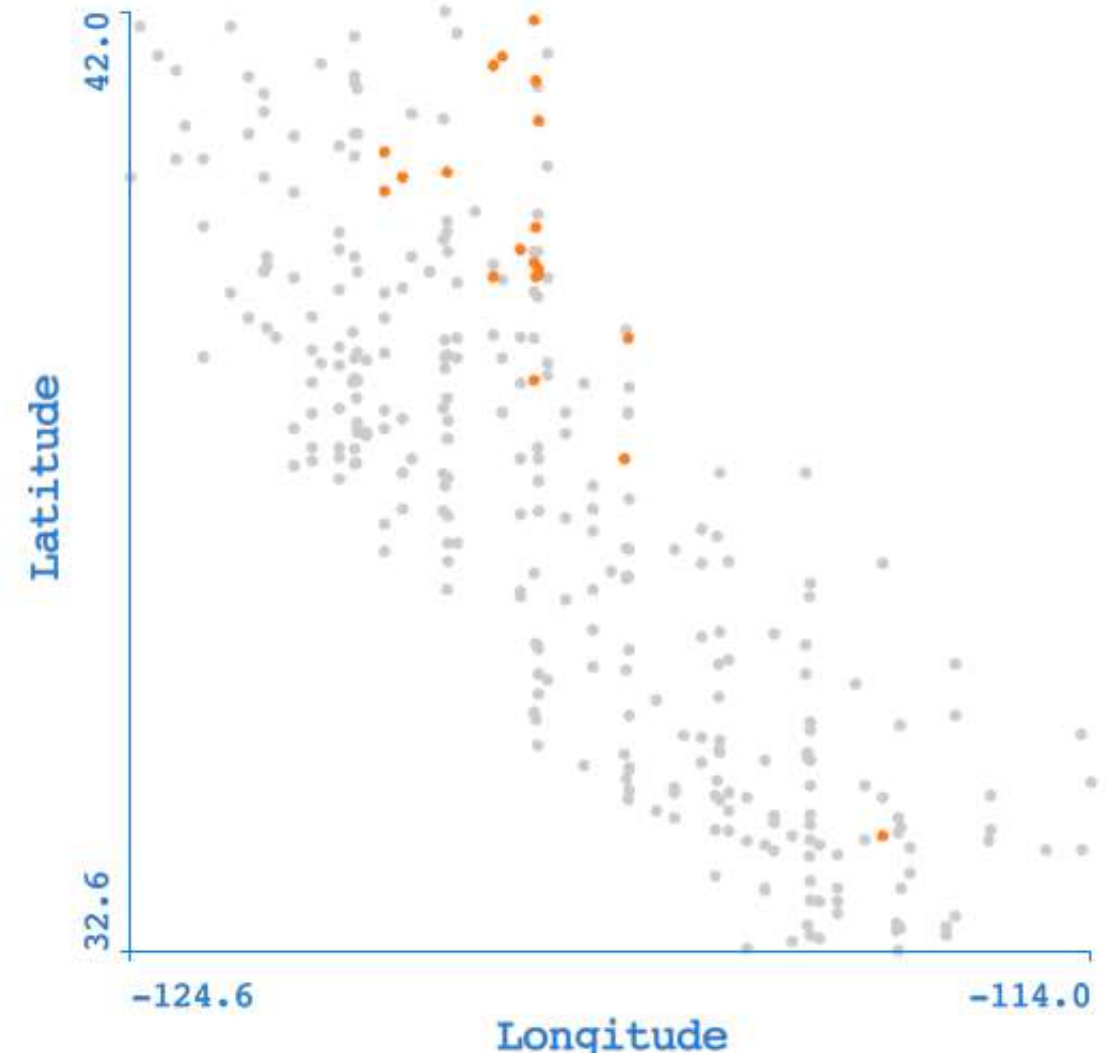
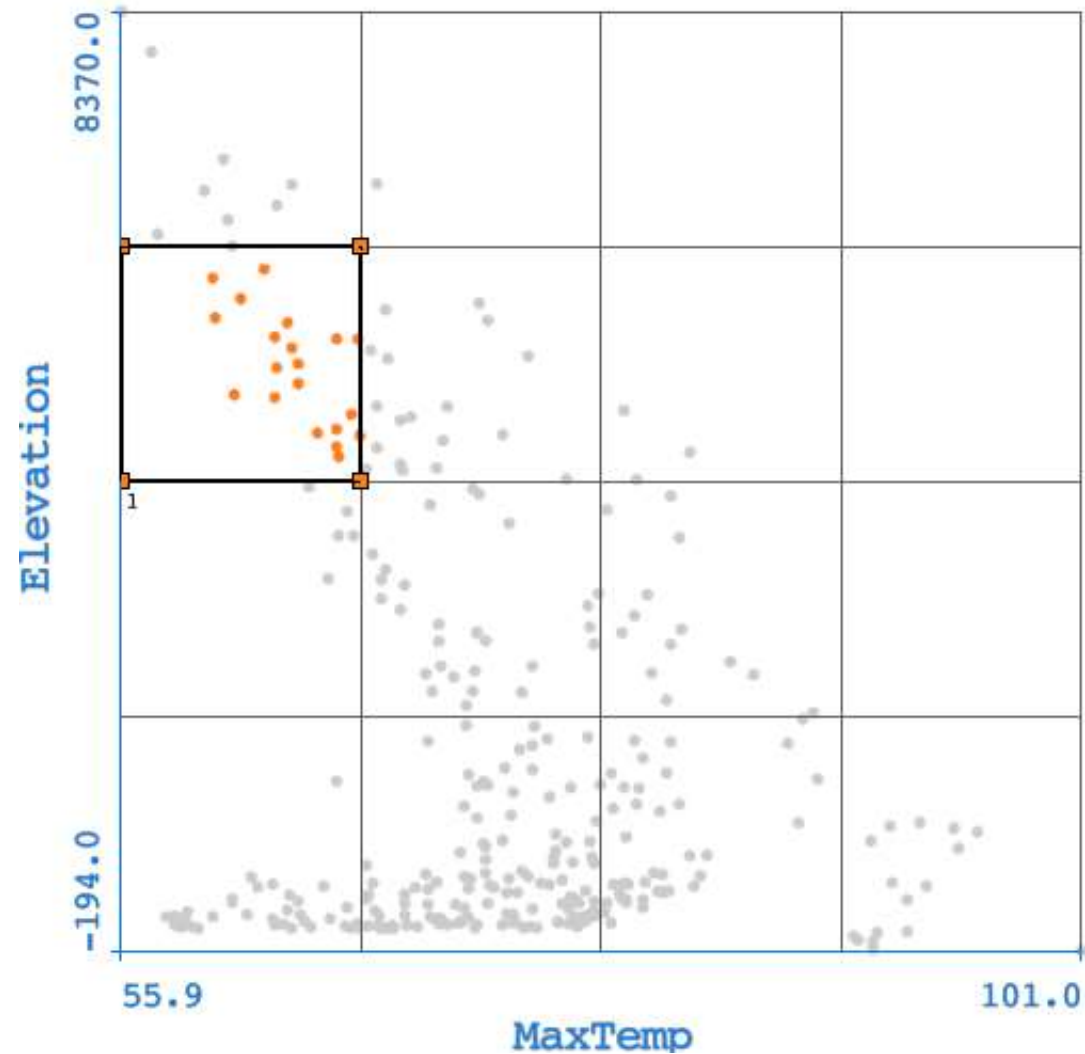
Snap-to-grid

- Constrained brush movement



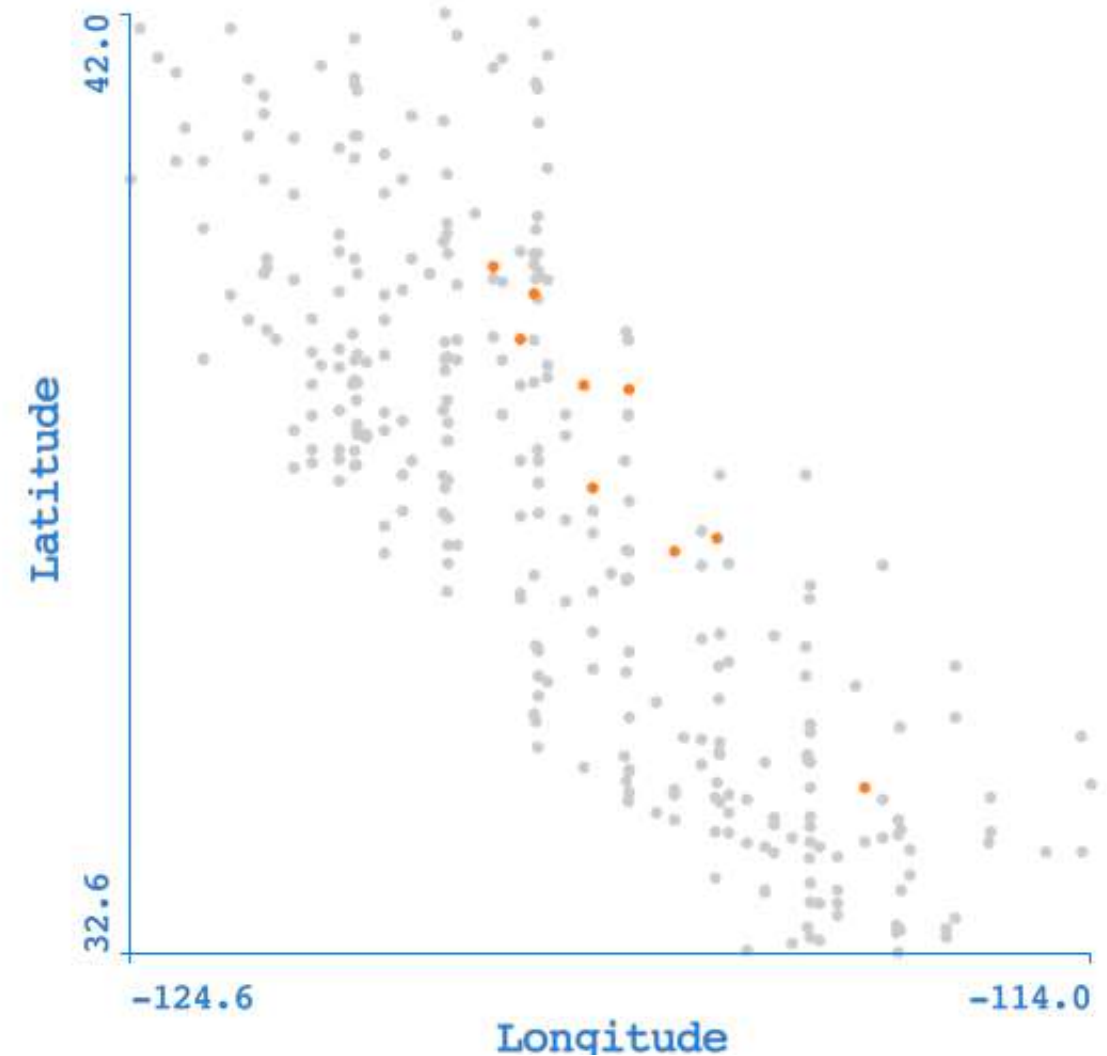
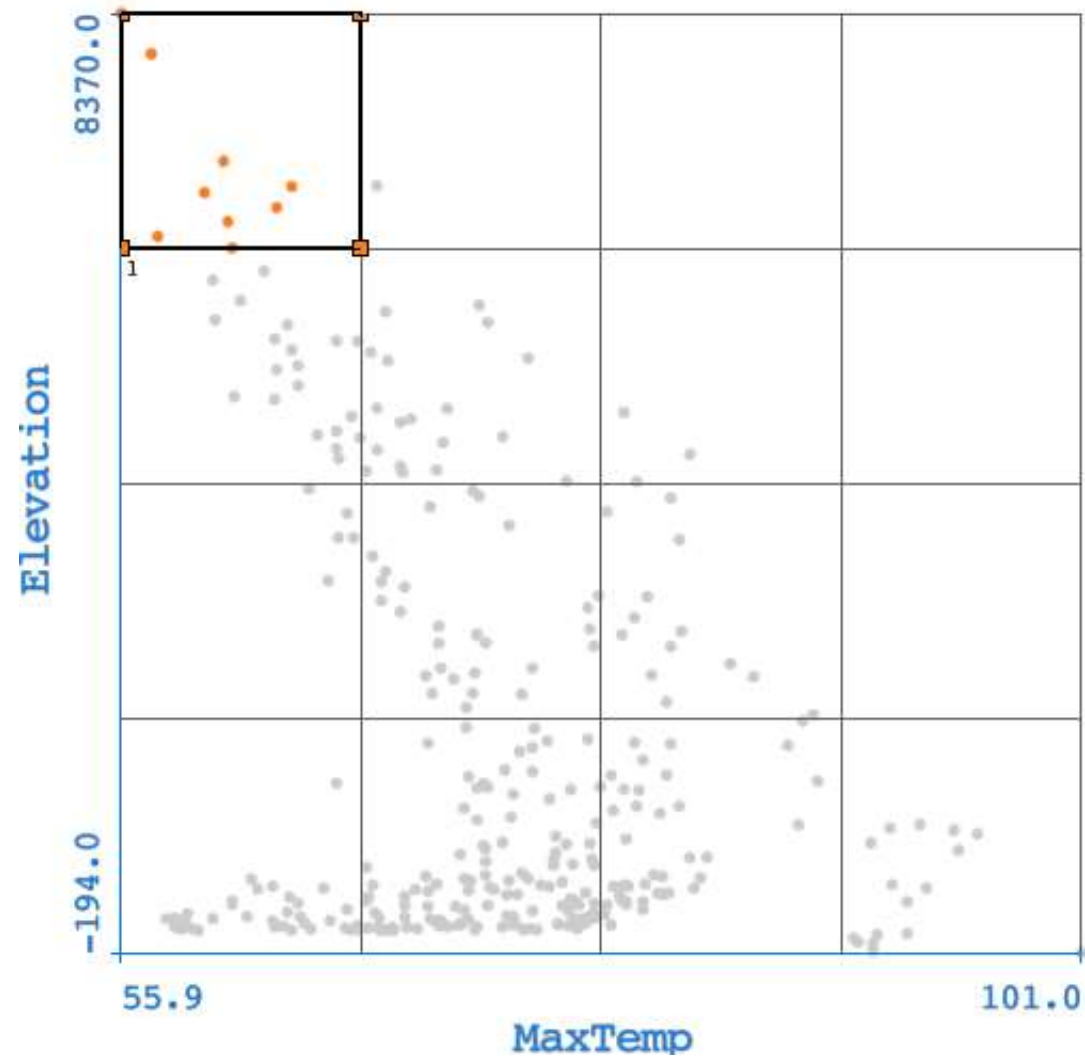
Snap-to-grid

- Constrained brush movement



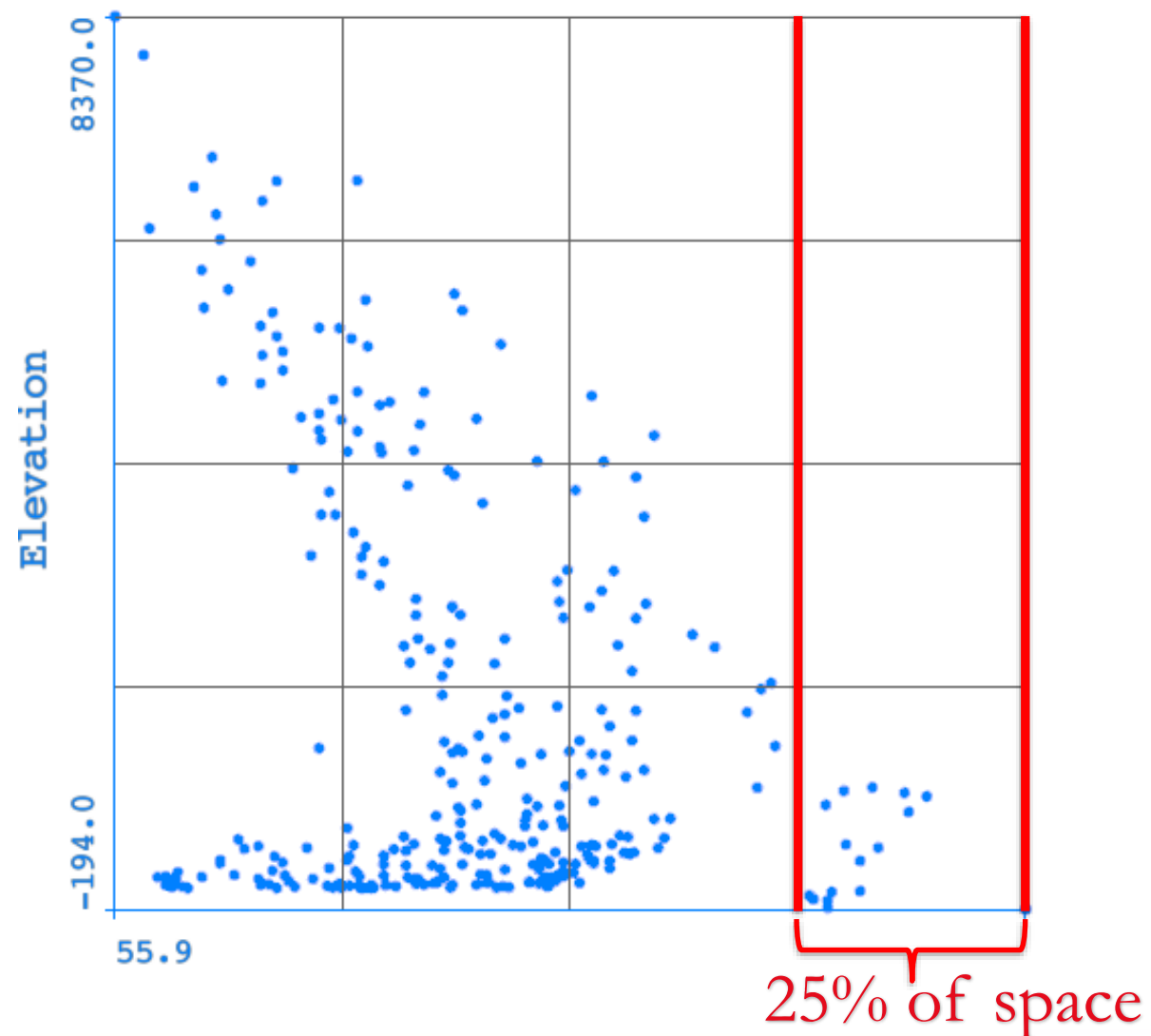
Snap-to-grid

- Constrained brush movement

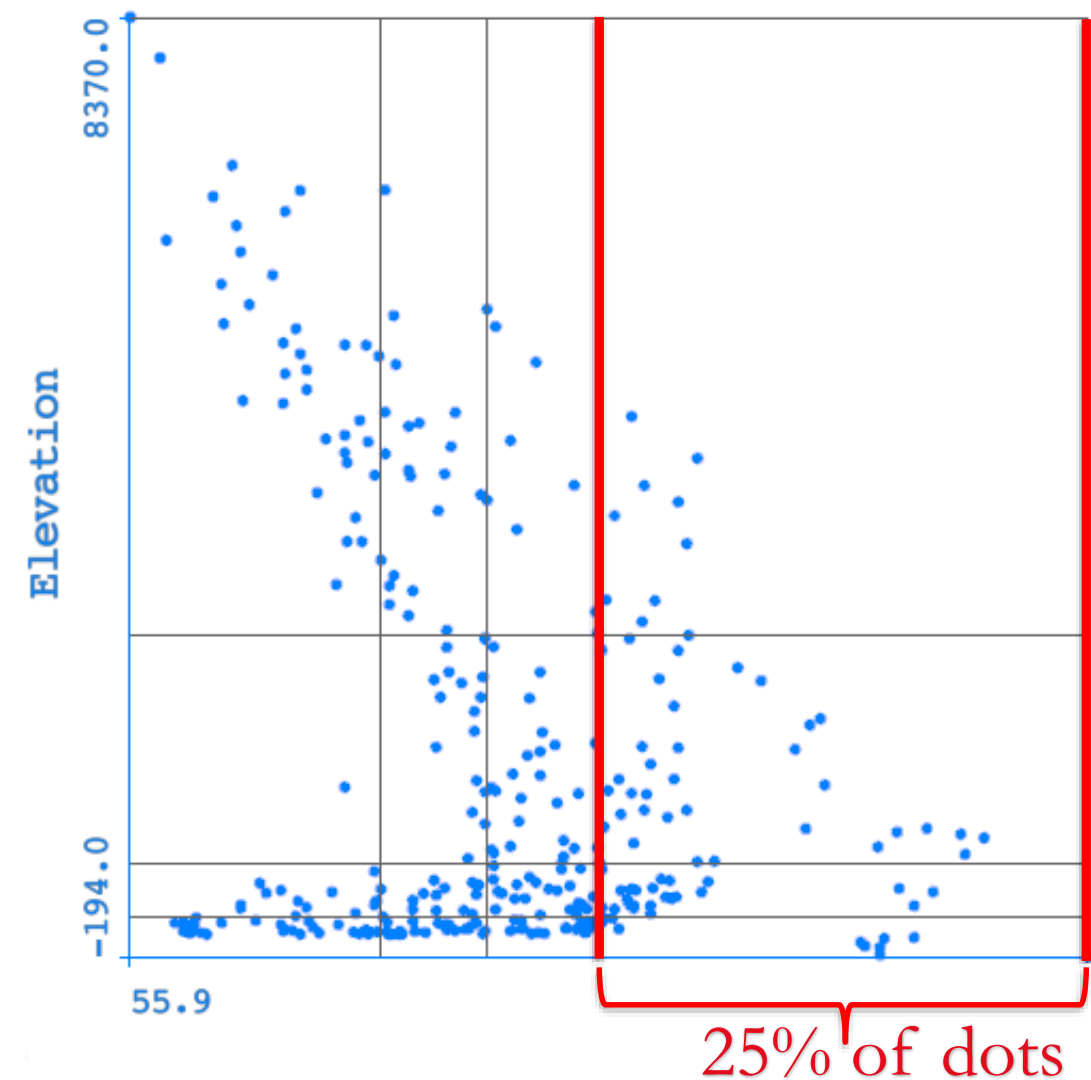


Rank-based Analysis

Regular grid (value-based)

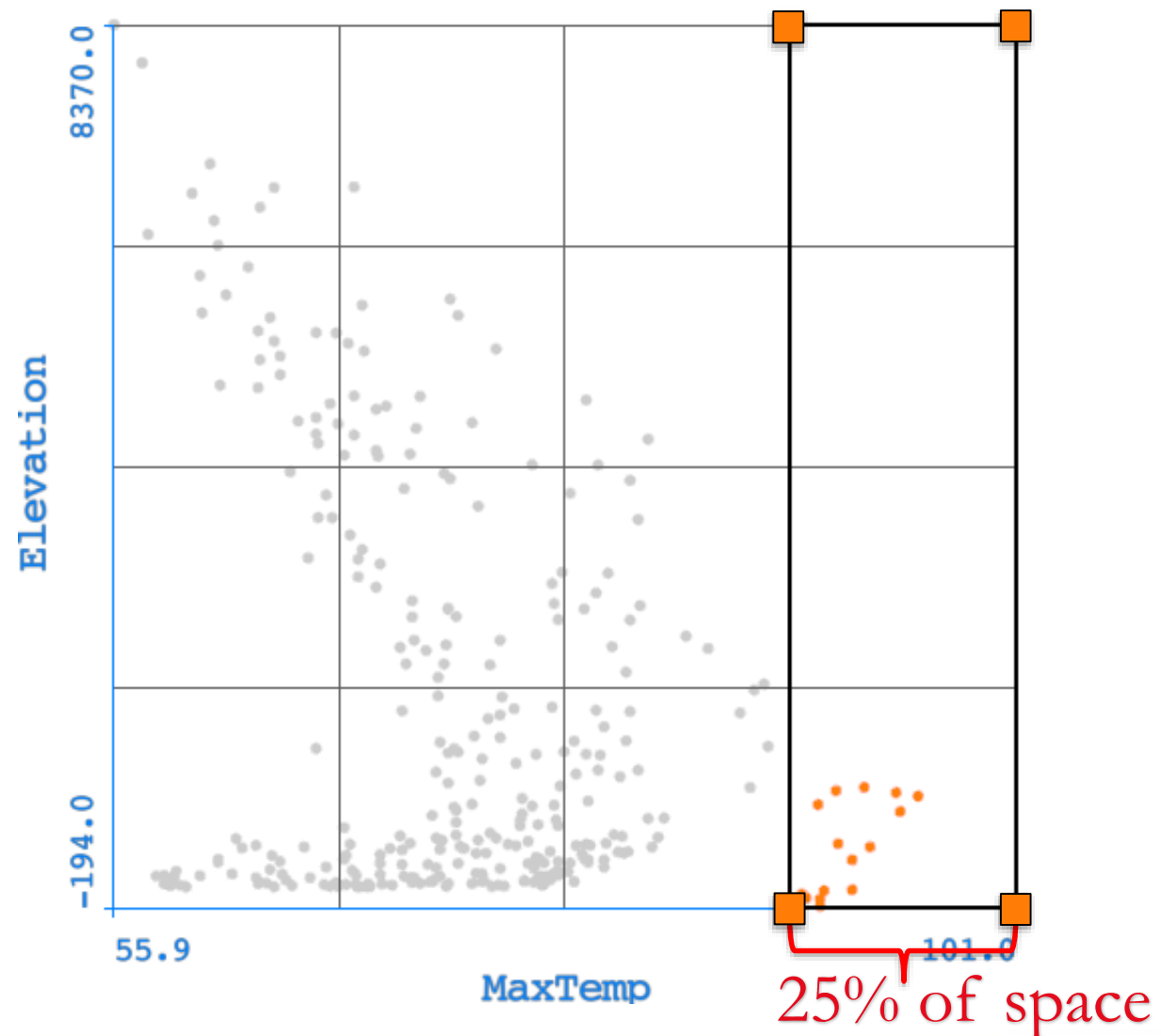


Percentile grid (rank-based)

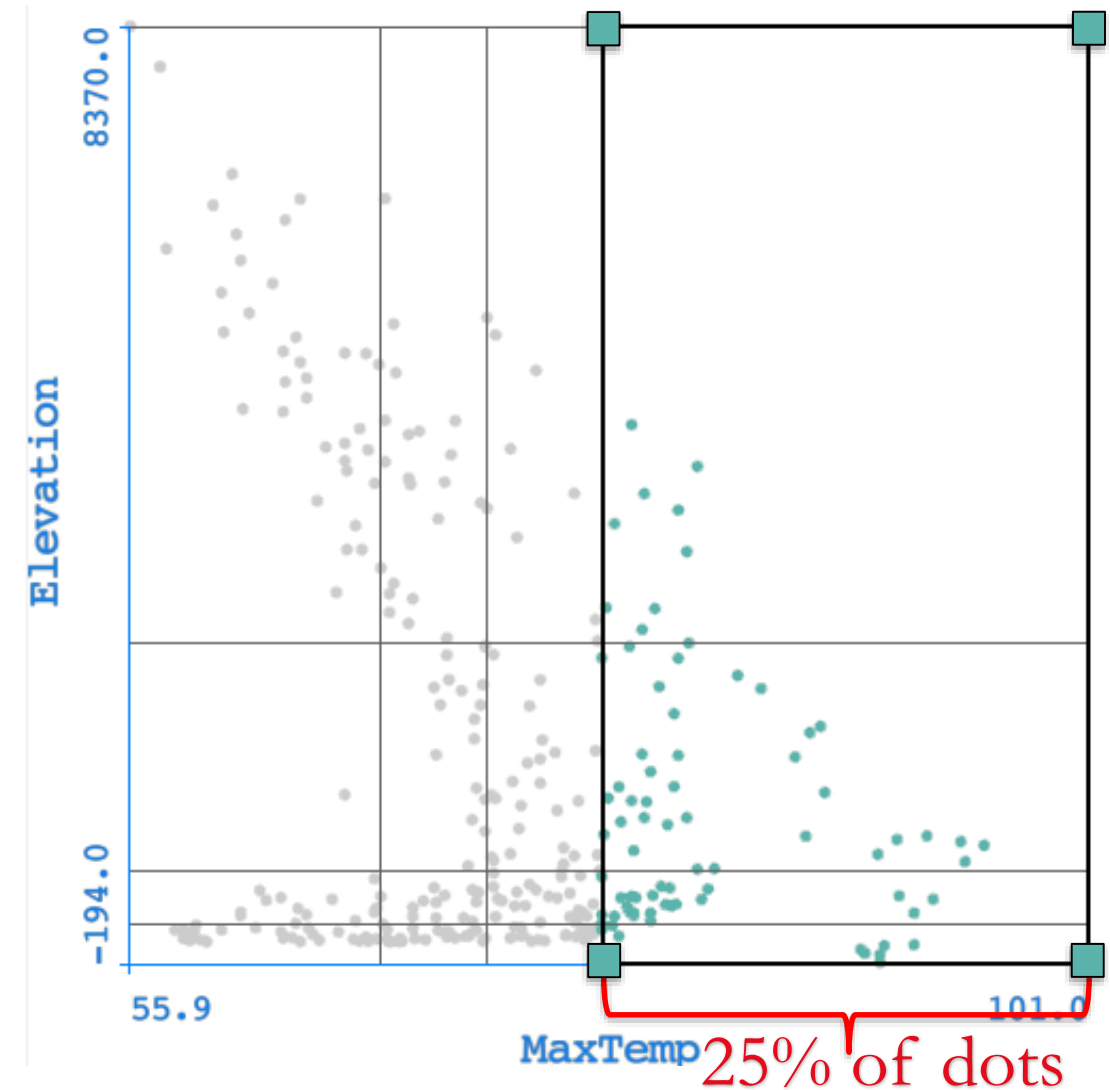


Snap-to-grid

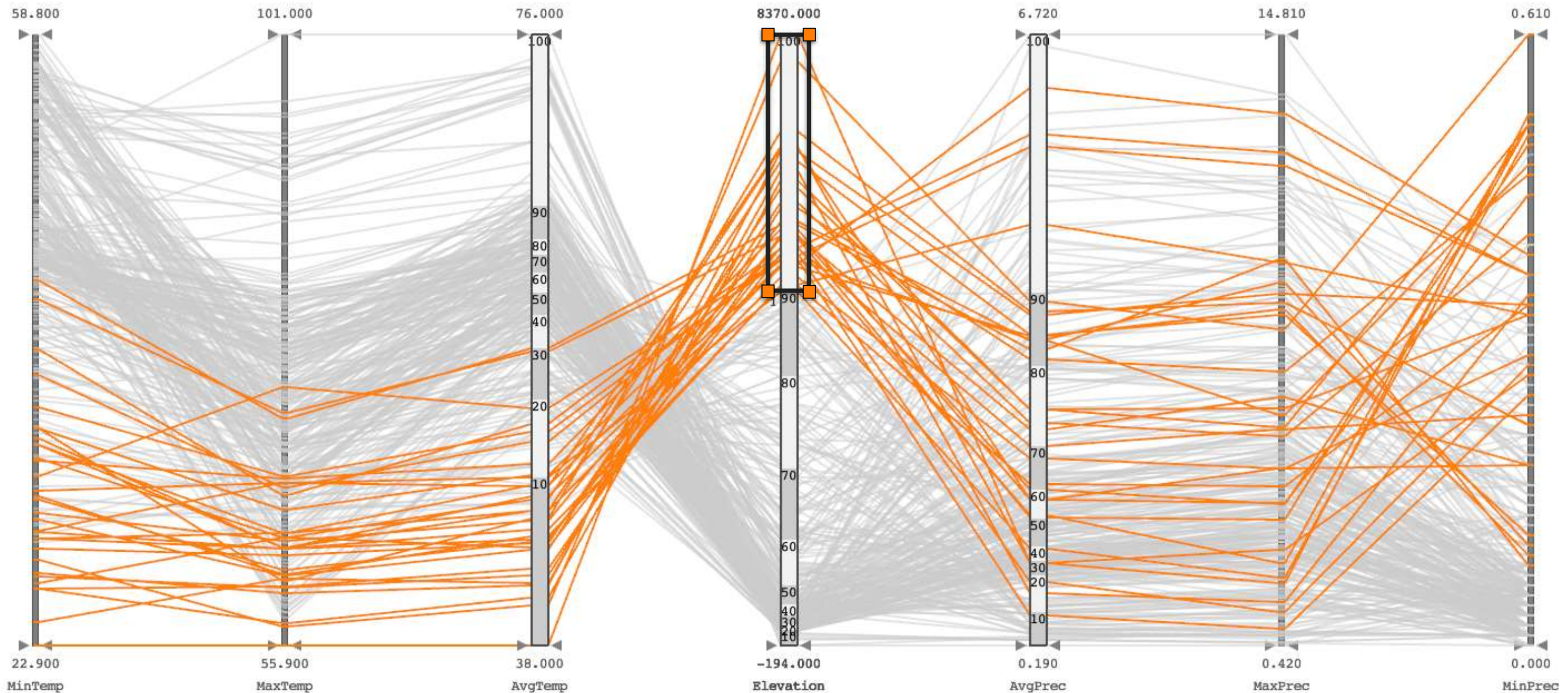
Regular grid (value-based)



Percentile grid (rank-based)



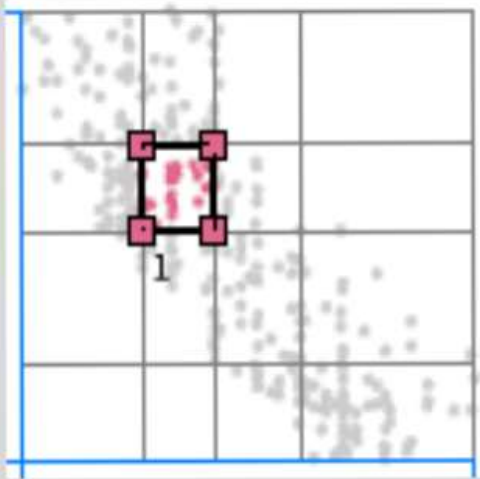
Percentile Grid - Parallel Coordinates



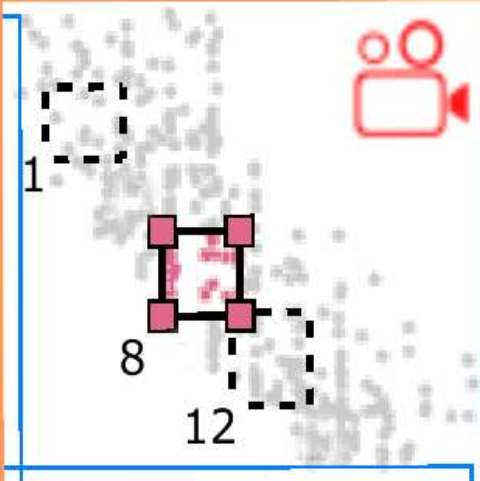
Extensions for Reproducible and Quantitative Visual Analytics

Structured Brushing

Constrained

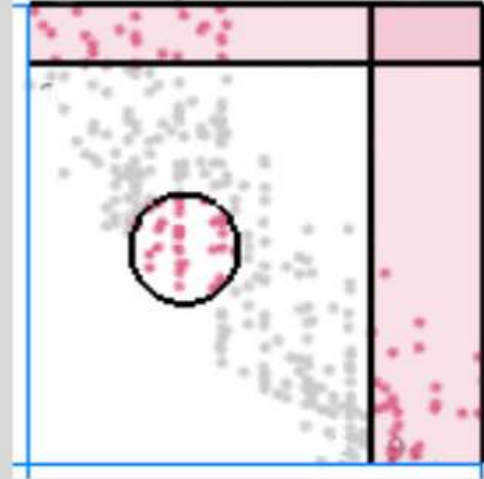


Automatic

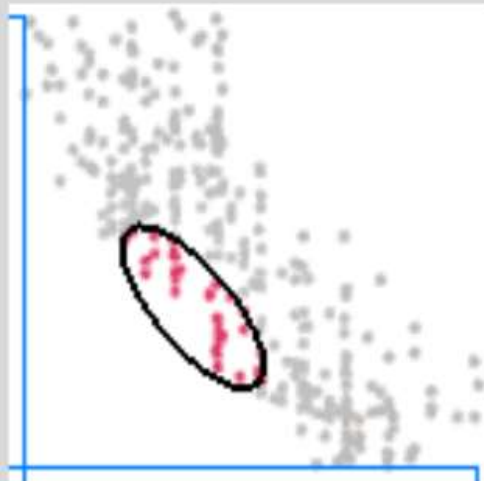


Percentile Brushes

Circle/Rectangle

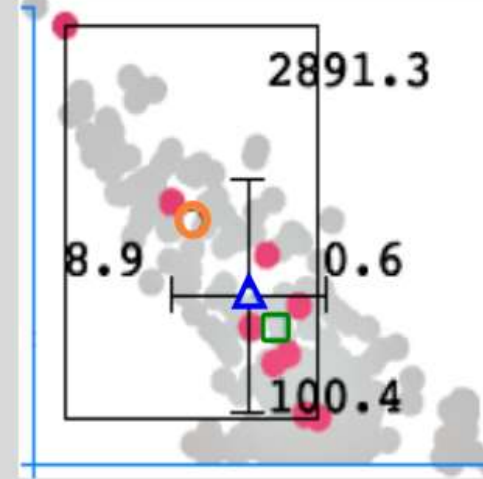


Mahalanobis



Linked Statistics

Overlays

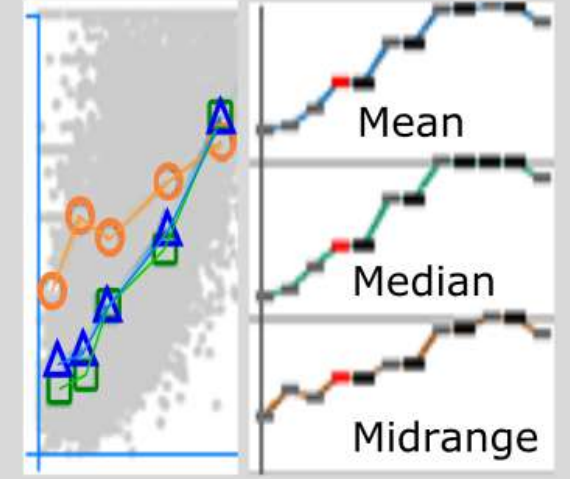


Numerical Tables

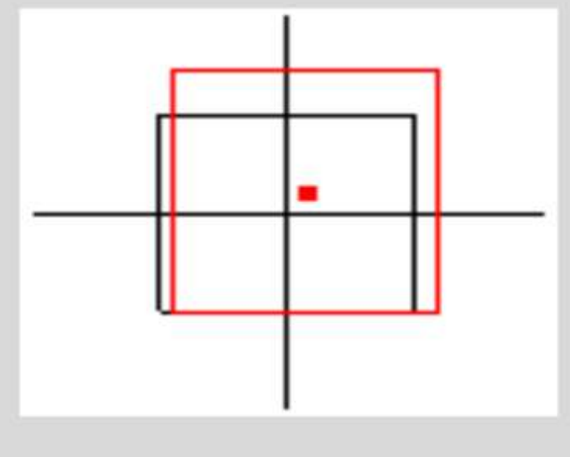
	X	Y
△ Mean	55.8	2933.6
□ Median	57.9	2355.0
○ Midrange	51.0	4329.0
Spread	6.4	2175.1

Change Visualization

Traces/Graphs



Difference-Plot



Scatterplot 2D

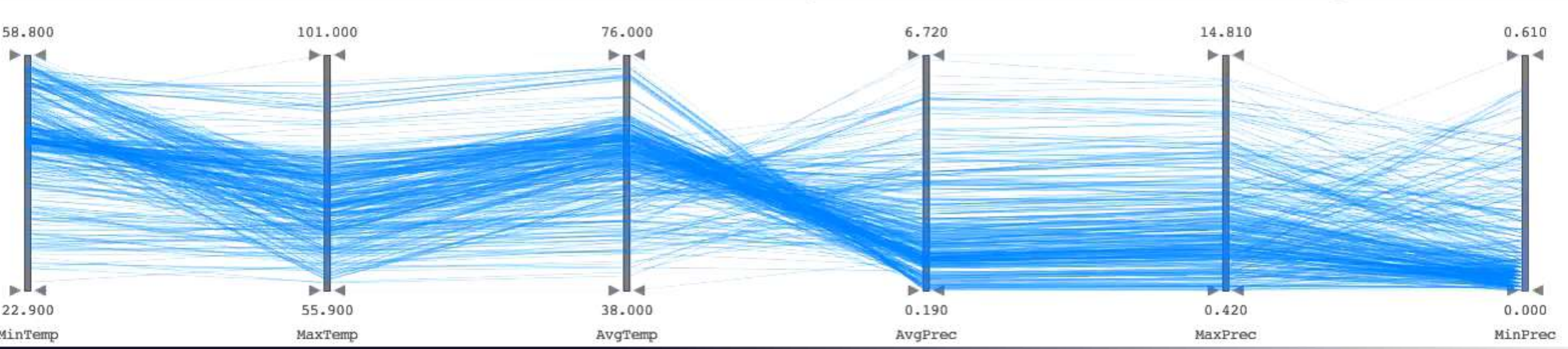
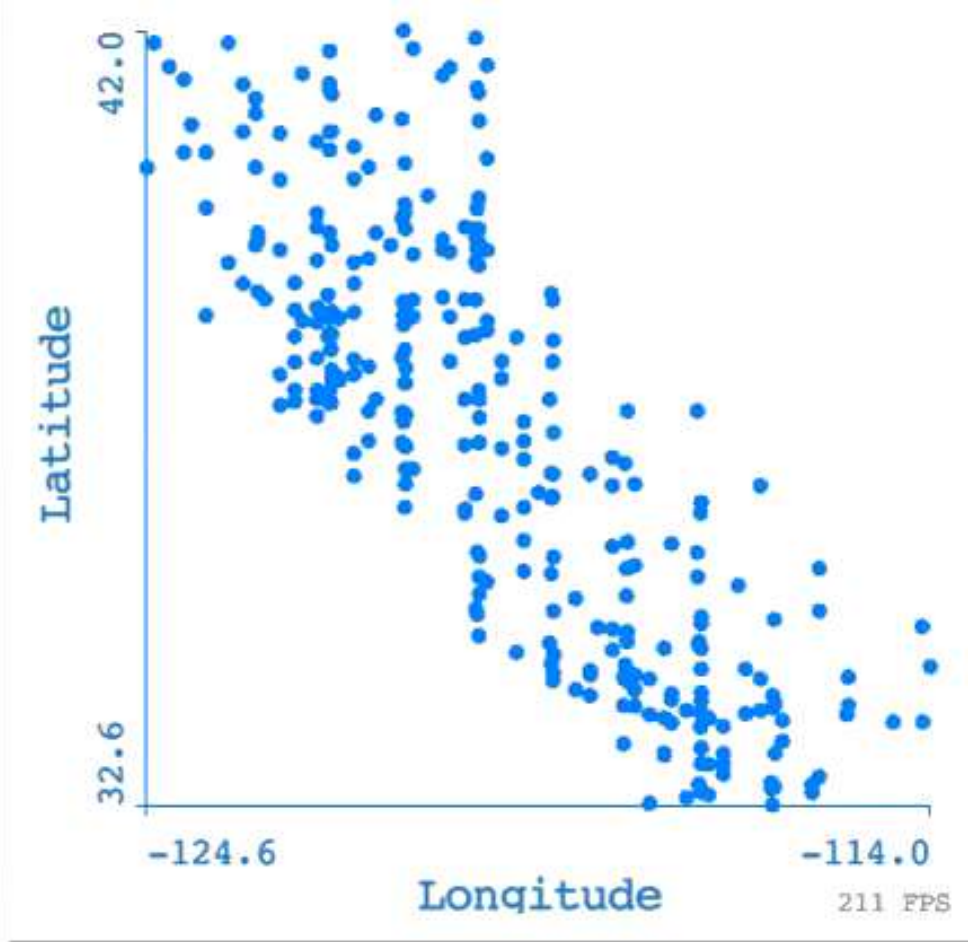
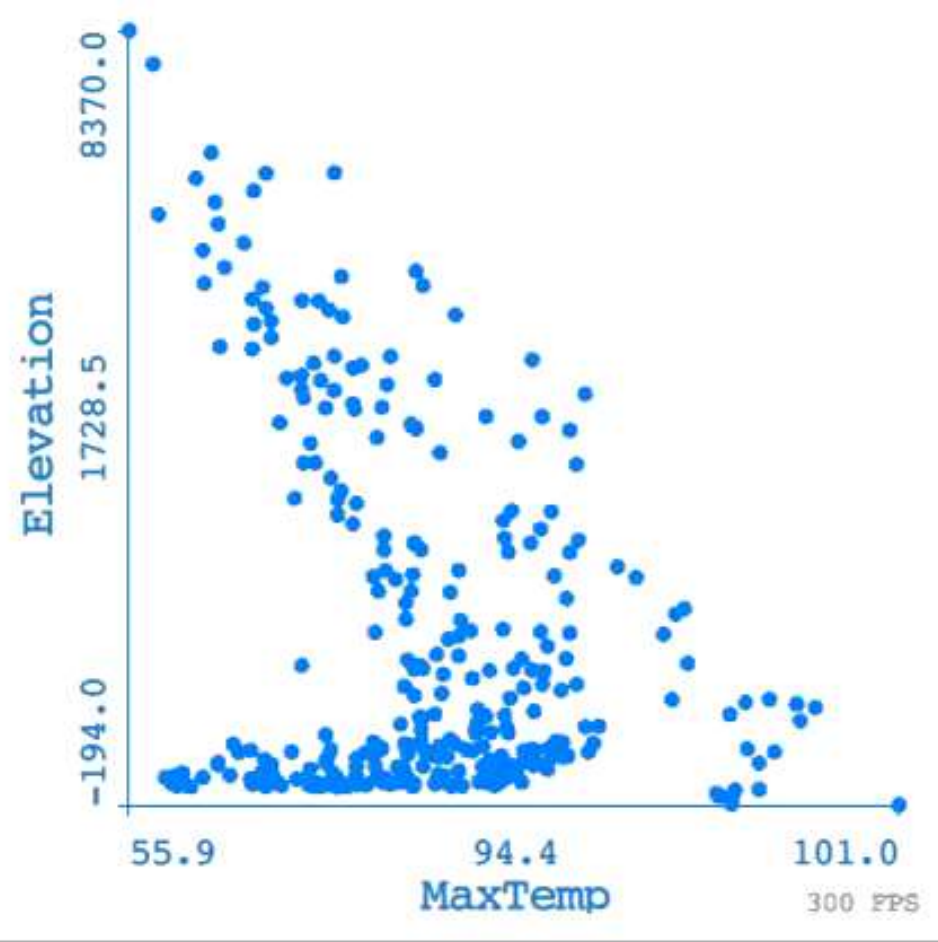
X Axis: MaxTemp

Y Axis: Elevation

☐ ☒ ☐ ☐ ☐

☐ ☒ M G

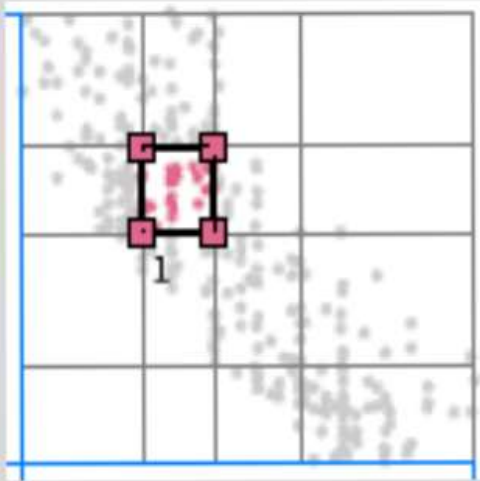
Rectangle



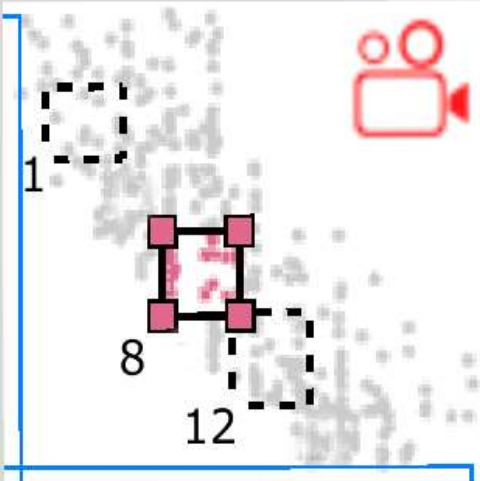
Extensions for Reproducible and Quantitative Visual Analytics

Structured Brushing

Constrained

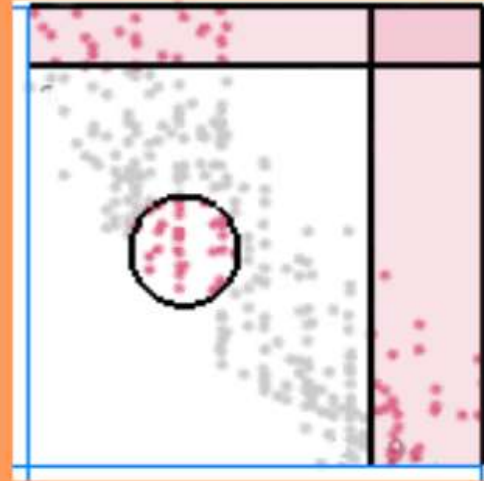


Automatic

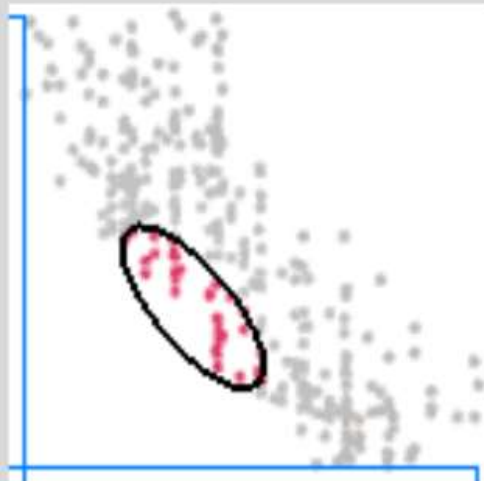


Percentile Brushes

Circle/Rectangle

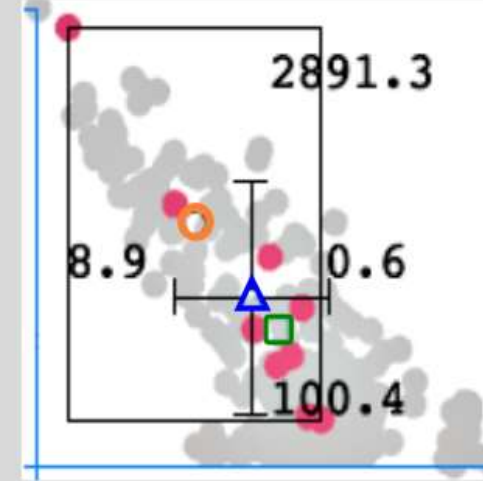


Mahalanobis



Linked Statistics

Overlays

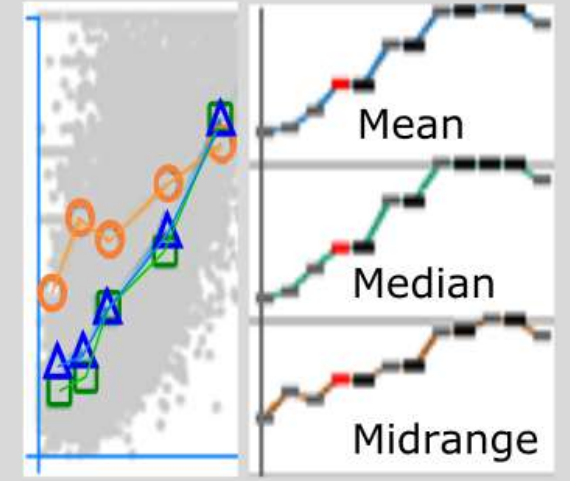


Numerical Tables

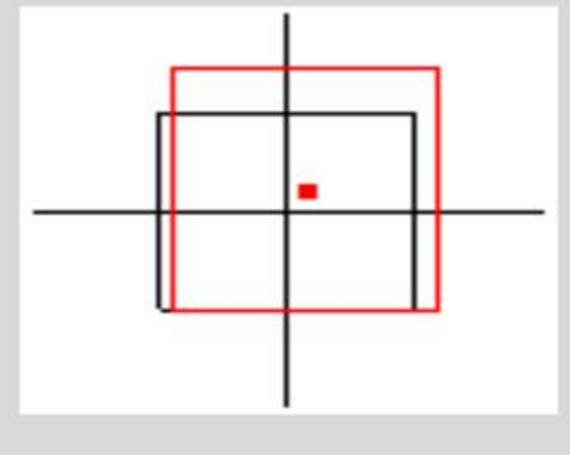
	X	Y
△ Mean	55.8	2933.6
□ Median	57.9	2355.0
○ Midrange	51.0	4329.0
Spread	6.4	2175.1

Change Visualization

Traces/Graphs



Difference-Plot



Scatterplot 2D

X Axis

MaxTemp

Y Axis:

Elevation



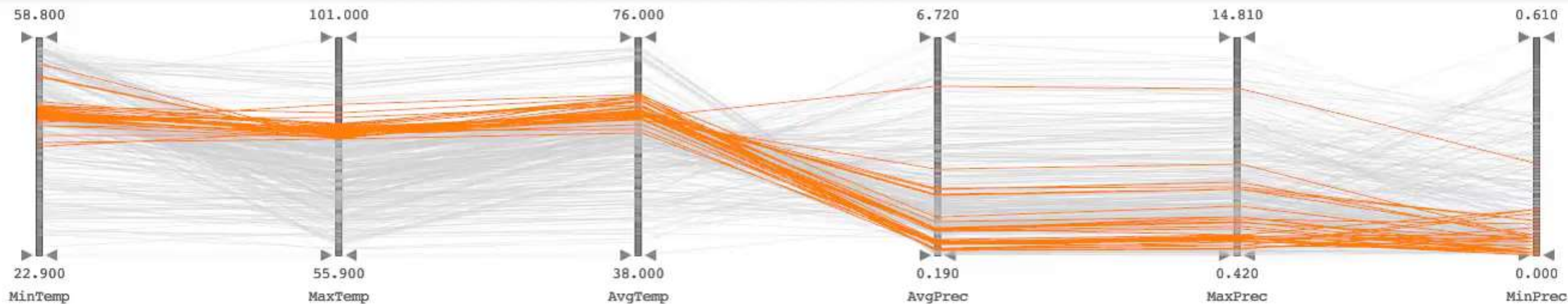
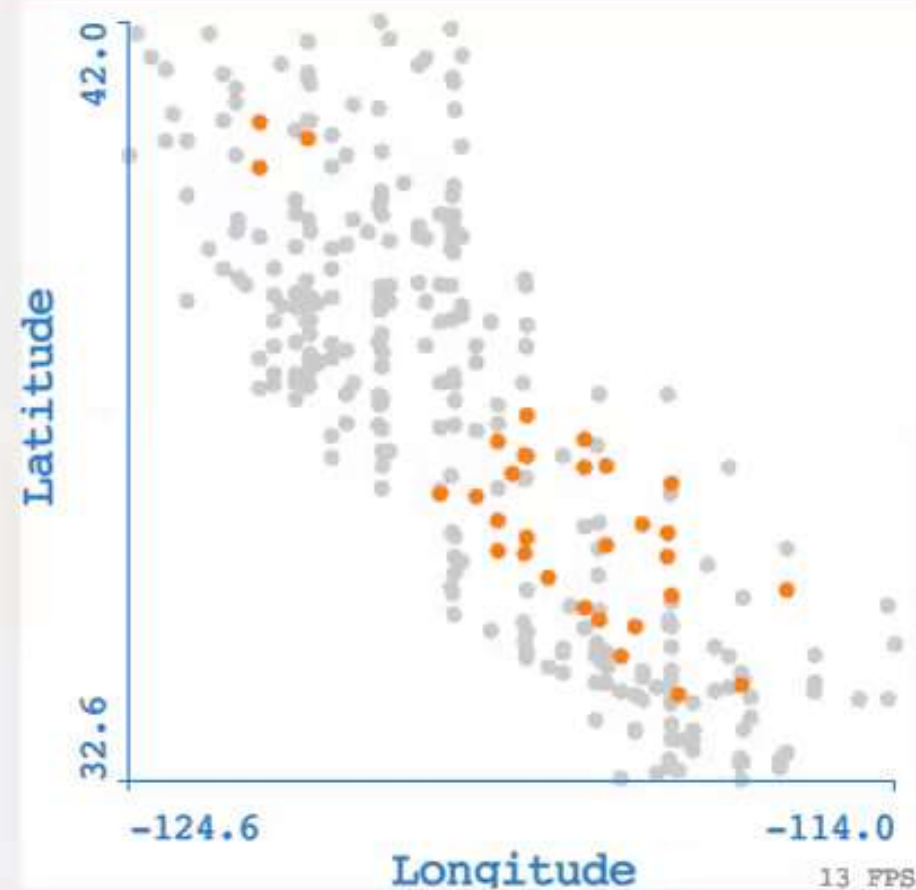
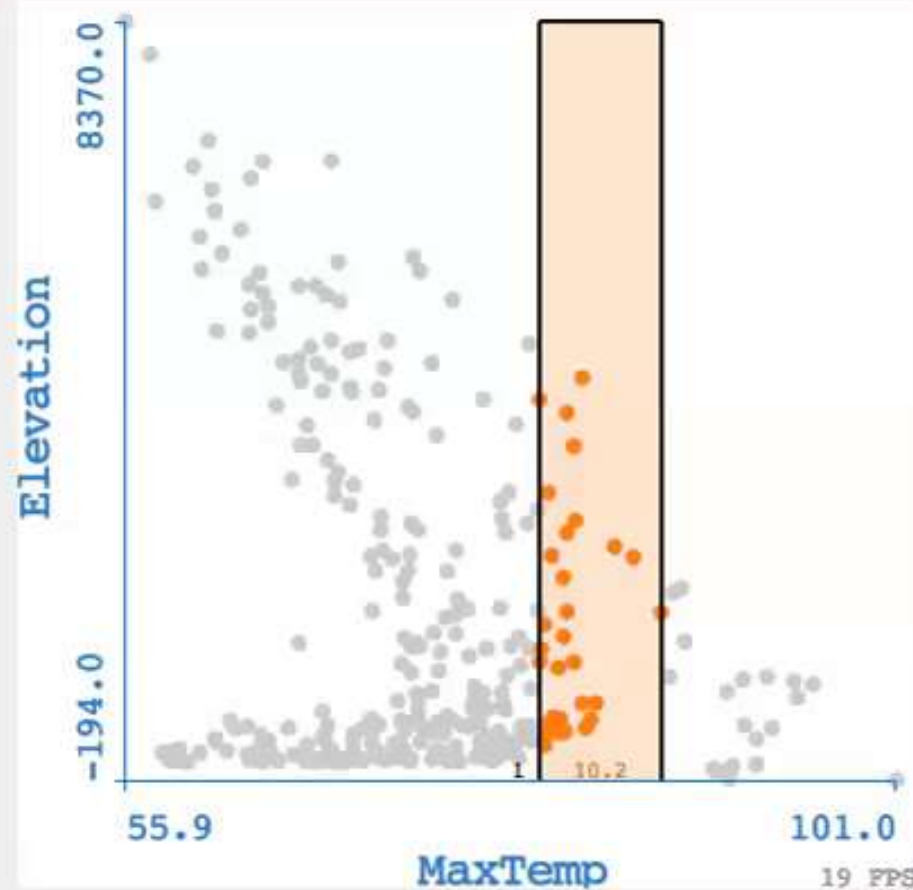
Brush size

To specify the size of brush,
enter percentage of the data
items to select

10

Seeding line

☐ Show seeding line for selected brus



Scatterplot 2D

X Axis

MaxTemp

Y Axis:

Elevation



M G

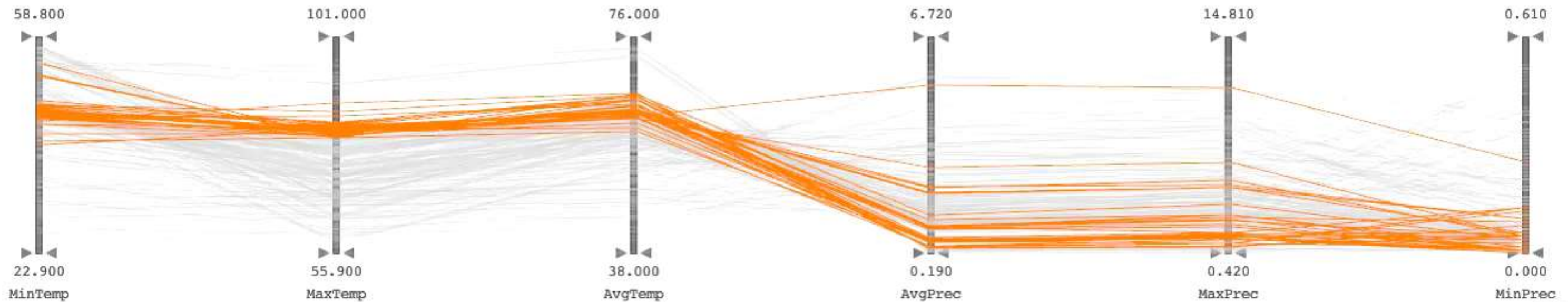
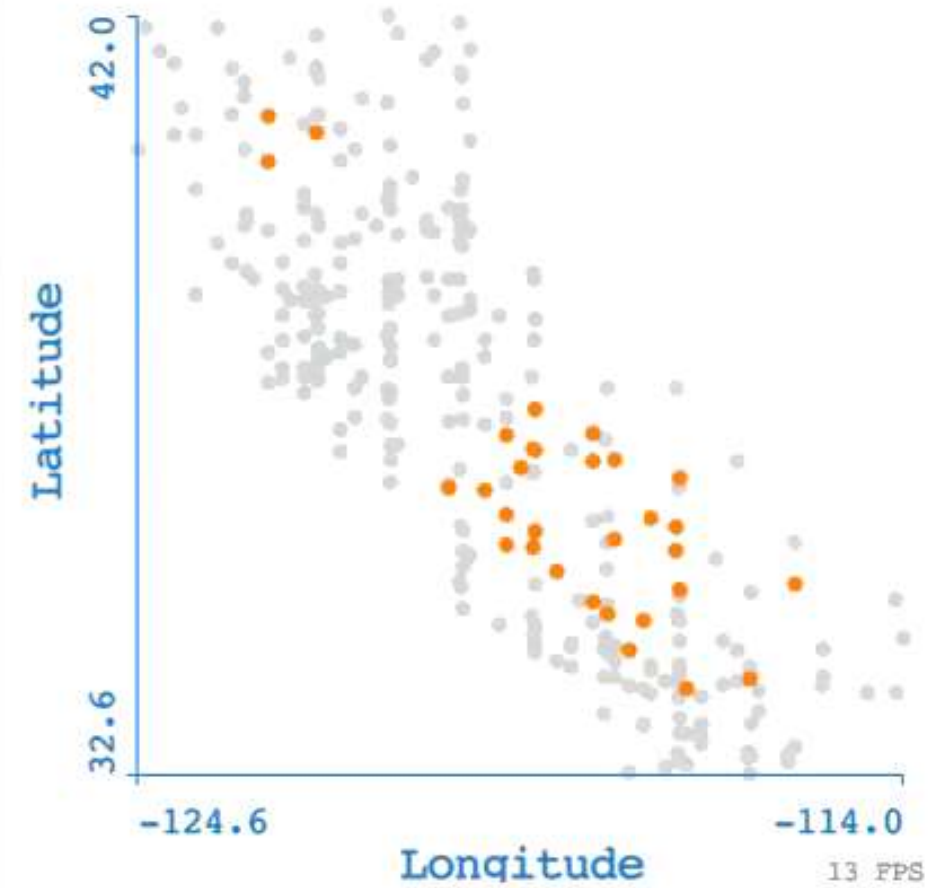
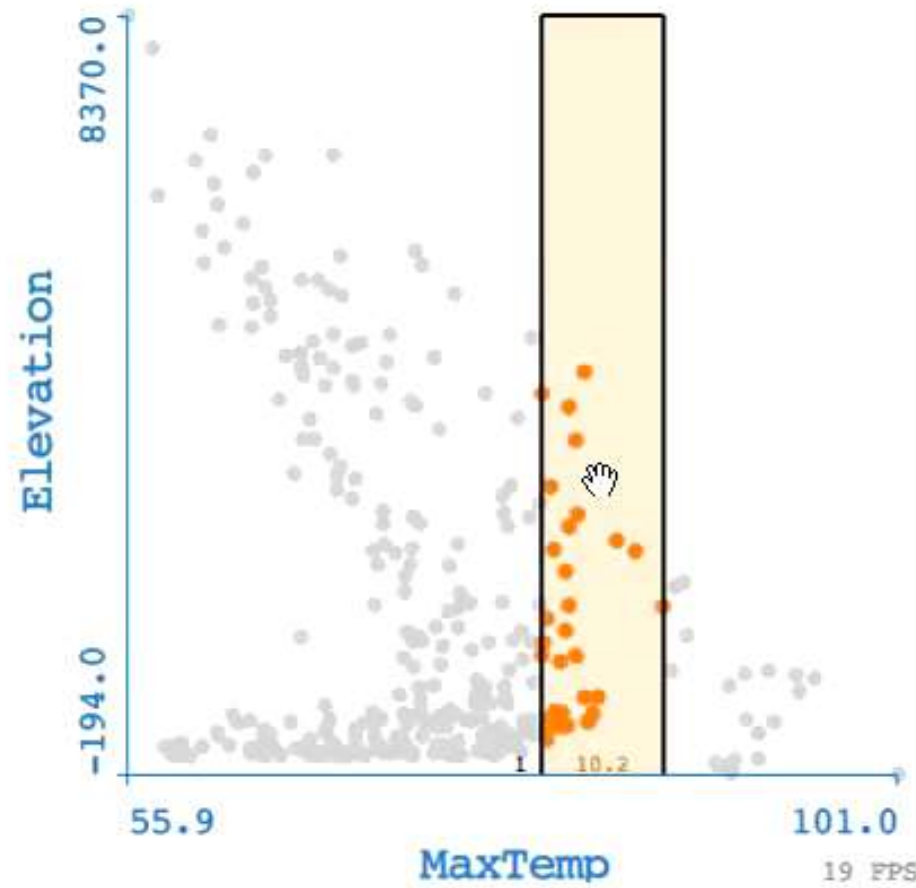
Brush size

To specify the size of brush,
enter percentage of the data
items to select

10

Seeding line

☐ Show seeding line for selected brus



Scatterplot 2D

X Axis

MaxTemp

Y Axis:

Elevation



M G

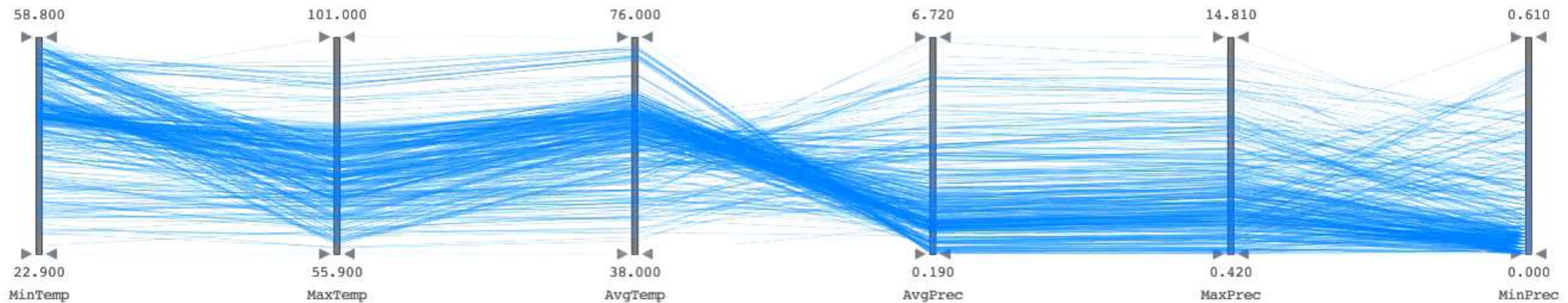
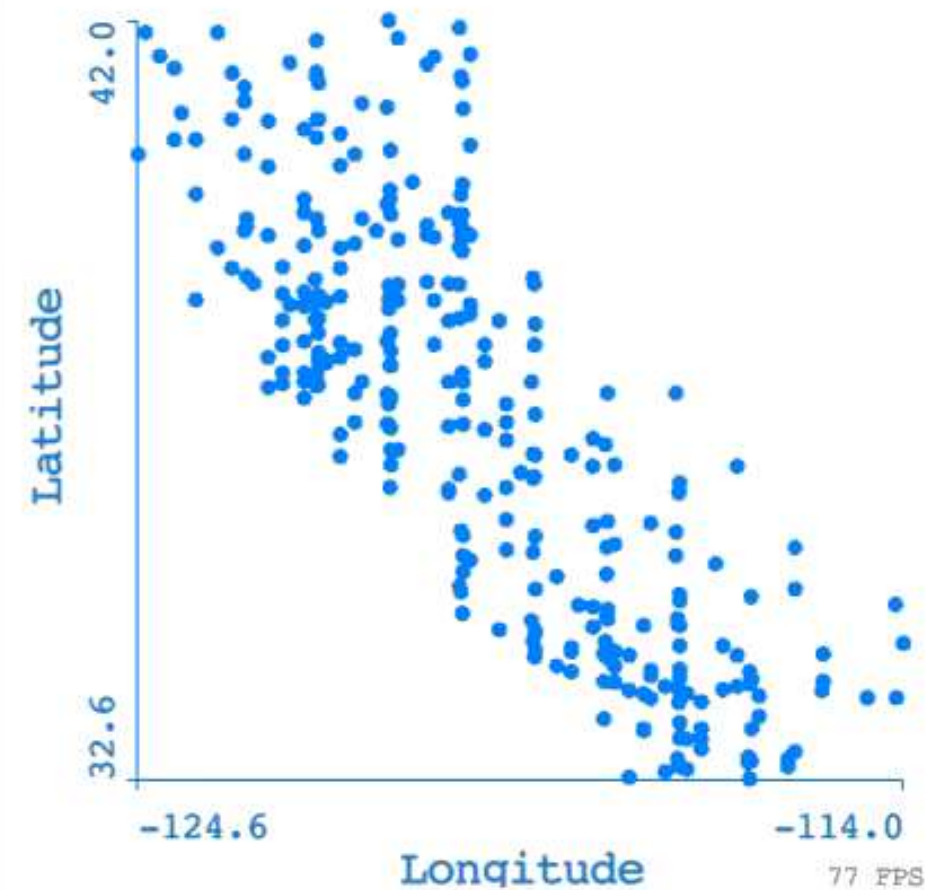
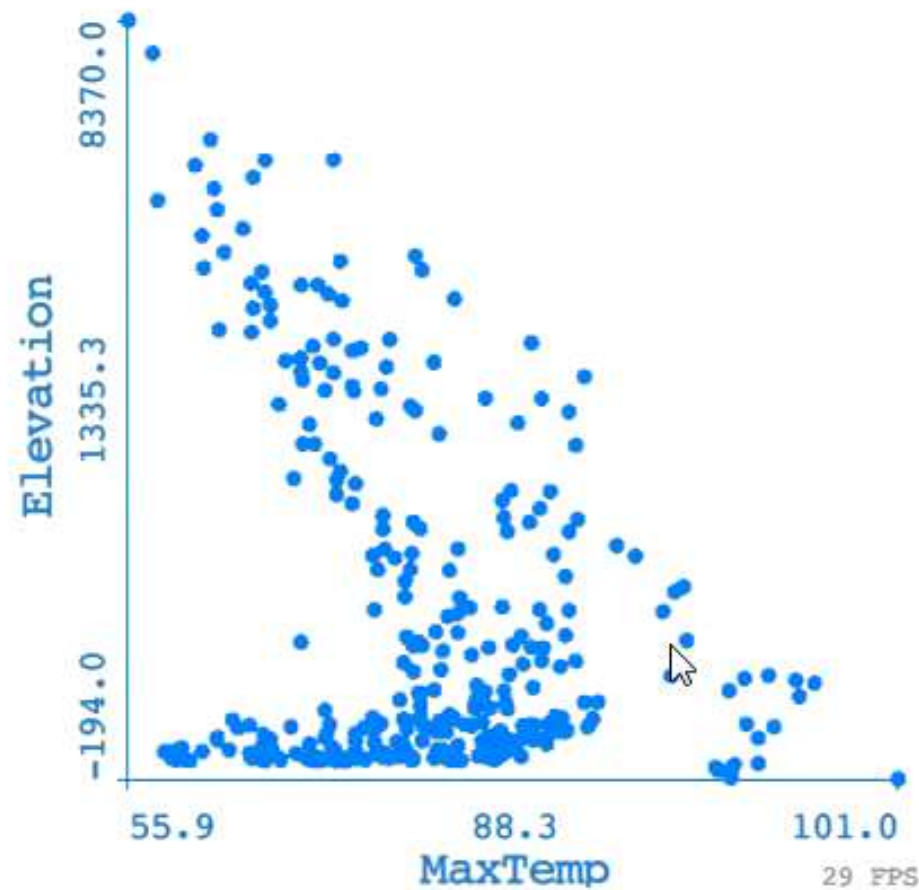
Brush size

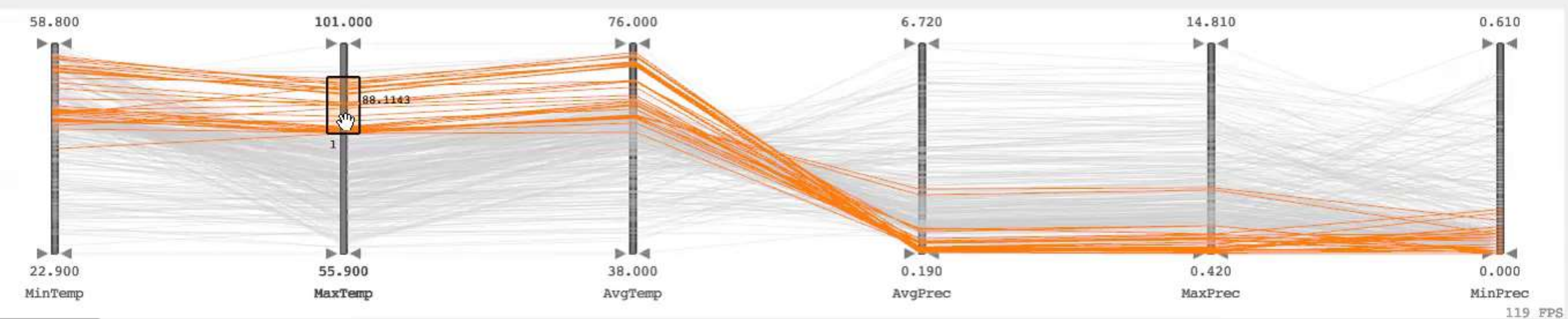
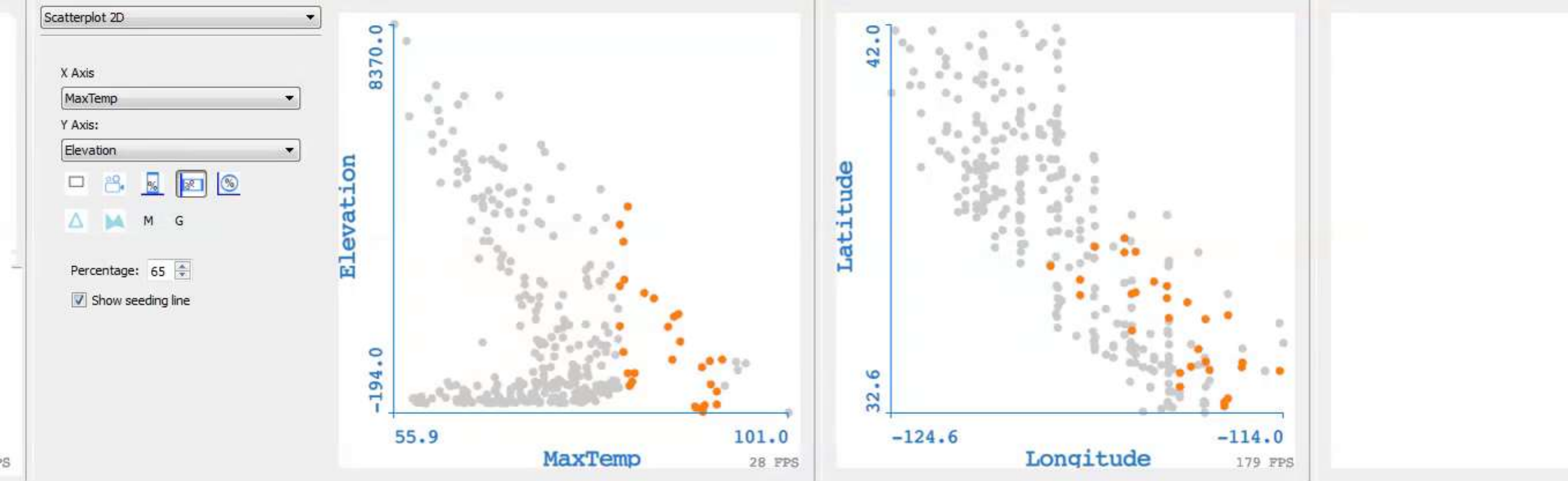
To specify the size of brush,
enter percentage of the data
items to select

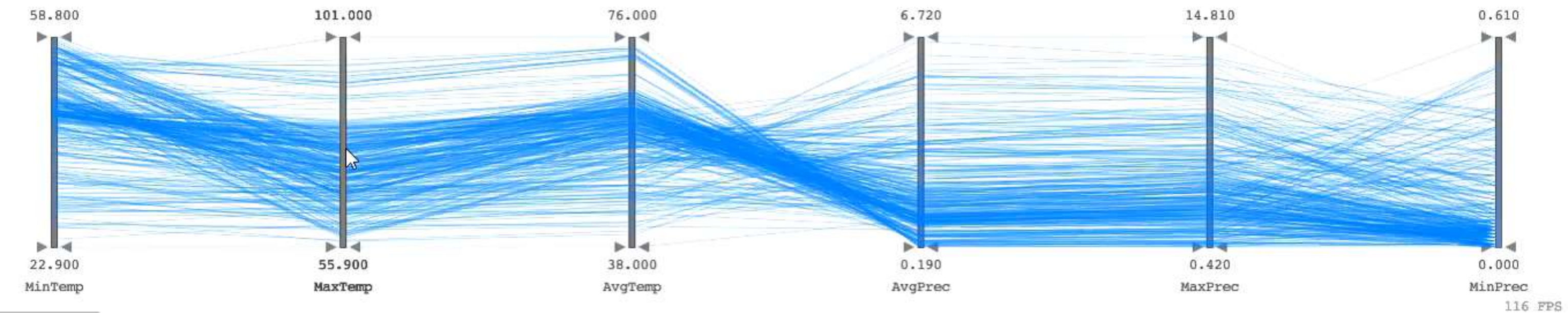
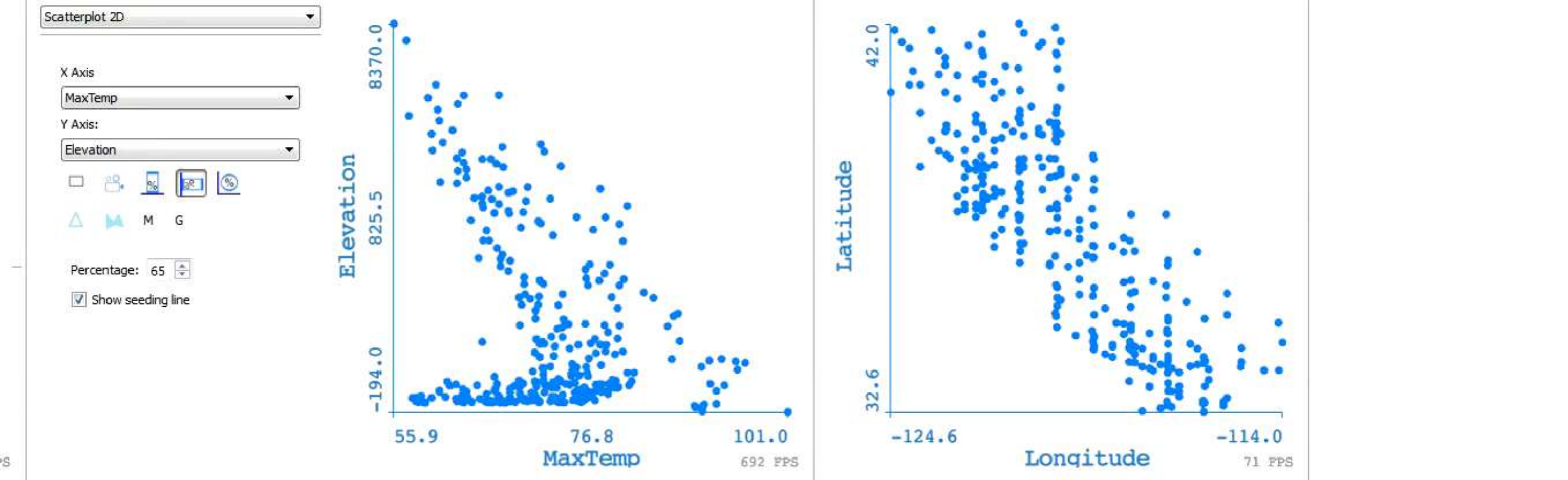
10

Seeding line

☐ Show seeding line for selected brus







Scatterplot 2D

X Axis

MaxTemp

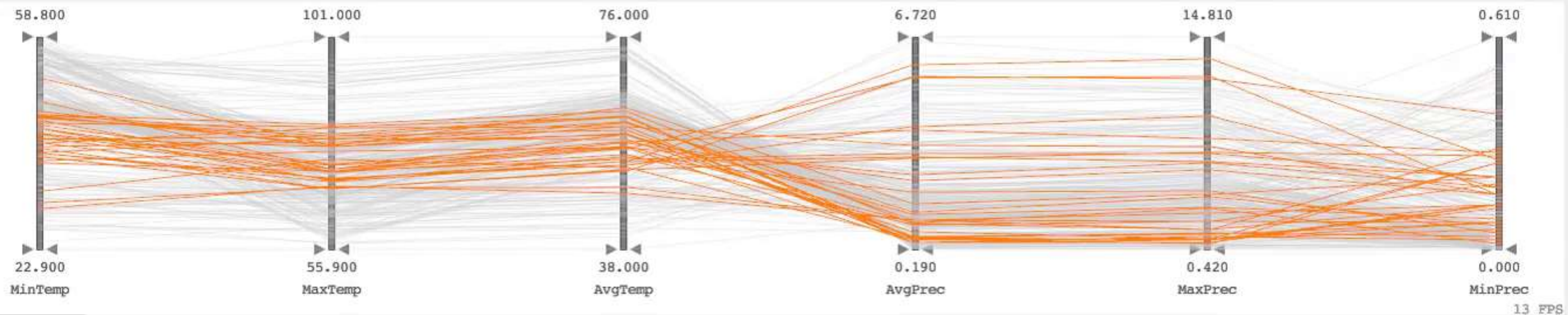
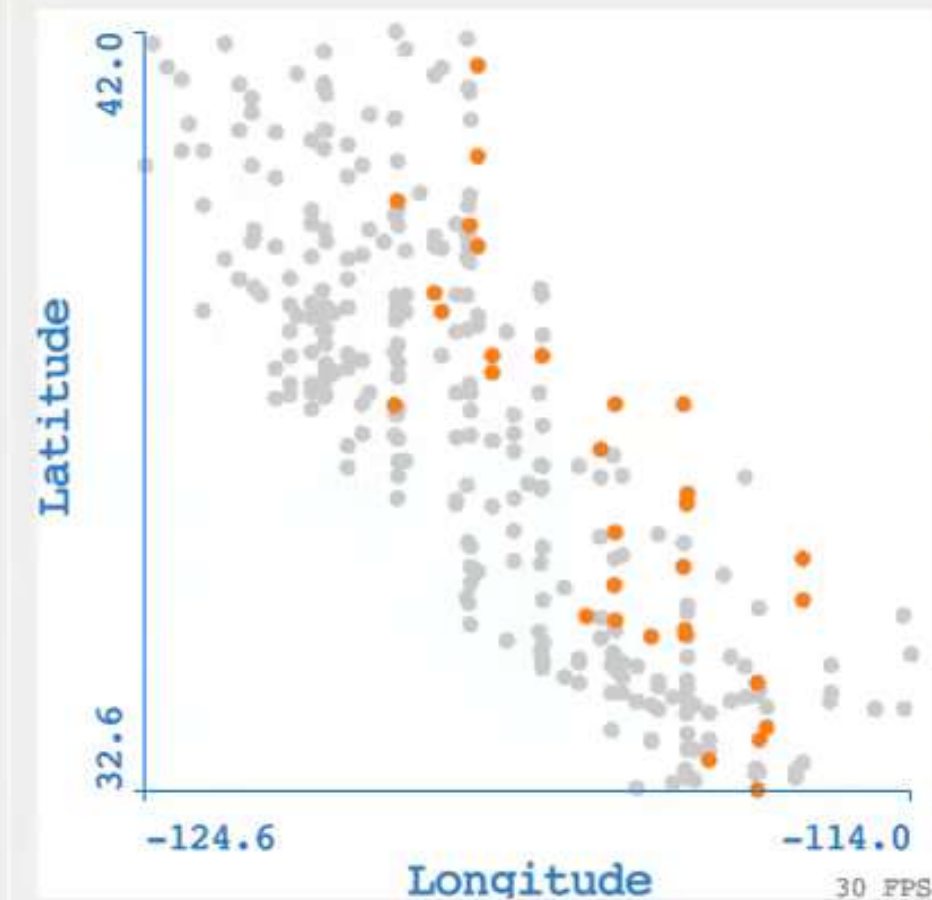
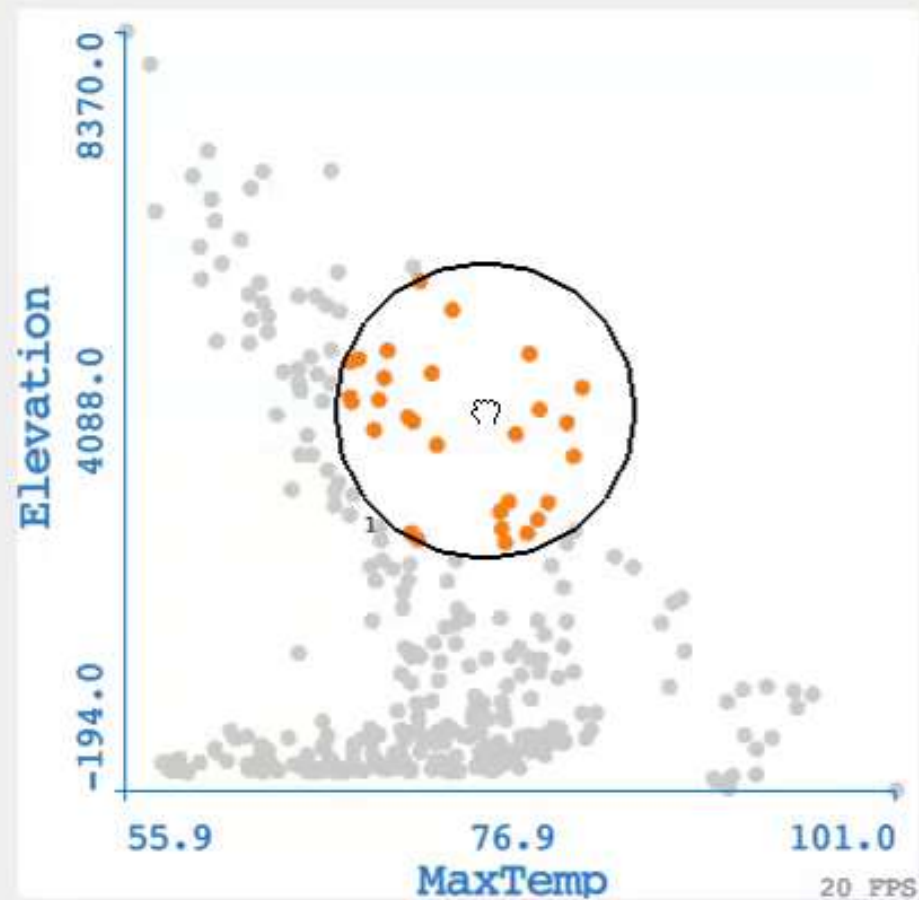
Y Axis:

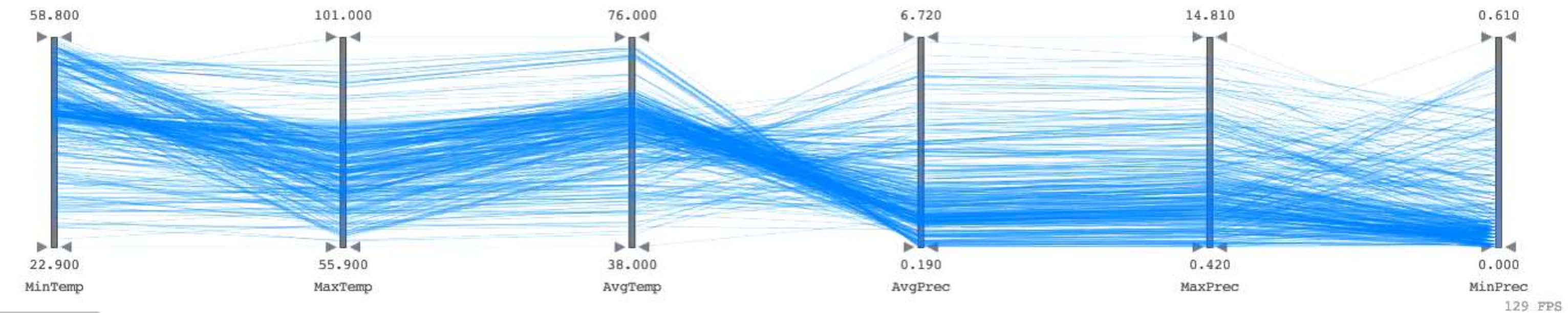
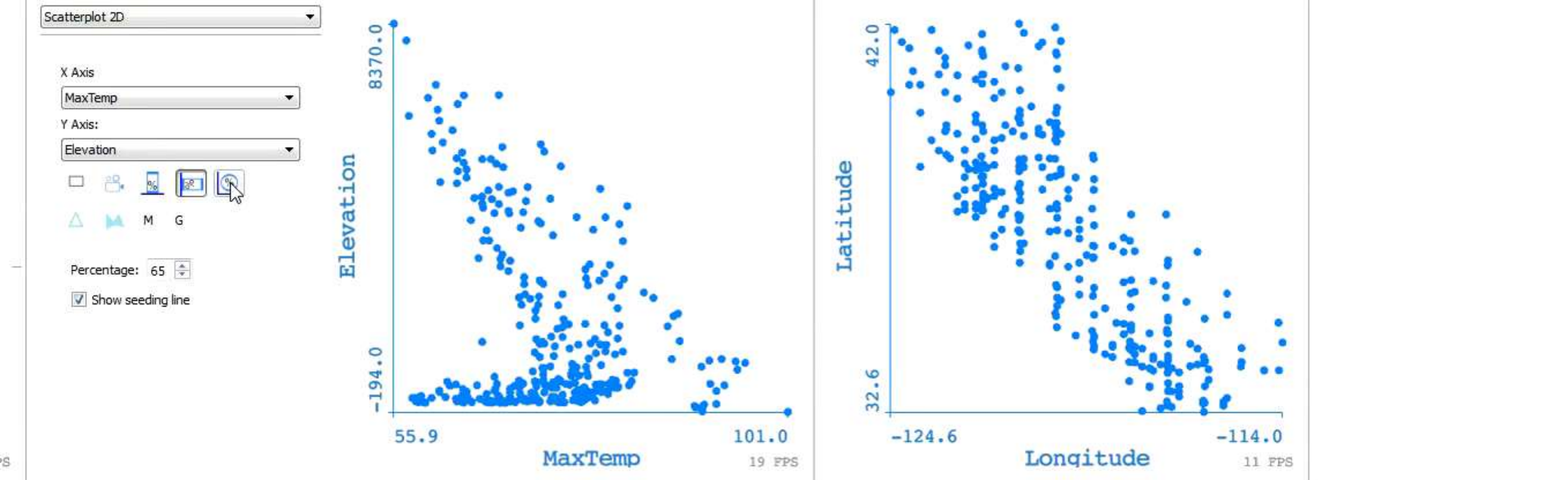
Elevation



M G

Percentage: 10

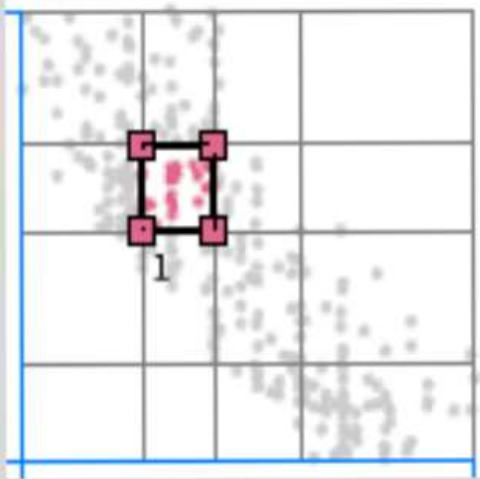




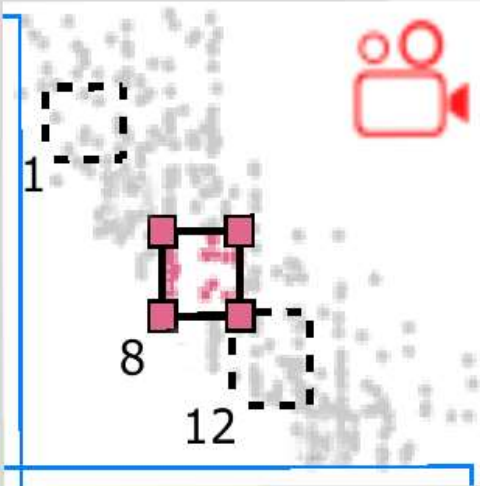
Extensions for Reproducible and Quantitative Visual Analytics

Structured Brushing

Constrained

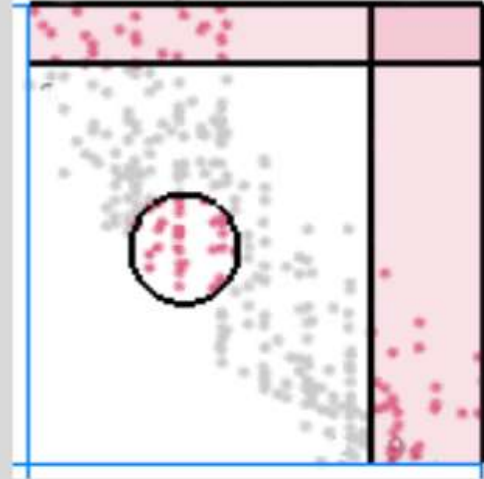


Automatic

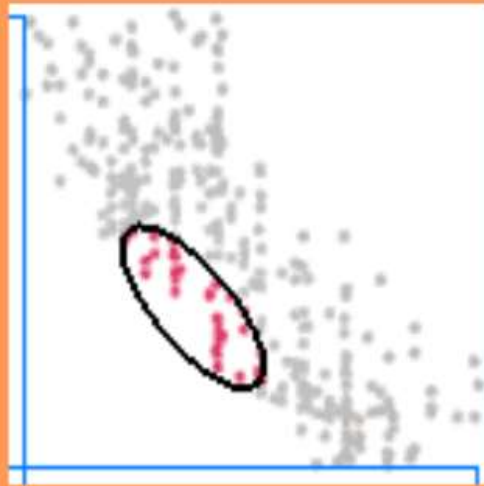


Percentile Brushes

Circle/Rectangle

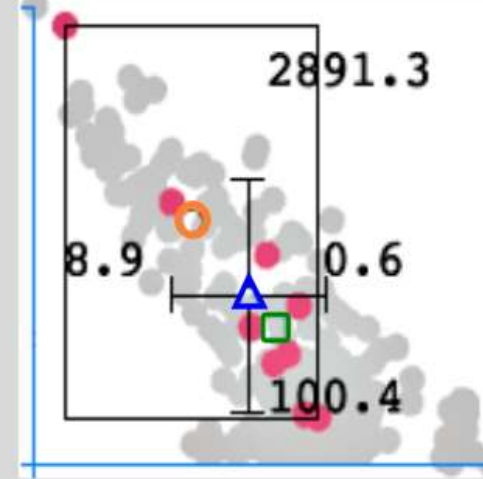


Mahalanobis



Linked Statistics

Overlays

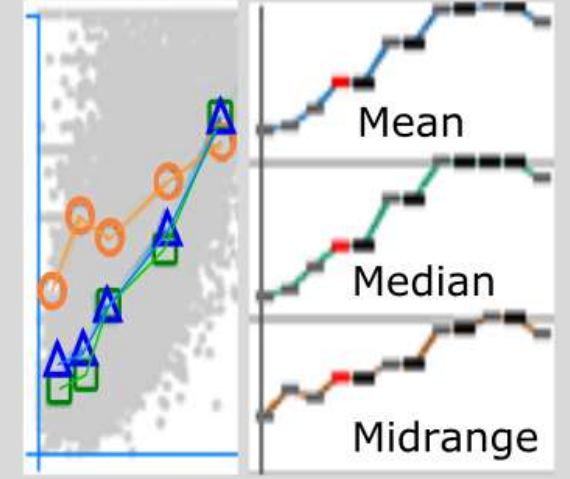


Numerical Tables

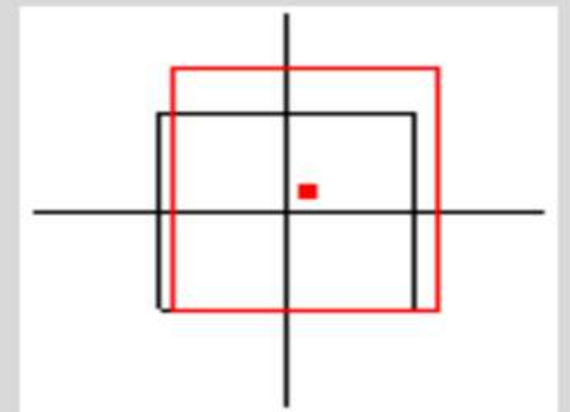
	X	Y
△ Mean	55.8	2933.6
□ Median	57.9	2355.0
○ Midrange	51.0	4329.0
Spread	6.4	2175.1

Change Visualization

Traces/Graphs



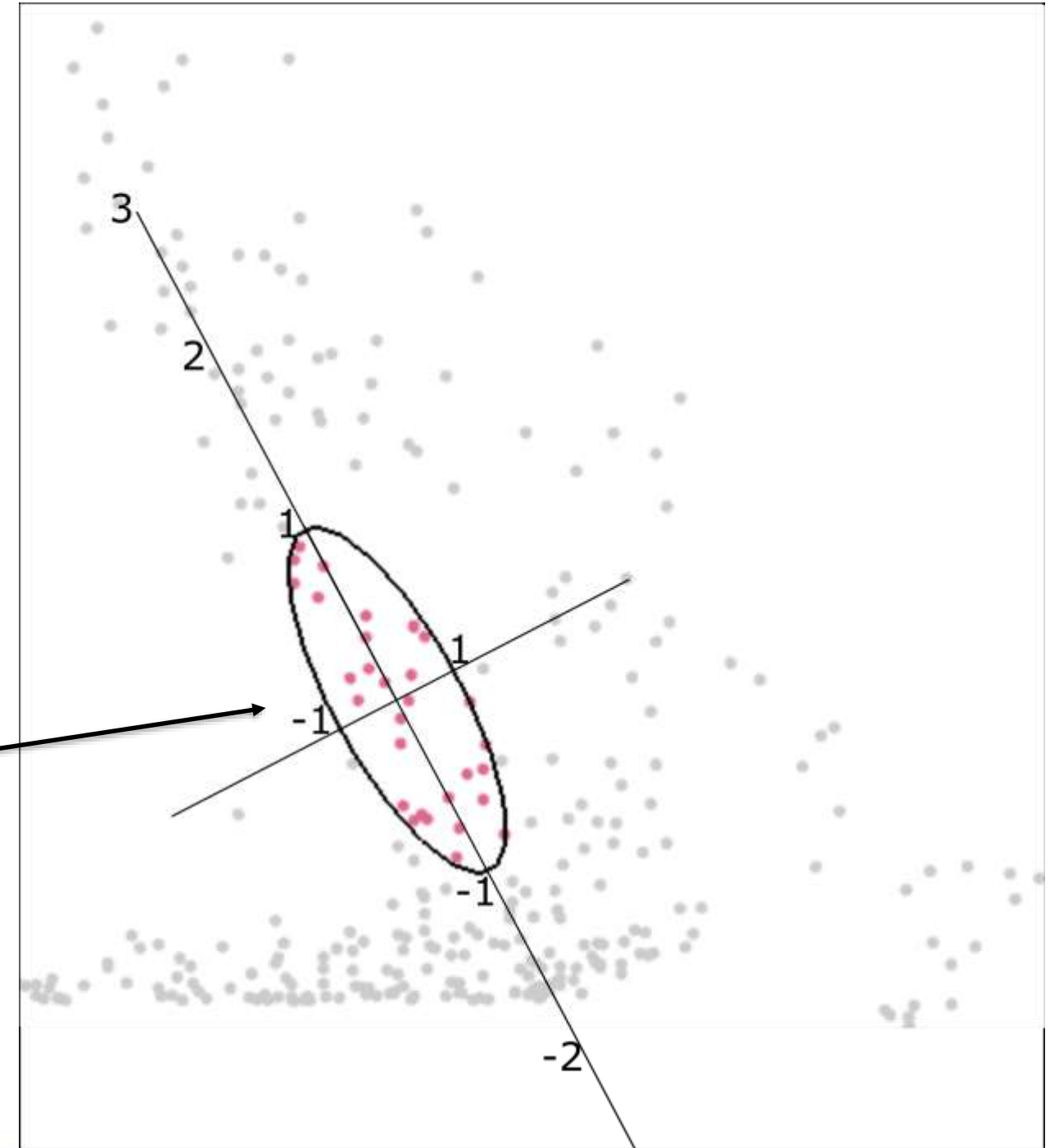
Difference-Plot



Mahalanobis Brush

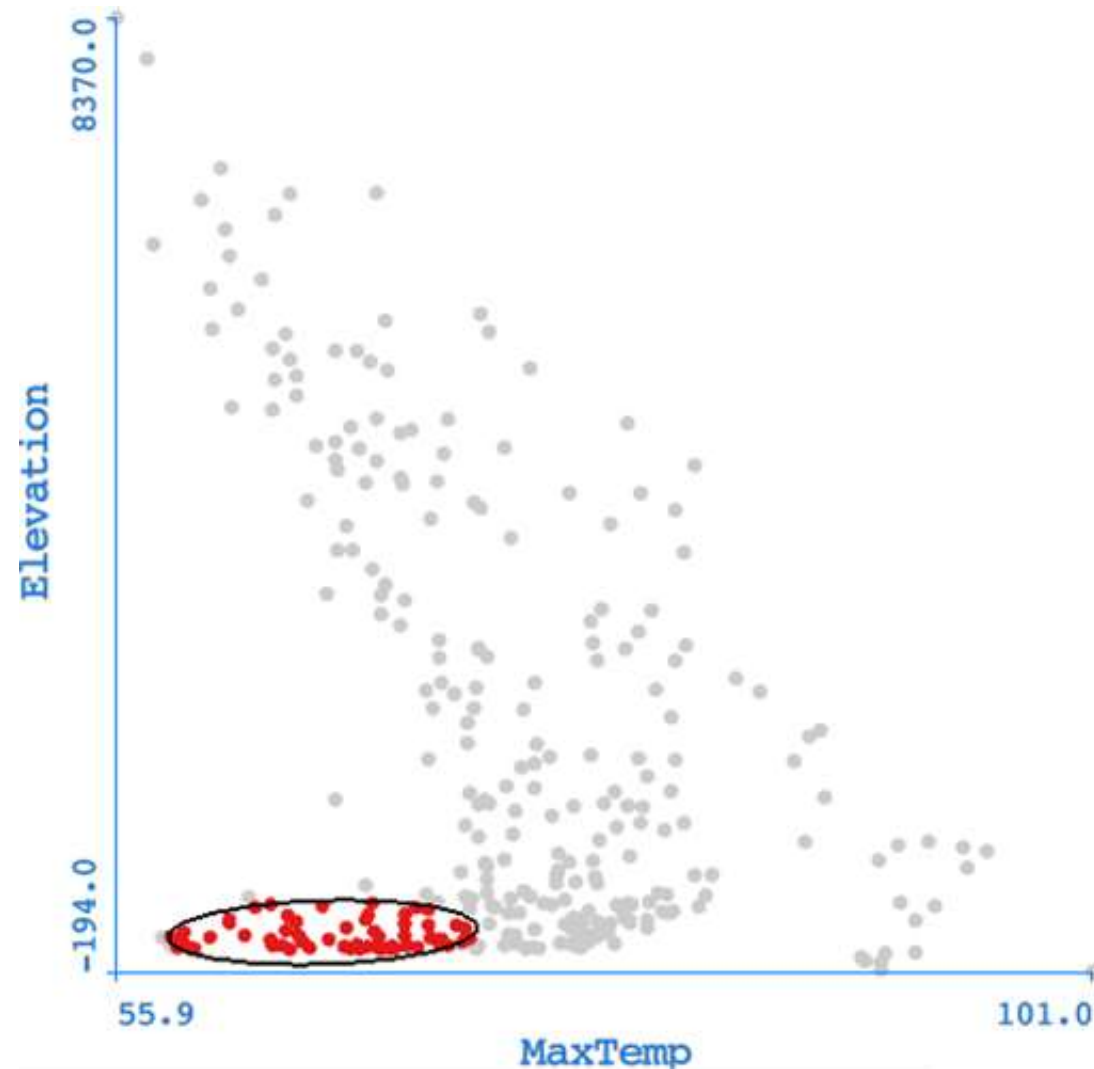
- Mahalanobis distance [Mah36]
- takes the data distribution into account

Mahalanobis brush

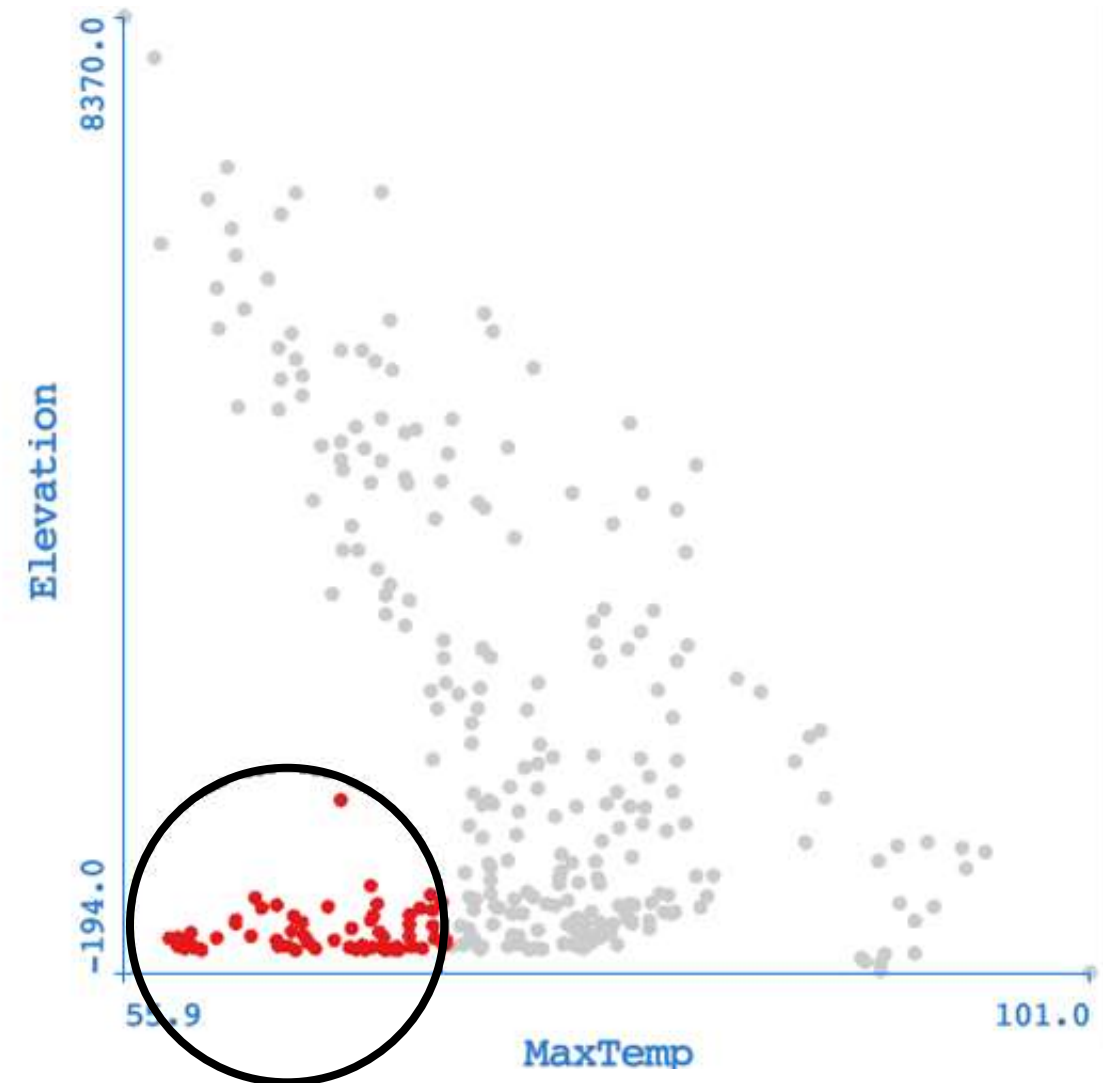


Mahalanobis Brush

Mahalanobis brush (20%)



Circle brush (20%)



Scatterplot 2D

X Axis

MaxTemp

Y Axis:

Elevation

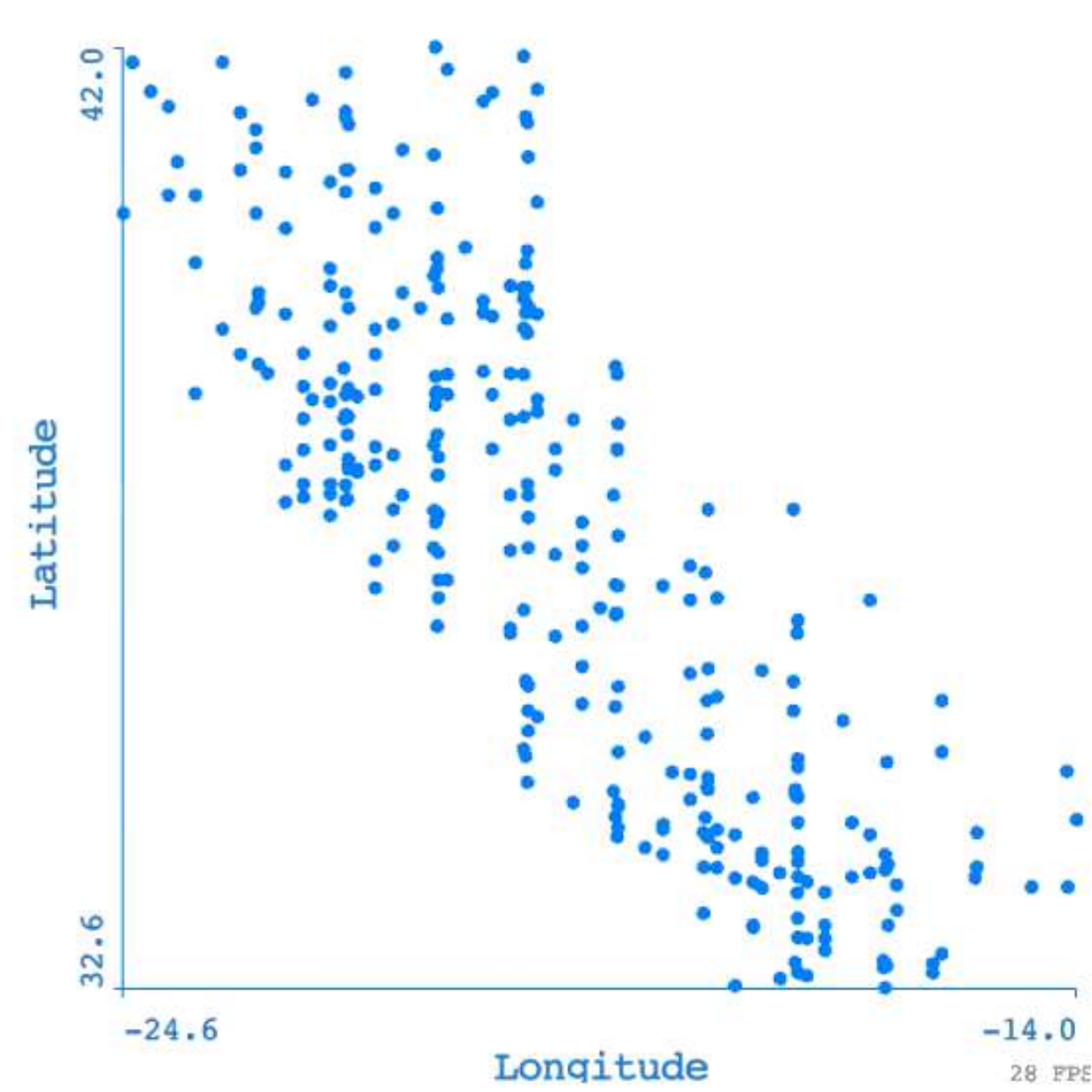
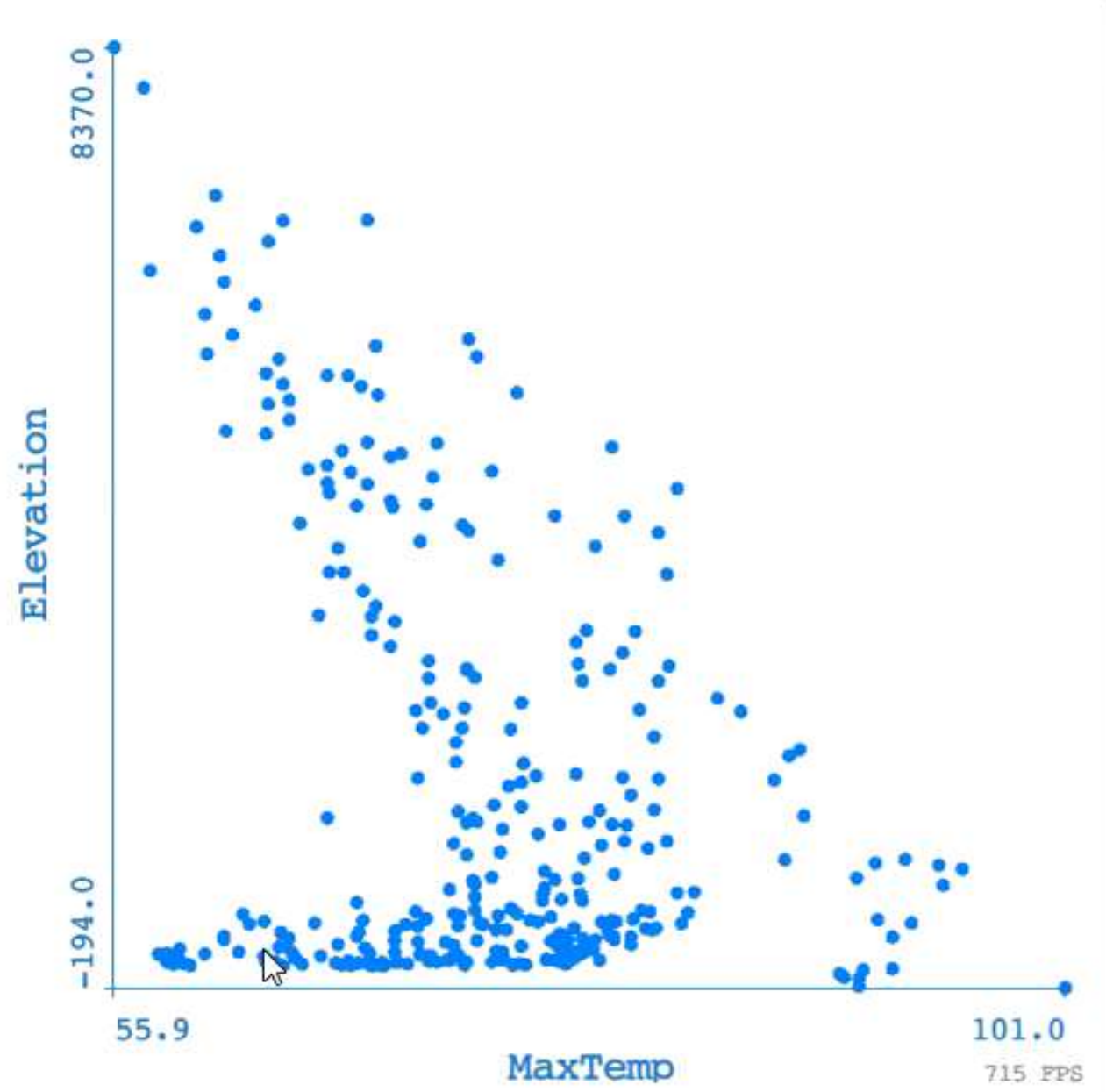


M G

Percentage: 20

Percentage md: 20

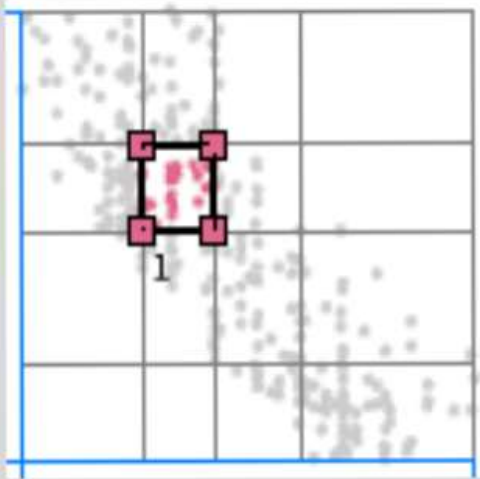
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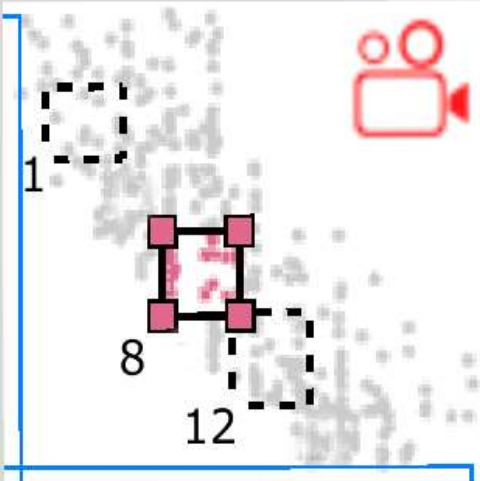
Extensions for Reproducible and Quantitative Visual Analytics

Structured Brushing

Constrained

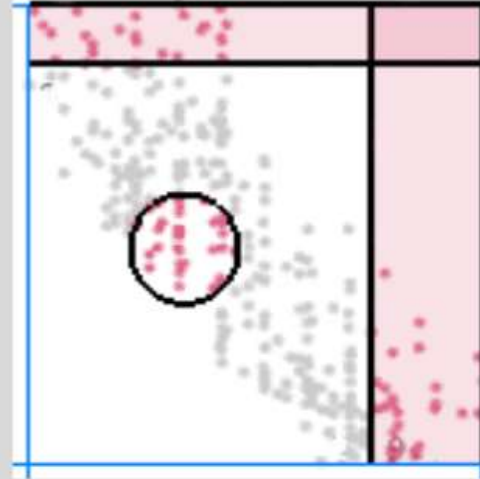


Automatic

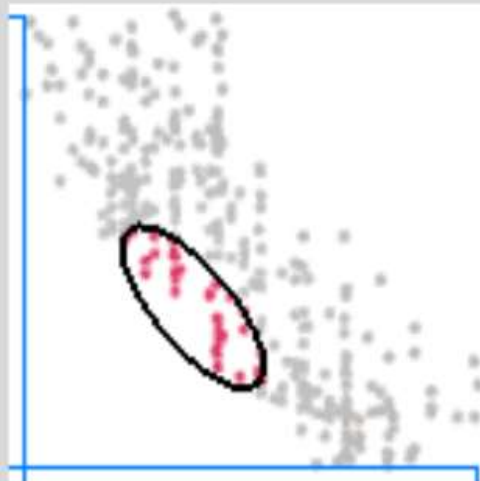


Percentile Brushes

Circle/Rectangle

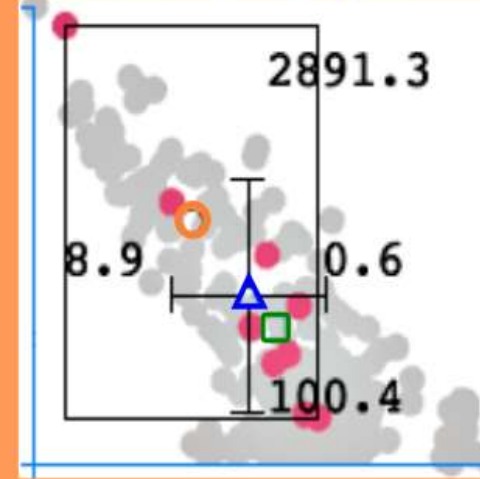


Mahalanobis



Linked Statistics

Overlays

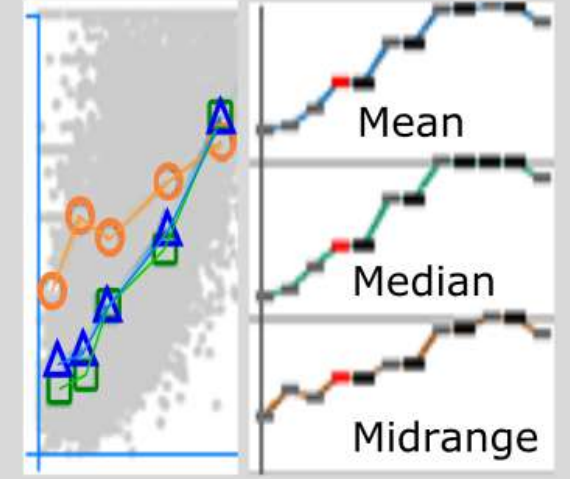


Numerical Tables

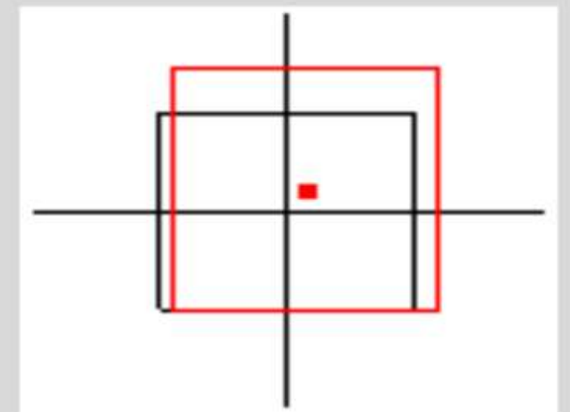
	X	Y
△ Mean	55.8	2933.6
□ Median	57.9	2355.0
○ Midrange	51.0	4329.0
Spread	6.4	2175.1

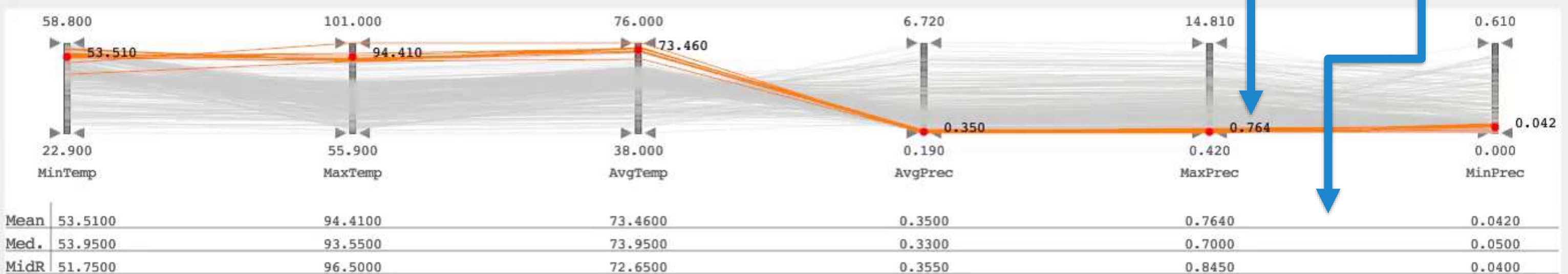
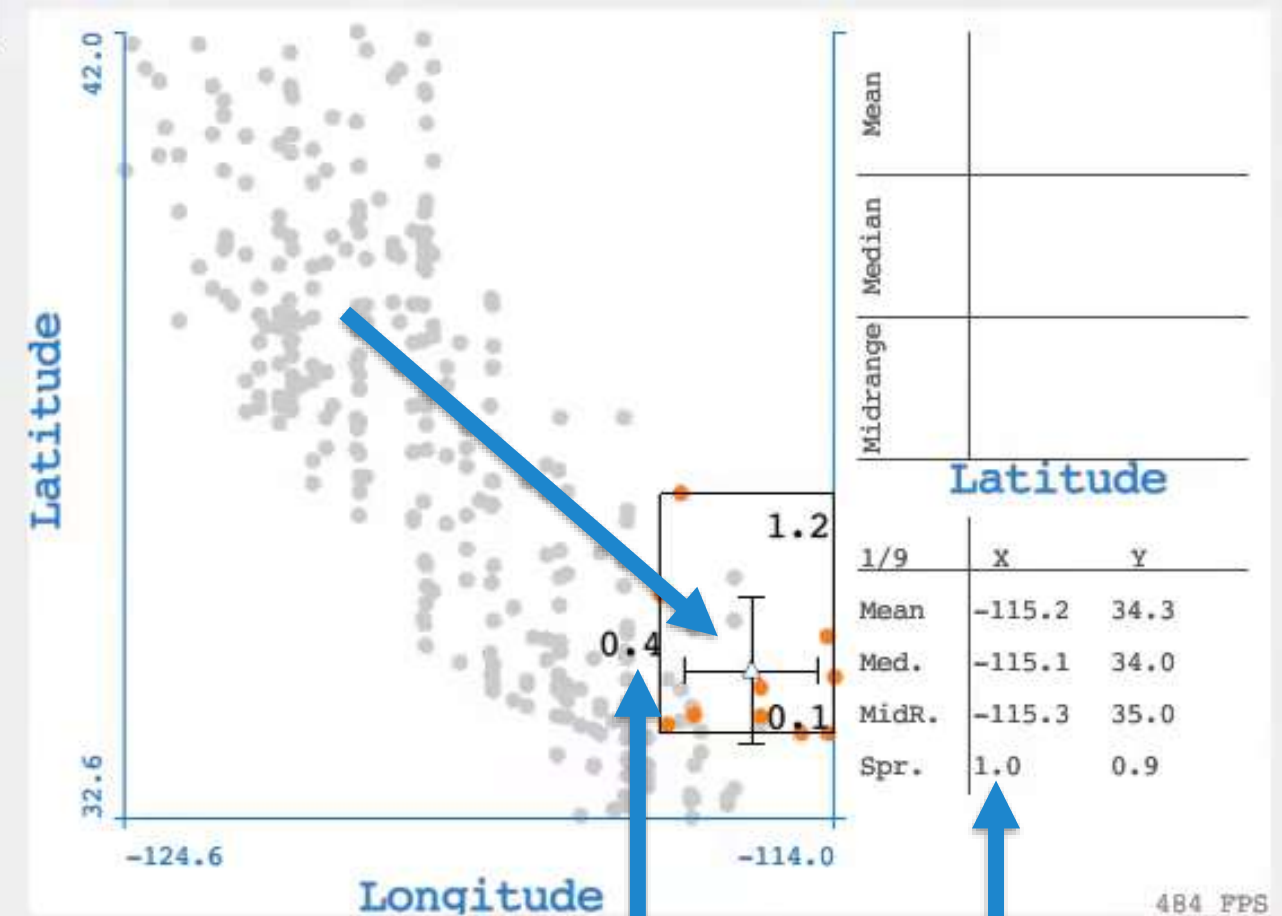
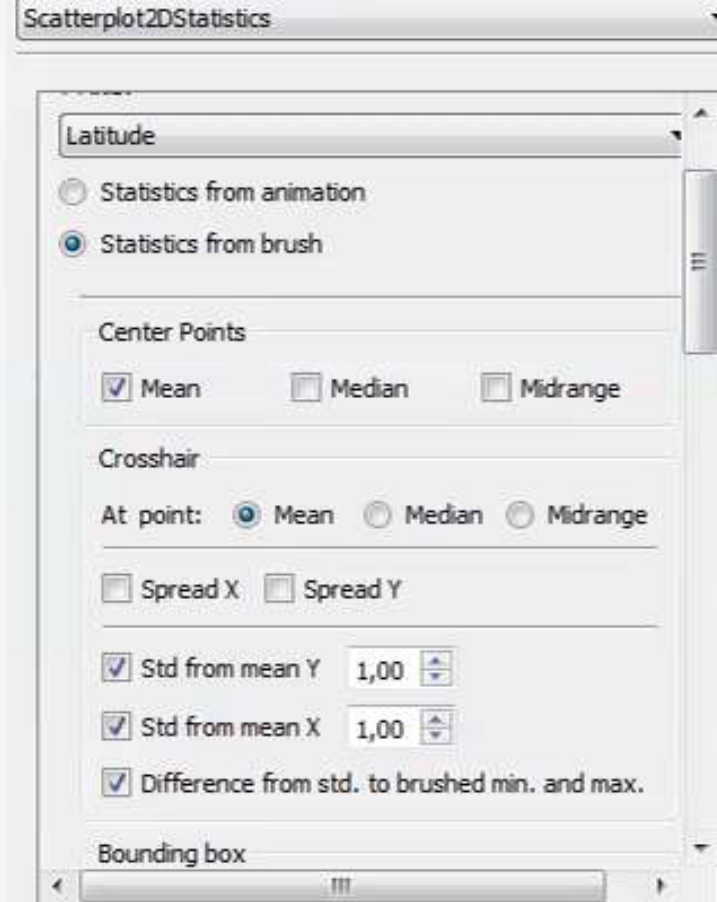
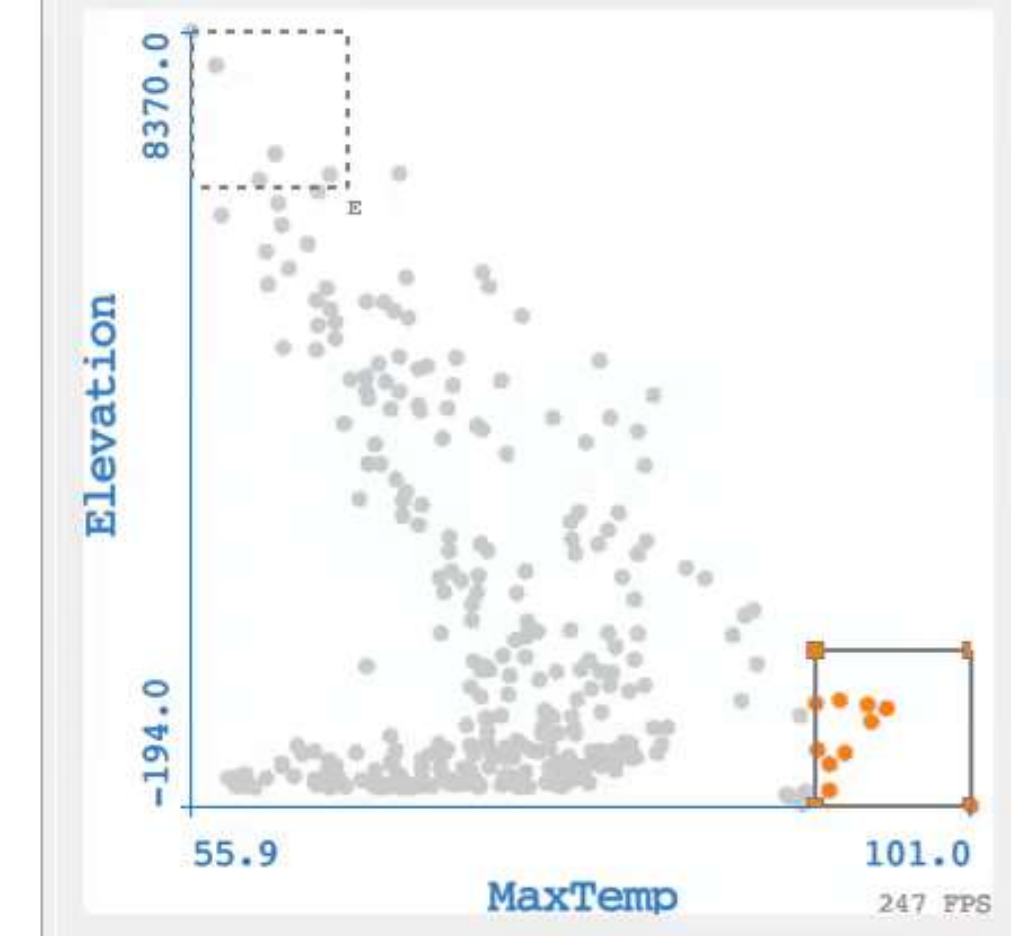
Change Visualization

Traces/Graphs



Difference-Plot

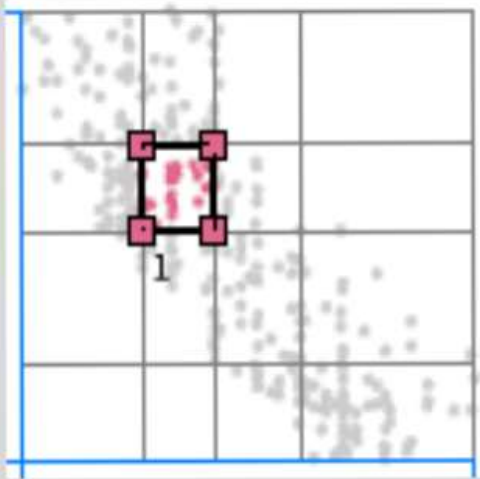




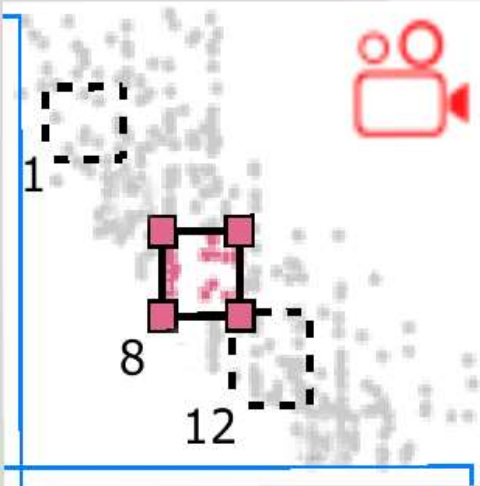
Extensions for Reproducible and Quantitative Visual Analytics

Structured Brushing

Constrained

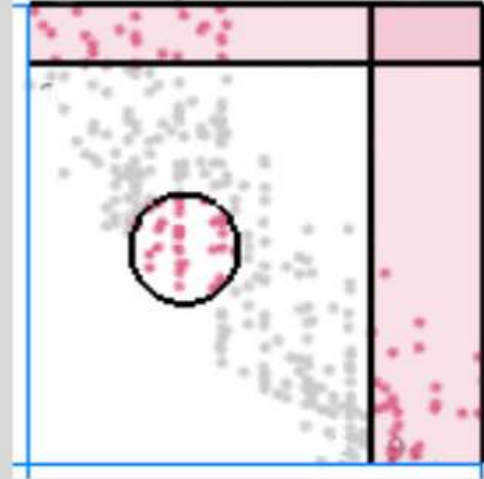


Automatic

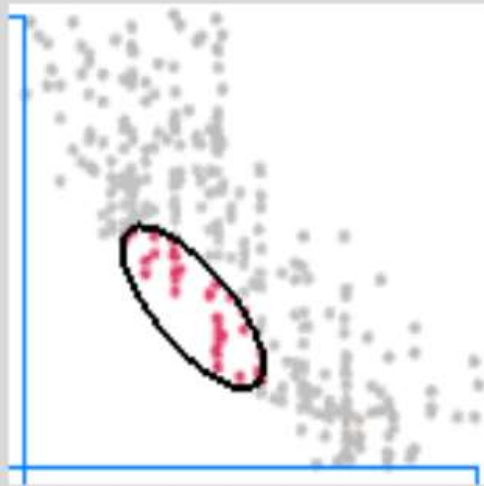


Percentile Brushes

Circle/Rectangle

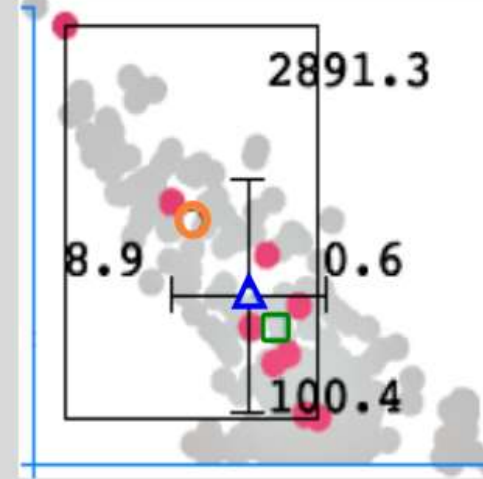


Mahalanobis



Linked Statistics

Overlays

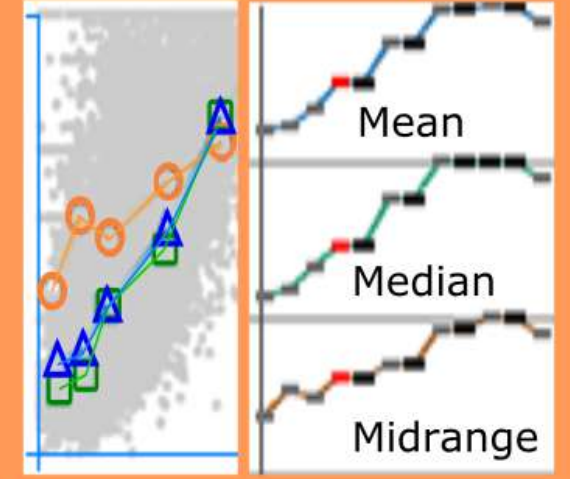


Numerical Tables

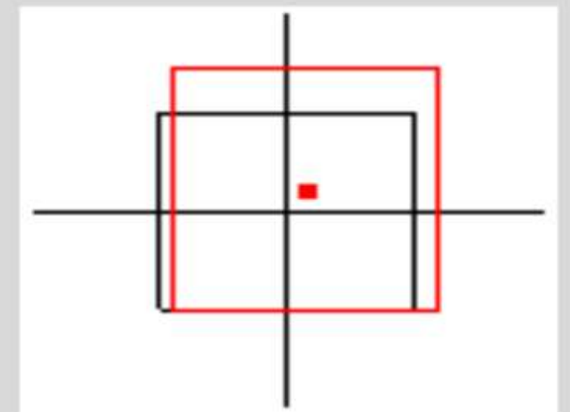
	X	Y
△ Mean	55.8	2933.6
□ Median	57.9	2355.0
○ Midrange	51.0	4329.0
Spread	6.4	2175.1

Change Visualization

Traces/Graphs

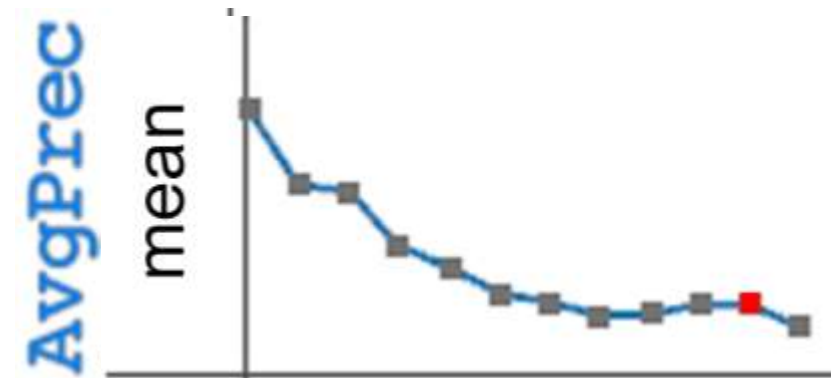
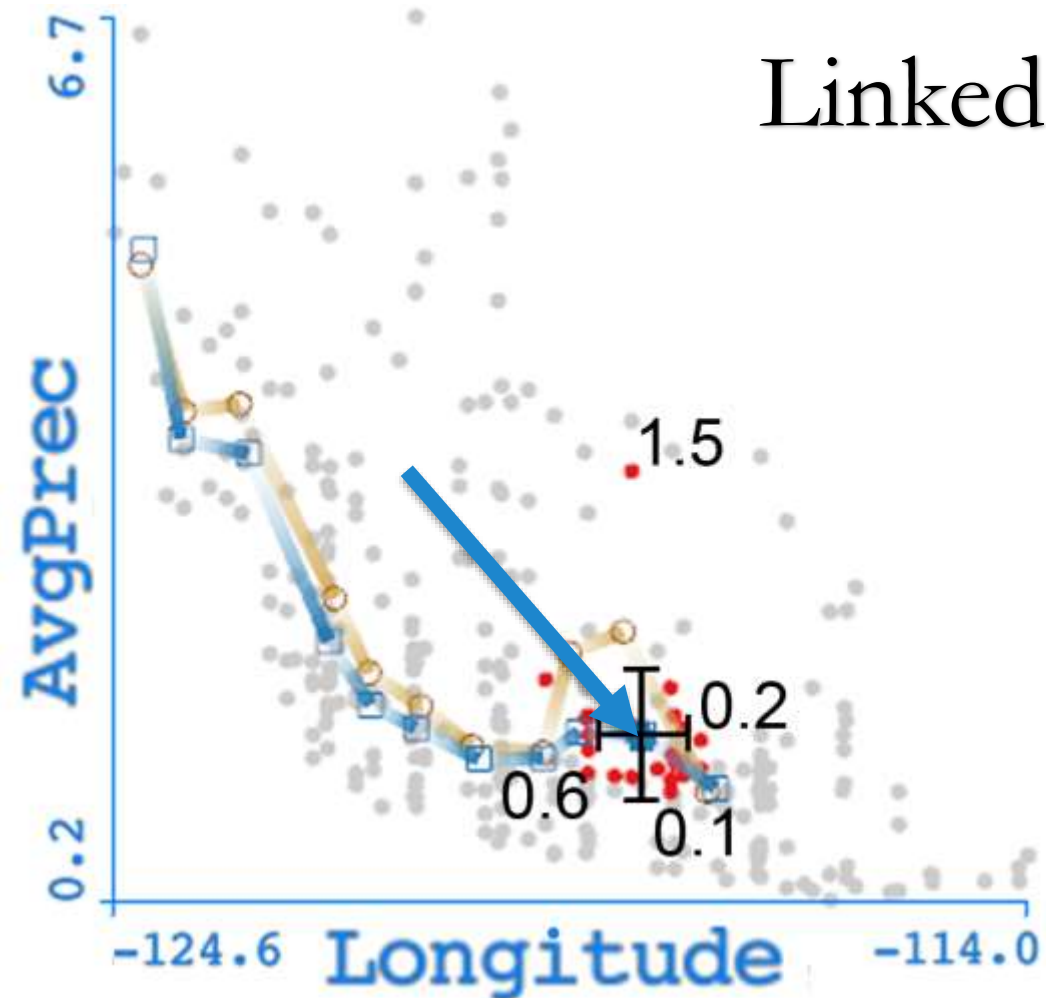


Difference-Plot



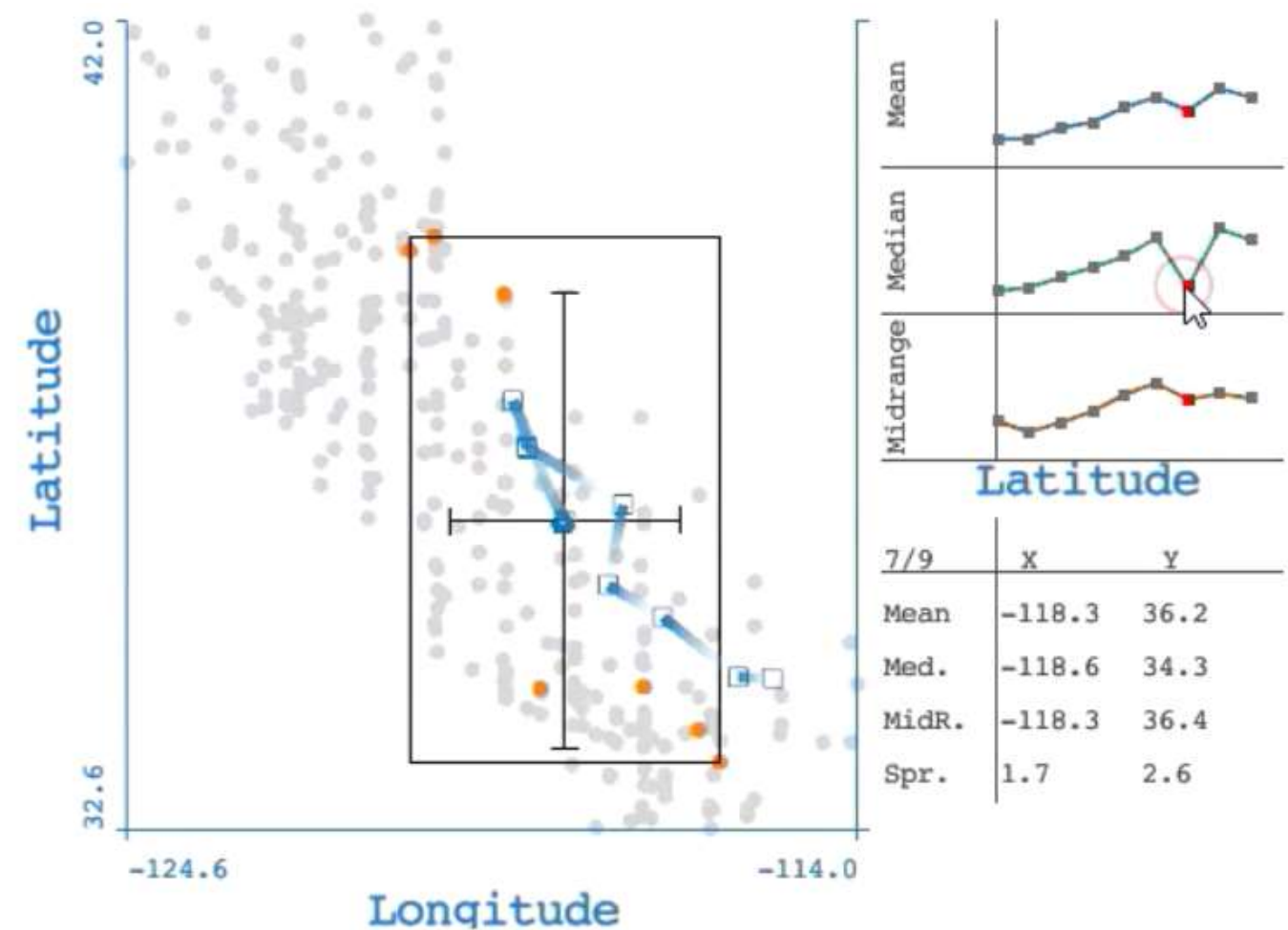
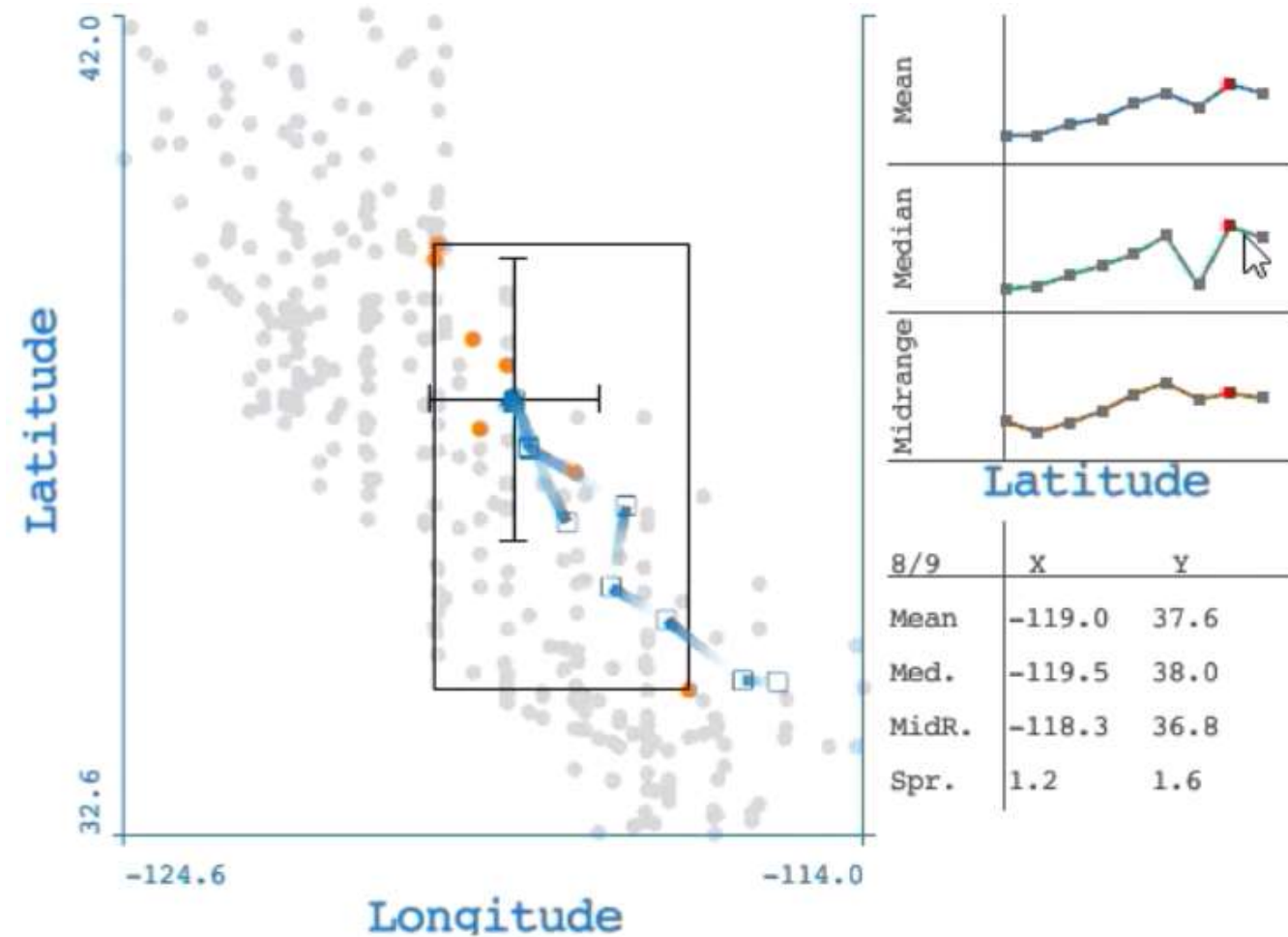
Traces/Graphs

- Summarization of the brushed data along the path



Traces/Graphs

- Summarization of the brushed data along the path



Innovations in Terms of Linking and Brushing

- Reproducibility is supported through **constrained brushing**

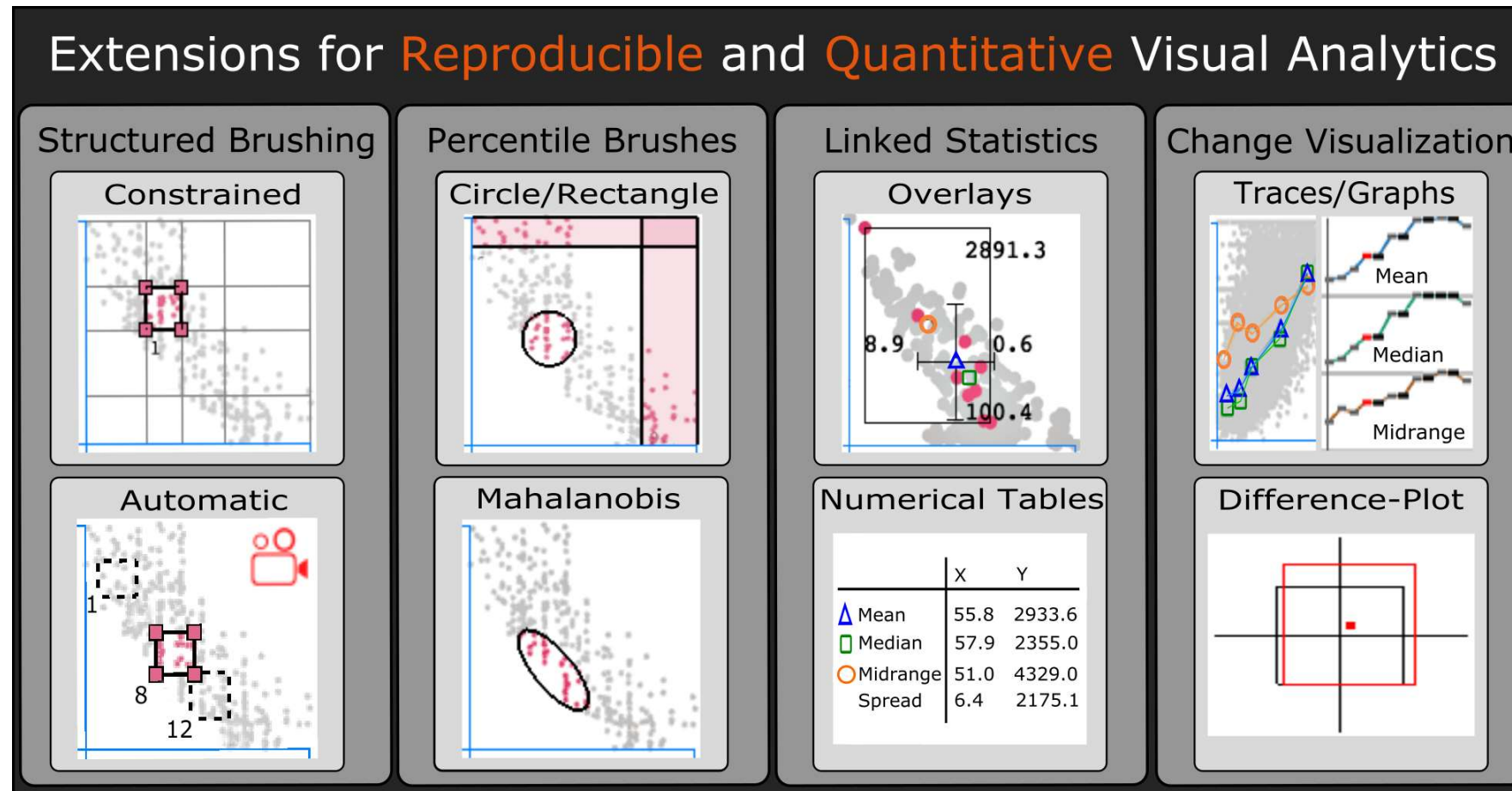
Innovations in Terms of Linking and Brushing

- Reproducibility is supported through **constrained brushing**
- Interpretability and quantitative insight are enhanced with **percentile brushing**
 - percentile grid
 - percentile brushes
 - Mahalanobis brush

Innovations in Terms of Linking and Brushing

- Reproducibility is supported through **constrained brushing**
- Interpretability and quantitative insight are enhanced with **percentile brushing**
 - percentile grid
 - percentile brushes
 - Mahalanobis brush
- Focus only on interpretation of linked views using **animated brushing** and **overly statistics**

Towards Quantitative Visual Analytics with Structured Brushing and Linked Statistics



Thank you for your attention!

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